Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-23 12:02:44
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 10
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
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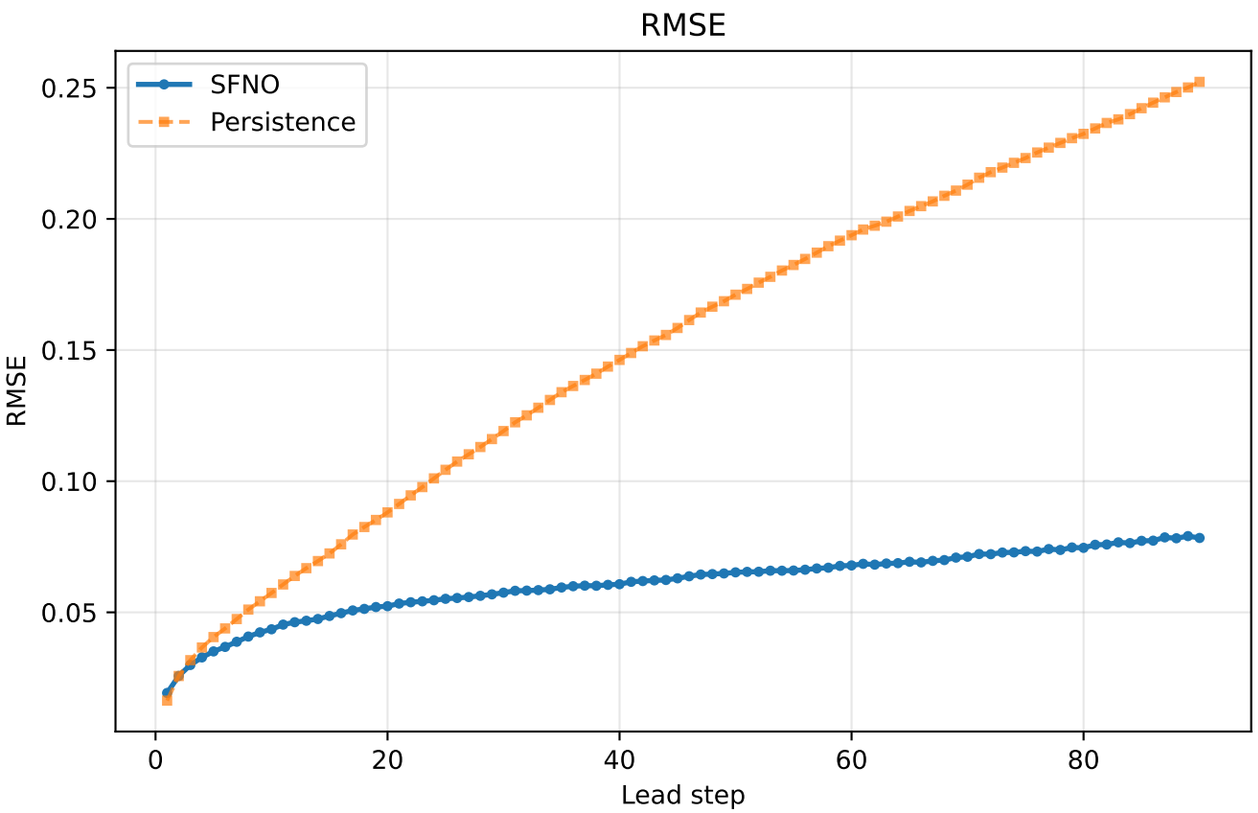
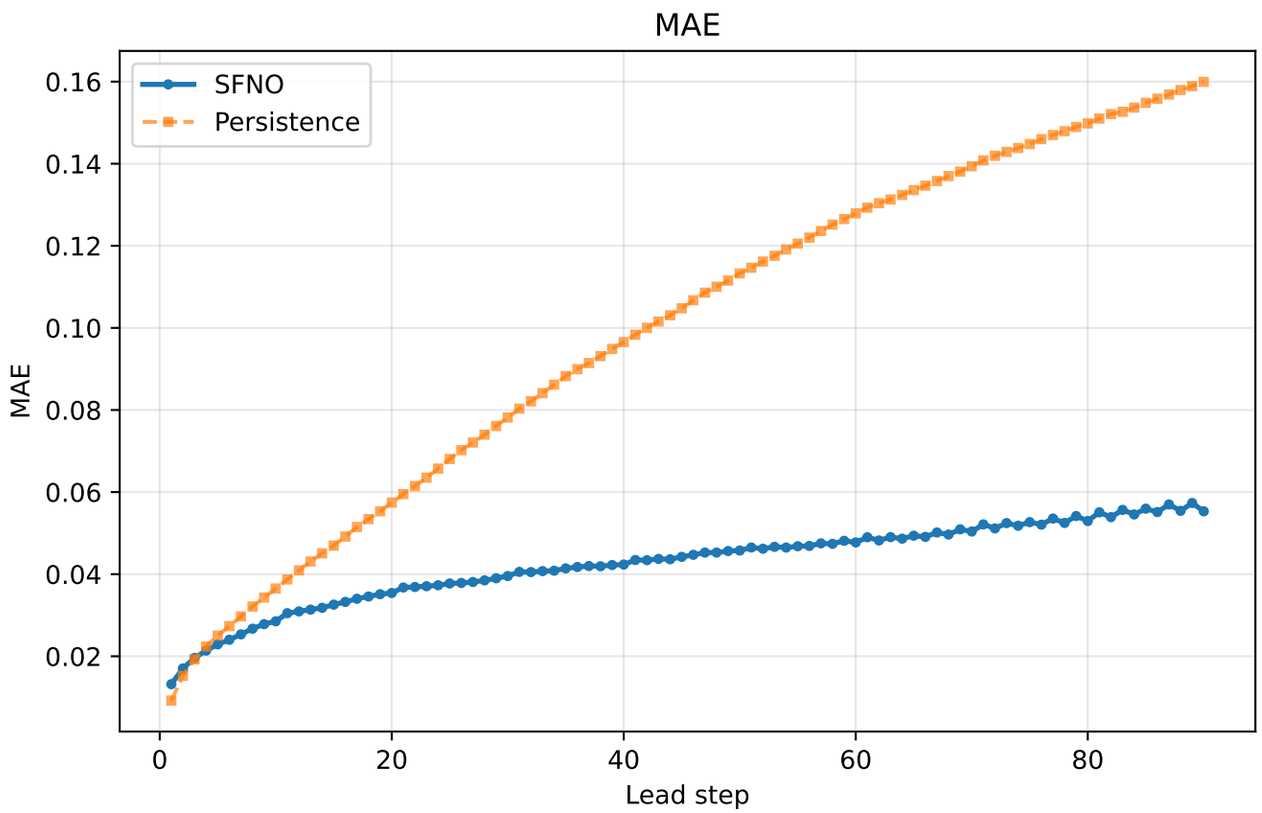
Metrics by Lead Step — Validation

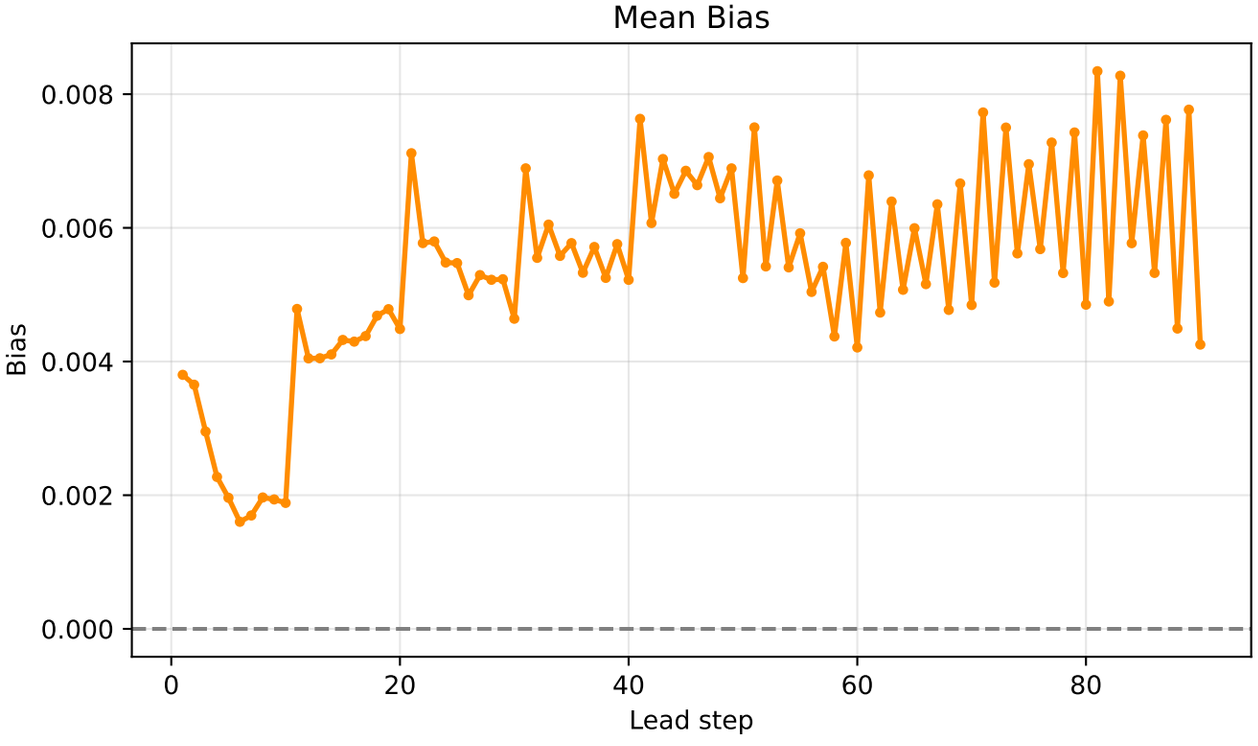
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0193	0.0164	-17.8%	0.0132	0.0038	0.9482	0.9592	0.9996
2	0.0257	0.0257	-0.1%	0.0170	0.0037	0.9011	0.8991	0.9994
3	0.0299	0.0318	5.8%	0.0196	0.0030	0.8604	0.8481	0.9991
4	0.0328	0.0366	10.3%	0.0214	0.0023	0.8286	0.8008	0.9990
5	0.0351	0.0406	13.5%	0.0229	0.0020	0.7993	0.7574	0.9988
6	0.0368	0.0439	16.1%	0.0240	0.0016	0.7757	0.7202	0.9987
7	0.0388	0.0474	18.3%	0.0253	0.0017	0.7512	0.6815	0.9985
8	0.0408	0.0510	20.1%	0.0267	0.0020	0.7255	0.6425	0.9984
9	0.0424	0.0542	21.9%	0.0278	0.0019	0.7015	0.6072	0.9983
10	0.0435	0.0574	24.1%	0.0285	0.0019	0.6812	0.5705	0.9982
11	0.0453	0.0606	25.2%	0.0305	0.0048	0.6551	0.5345	0.9980
12	0.0462	0.0639	27.6%	0.0309	0.0040	0.6372	0.4984	0.9979
13	0.0468	0.0668	30.0%	0.0314	0.0040	0.6184	0.4642	0.9979
14	0.0475	0.0695	31.8%	0.0318	0.0041	0.6002	0.4363	0.9978
15	0.0486	0.0725	32.9%	0.0326	0.0043	0.5814	0.4112	0.9977
16	0.0497	0.0759	34.6%	0.0332	0.0043	0.5638	0.3812	0.9976
17	0.0507	0.0797	36.3%	0.0340	0.0044	0.5450	0.3482	0.9975
18	0.0513	0.0825	37.8%	0.0346	0.0047	0.5247	0.3242	0.9974
19	0.0520	0.0852	39.0%	0.0351	0.0048	0.5056	0.3058	0.9974
20	0.0523	0.0881	40.6%	0.0354	0.0045	0.4889	0.2821	0.9973
21	0.0533	0.0913	41.6%	0.0368	0.0071	0.4737	0.2589	0.9972
22	0.0538	0.0946	43.1%	0.0369	0.0058	0.4646	0.2377	0.9972
23	0.0542	0.0978	44.6%	0.0371	0.0058	0.4548	0.2157	0.9971
24	0.0546	0.1011	46.0%	0.0373	0.0055	0.4410	0.1932	0.9971
25	0.0552	0.1043	47.1%	0.0377	0.0055	0.4273	0.1725	0.9970
26	0.0555	0.1075	48.4%	0.0379	0.0050	0.4162	0.1551	0.9970
27	0.0558	0.1103	49.4%	0.0381	0.0053	0.4017	0.1418	0.9970
28	0.0563	0.1130	50.2%	0.0385	0.0052	0.3883	0.1313	0.9969
29	0.0569	0.1161	51.0%	0.0390	0.0052	0.3720	0.1164	0.9969
30	0.0575	0.1191	51.7%	0.0396	0.0046	0.3568	0.1038	0.9968
31	0.0582	0.1225	52.5%	0.0406	0.0069	0.3470	0.0889	0.9967
32	0.0583	0.1251	53.4%	0.0405	0.0056	0.3420	0.0821	0.9967
33	0.0585	0.1280	54.3%	0.0407	0.0060	0.3334	0.0694	0.9967
34	0.0588	0.1310	55.1%	0.0408	0.0056	0.3264	0.0582	0.9966
35	0.0595	0.1339	55.6%	0.0414	0.0058	0.3176	0.0464	0.9966
36	0.0600	0.1363	56.0%	0.0418	0.0053	0.3036	0.0400	0.9965
37	0.0602	0.1386	56.6%	0.0420	0.0057	0.2932	0.0378	0.9965
38	0.0601	0.1410	57.4%	0.0419	0.0053	0.2842	0.0334	0.9965
39	0.0605	0.1437	57.9%	0.0422	0.0058	0.2726	0.0265	0.9964
40	0.0607	0.1462	58.5%	0.0423	0.0052	0.2615	0.0227	0.9964
41	0.0616	0.1489	58.6%	0.0435	0.0076	0.2441	0.0178	0.9963
42	0.0619	0.1514	59.1%	0.0434	0.0061	0.2373	0.0118	0.9963
43	0.0622	0.1536	59.5%	0.0437	0.0070	0.2270	0.0084	0.9962
44	0.0623	0.1558	60.0%	0.0436	0.0065	0.2168	0.0082	0.9962
45	0.0630	0.1584	60.3%	0.0442	0.0069	0.2051	-0.0016	0.9961
46	0.0637	0.1614	60.5%	0.0447	0.0066	0.1916	-0.0150	0.9960
47	0.0644	0.1643	60.8%	0.0453	0.0071	0.1793	-0.0262	0.9960
48	0.0646	0.1665	61.2%	0.0453	0.0064	0.1718	-0.0296	0.9959
49	0.0648	0.1686	61.6%	0.0456	0.0069	0.1649	-0.0316	0.9959
50	0.0652	0.1711	61.9%	0.0458	0.0052	0.1584	-0.0359	0.9958
51	0.0654	0.1733	62.2%	0.0465	0.0075	0.1516	-0.0392	0.9958
52	0.0655	0.1757	62.7%	0.0462	0.0054	0.1447	-0.0454	0.9958
53	0.0659	0.1779	63.0%	0.0467	0.0067	0.1386	-0.0488	0.9958
54	0.0659	0.1803	63.5%	0.0465	0.0054	0.1381	-0.0525	0.9958
55	0.0659	0.1824	63.9%	0.0468	0.0059	0.1392	-0.0540	0.9958
56	0.0663	0.1847	64.1%	0.0469	0.0050	0.1327	-0.0585	0.9957
57	0.0667	0.1871	64.3%	0.0475	0.0054	0.1314	-0.0631	0.9957
58	0.0670	0.1896	64.7%	0.0474	0.0044	0.1238	-0.0696	0.9956
59	0.0677	0.1917	64.7%	0.0481	0.0058	0.1201	-0.0708	0.9955
60	0.0679	0.1938	65.0%	0.0477	0.0042	0.1144	-0.0716	0.9955
61	0.0685	0.1959	65.0%	0.0490	0.0068	0.1165	-0.0755	0.9954
62	0.0682	0.1974	65.5%	0.0482	0.0047	0.1120	-0.0678	0.9955
63	0.0686	0.1989	65.5%	0.0490	0.0064	0.1176	-0.0586	0.9954
64	0.0687	0.2009	65.8%	0.0487	0.0051	0.1113	-0.0567	0.9954
65	0.0693	0.2030	65.9%	0.0494	0.0060	0.1092	-0.0588	0.9953
66	0.0690	0.2049	66.3%	0.0491	0.0052	0.1045	-0.0583	0.9954
67	0.0696	0.2067	66.3%	0.0502	0.0064	0.1044	-0.0556	0.9953
68	0.0699	0.2088	66.5%	0.0497	0.0048	0.0978	-0.0542	0.9952
69	0.0709	0.2108	66.4%	0.0509	0.0067	0.0956	-0.0508	0.9951
70	0.0712	0.2131	66.6%	0.0504	0.0048	0.0840	-0.0529	0.9951
71	0.0722	0.2157	66.5%	0.0521	0.0077	0.0821	-0.0598	0.9949
72	0.0722	0.2178	66.9%	0.0512	0.0052	0.0700	-0.0609	0.9949
73	0.0728	0.2196	66.8%	0.0524	0.0075	0.0703	-0.0564	0.9948
74	0.0729	0.2214	67.1%	0.0518	0.0056	0.0622	-0.0508	0.9948
75	0.0733	0.2232	67.1%	0.0527	0.0070	0.0631	-0.0483	0.9948
76	0.0732	0.2253	67.5%	0.0521	0.0057	0.0578	-0.0507	0.9948
77	0.0741	0.2272	67.4%	0.0536	0.0073	0.0543	-0.0482	0.9946
78	0.0738	0.2290	67.8%	0.0525	0.0053	0.0476	-0.0477	0.9947
79	0.0747	0.2308	67.6%	0.0542	0.0074	0.0493	-0.0476	0.9946
80	0.0746	0.2324	67.9%	0.0530	0.0049	0.0413	-0.0445	0.9946
81	0.0758	0.2345	67.7%	0.0551	0.0083	0.0421	-0.0448	0.9944
82	0.0759	0.2366	67.9%	0.0539	0.0049	0.0298	-0.0443	0.9944
83	0.0767	0.2380	67.8%	0.0557	0.0083	0.0285	-0.0426	0.9943
84	0.0764	0.2399	68.2%	0.0546	0.0058	0.0224	-0.0439	0.9943
85	0.0773	0.2422	68.1%	0.0560	0.0074	0.0304	-0.0456	0.9942
86	0.0773	0.2443	68.4%	0.0551	0.0053	0.0238	-0.0445	0.9942
87	0.0786	0.2464	68.1%	0.0570	0.0076	0.0263	-0.0446	0.9940
88	0.0782	0.2484	68.5%	0.0554	0.0045	0.0173	-0.0447	0.9940
89	0.0791	0.2501	68.4%	0.0573	0.0078	0.0240	-0.0431	0.9939
90	0.0784	0.2522	68.9%	0.0553	0.0043	0.0184	-0.0441	0.9940

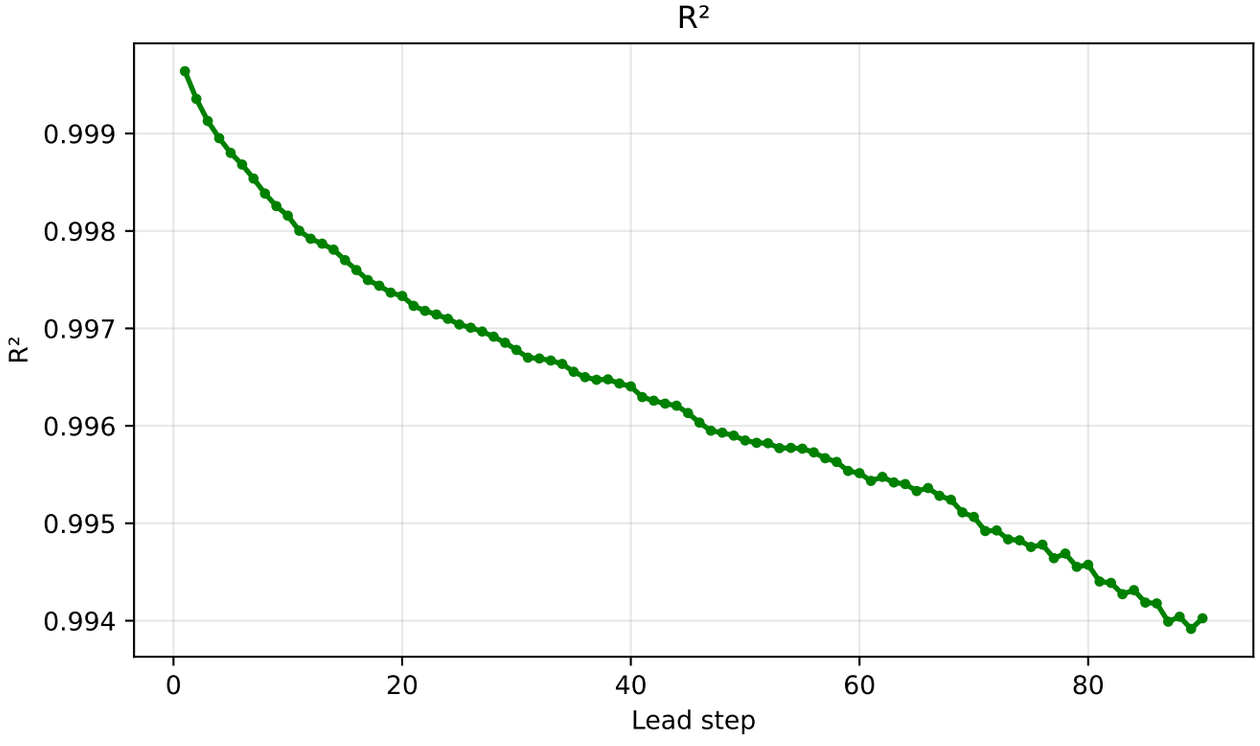
Forecast Skill vs Lead Step — Validation





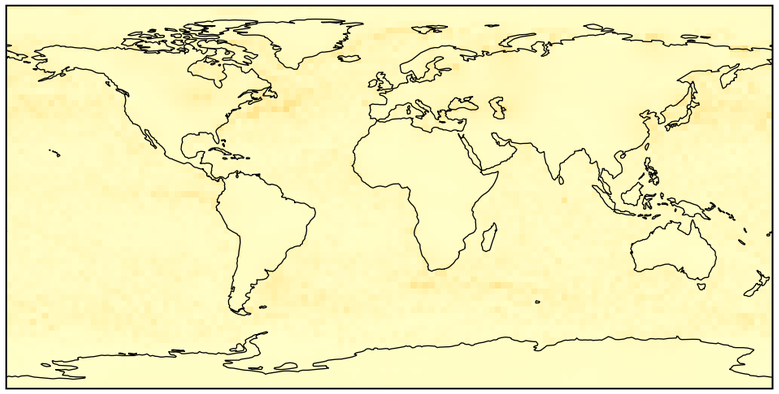
Bias and R² — Validation



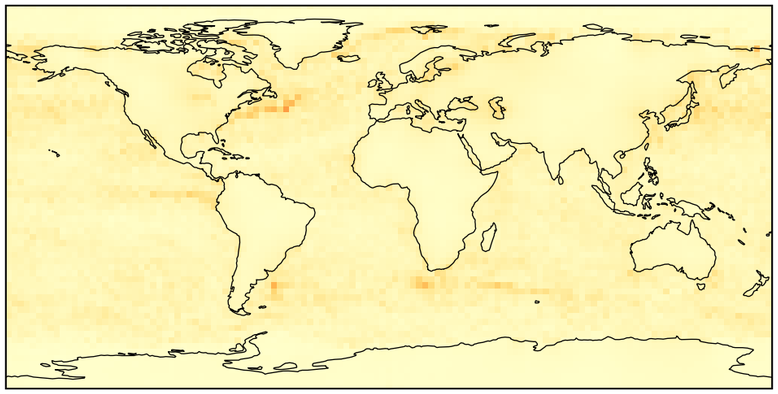


Spatial RMSE (all grid) — Validation

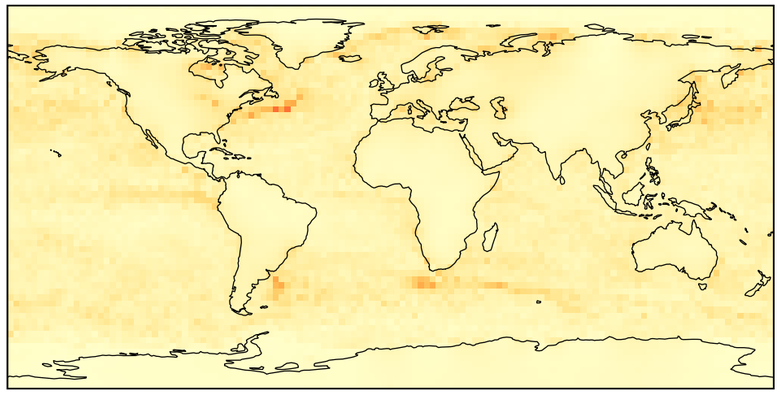
Step 1



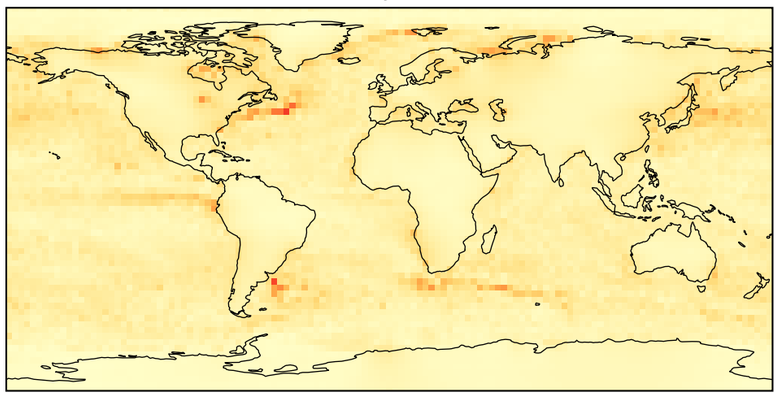
Step 3



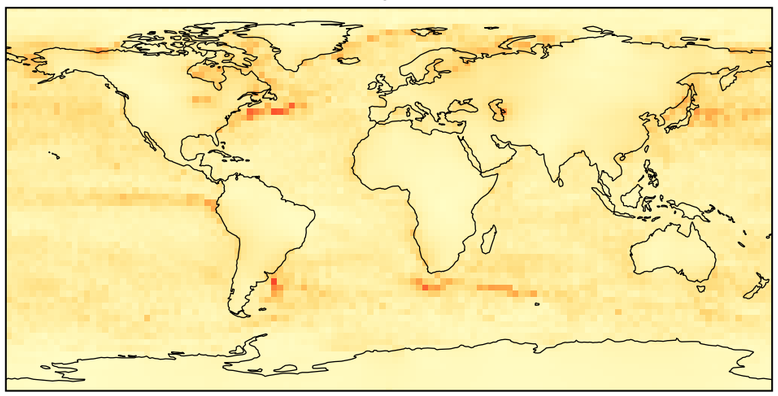
Step 7



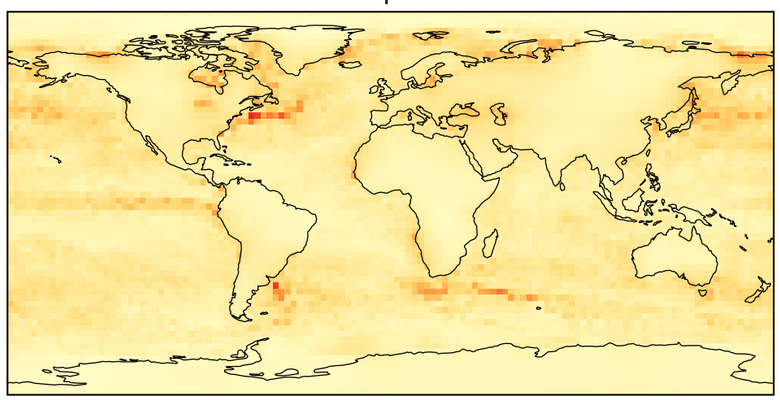
Step 14



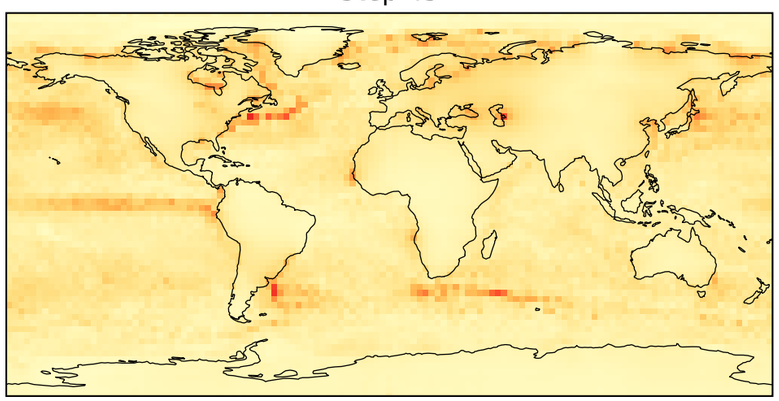
Step 21



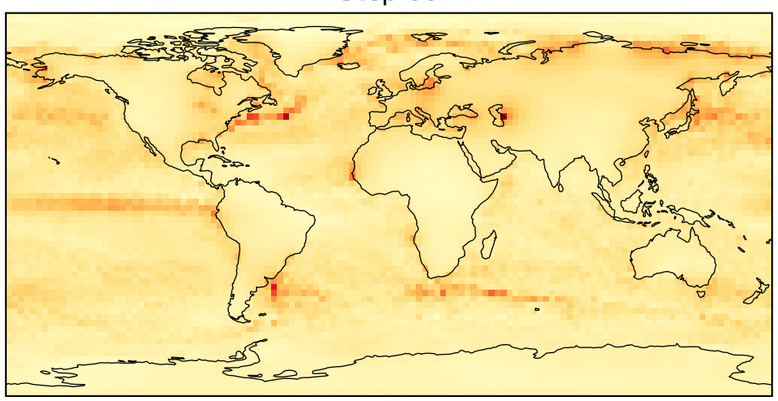
Step 30



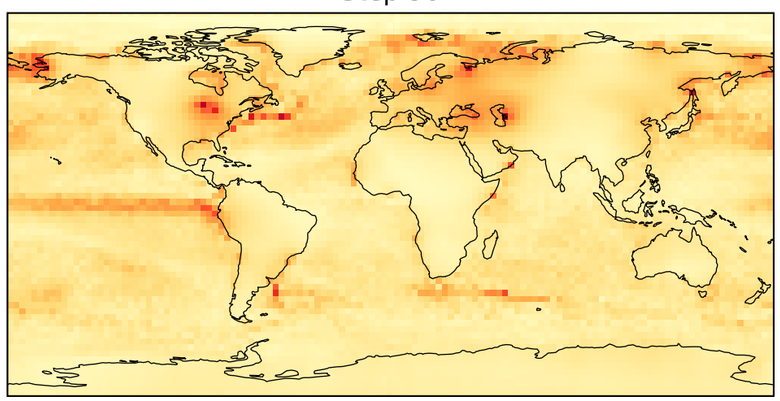
Step 45

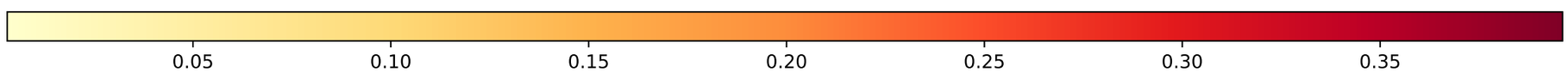


Step 60



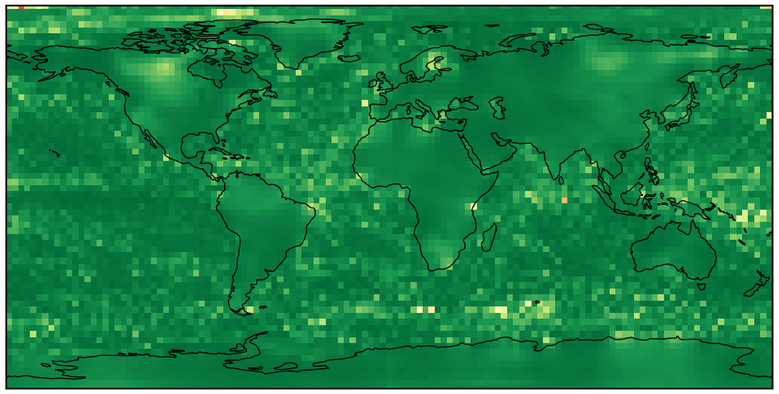
Step 90



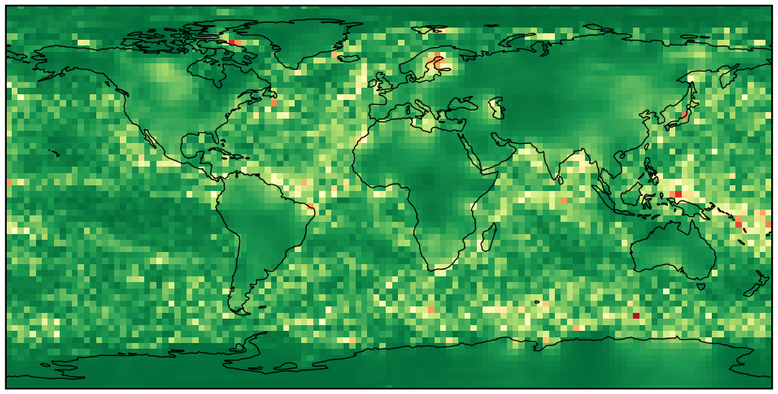


Spatial ACC (all grid) — Validation

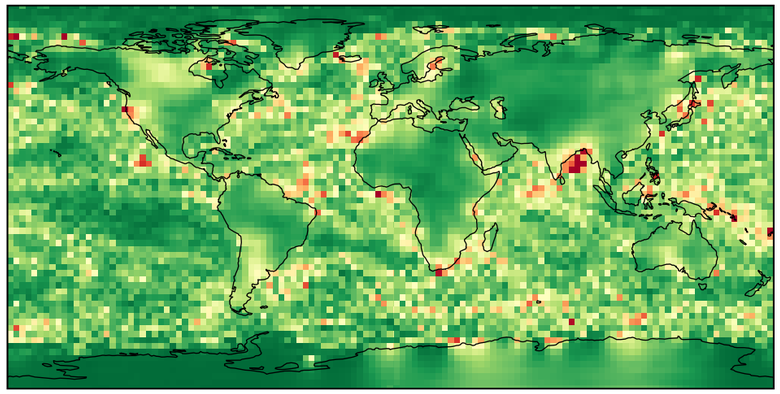
Step 1



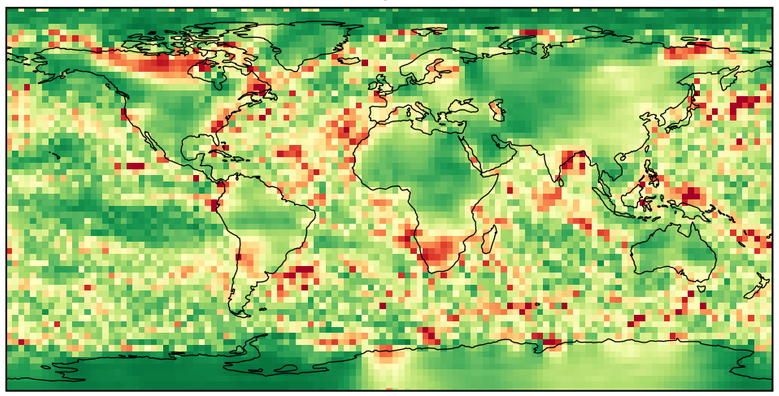
Step 3



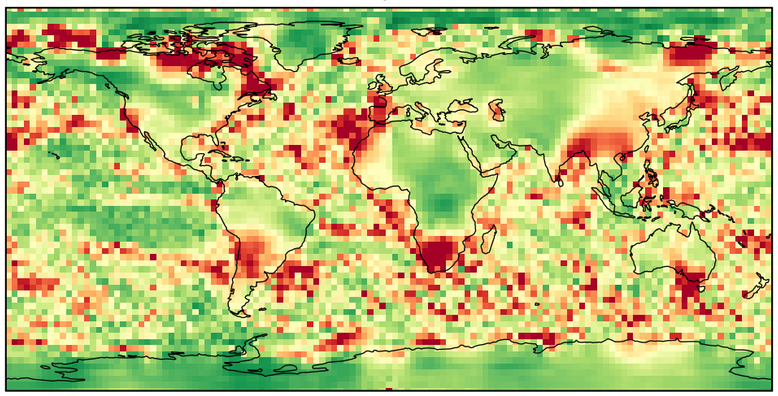
Step 7



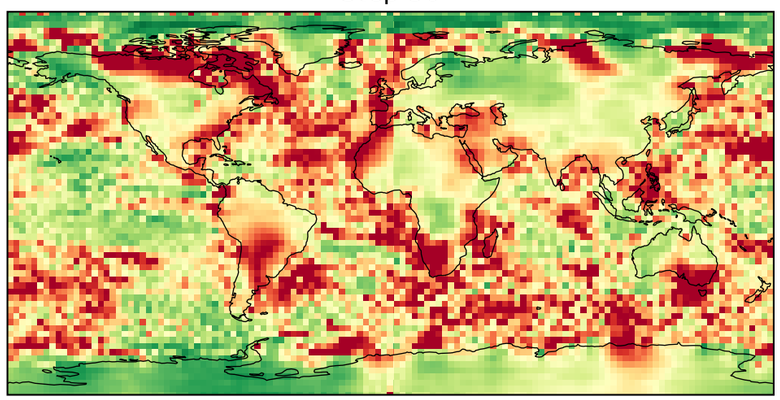
Step 14



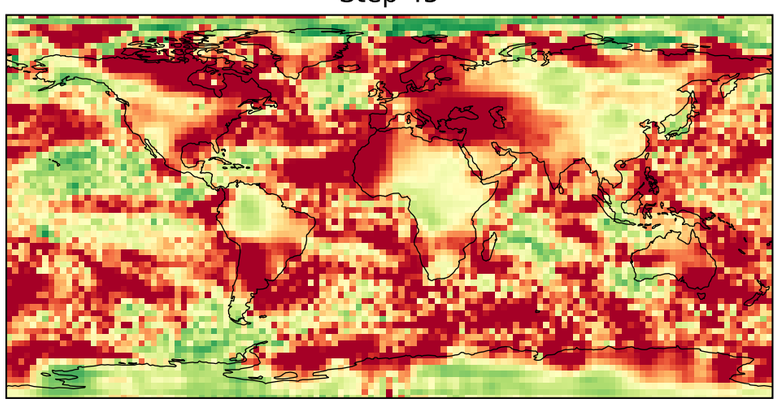
Step 21



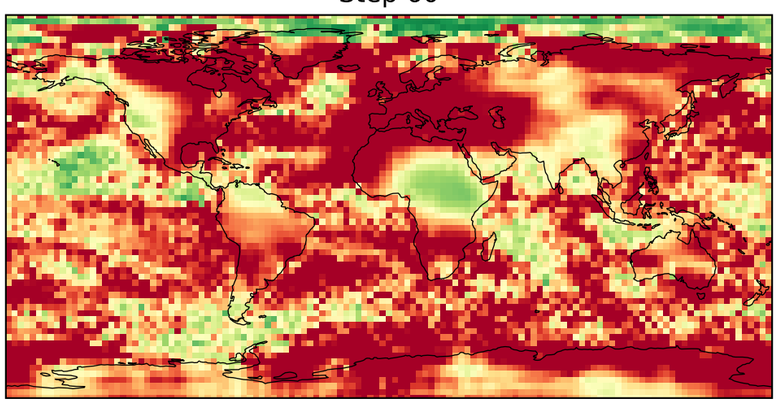
Step 30



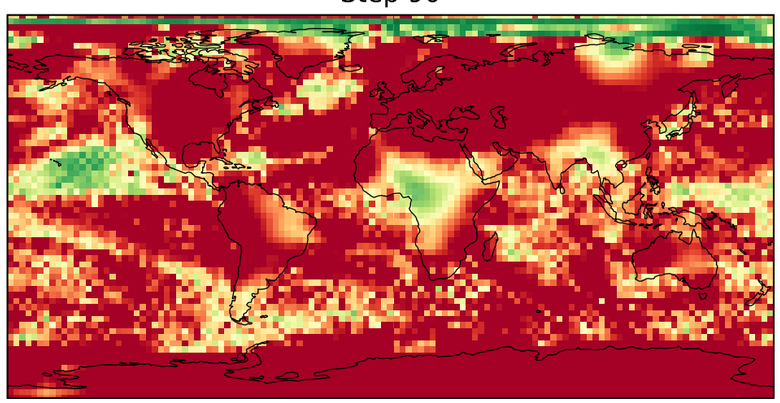
Step 45



Step 60

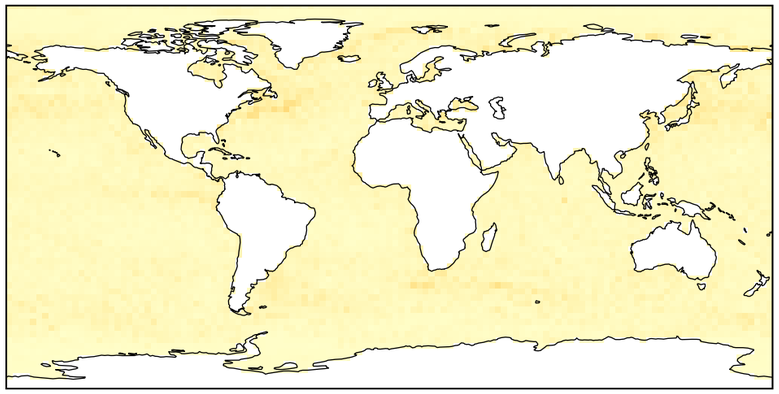


Step 90

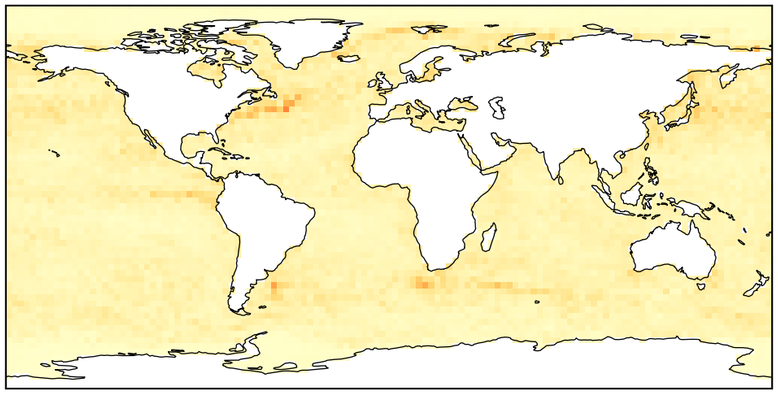


Spatial RMSE (ocean only) — Validation

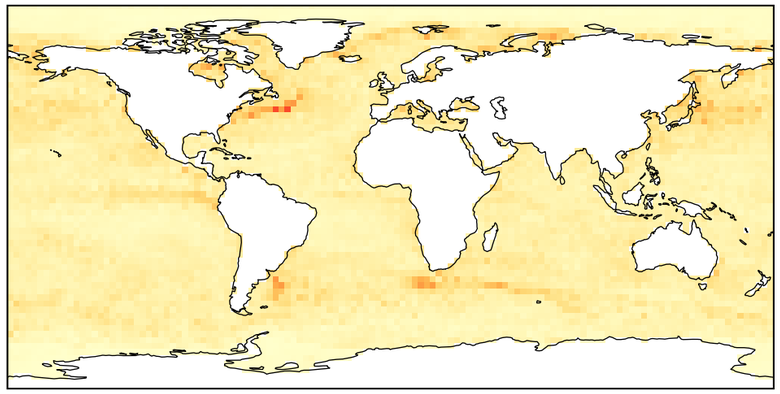
Step 1



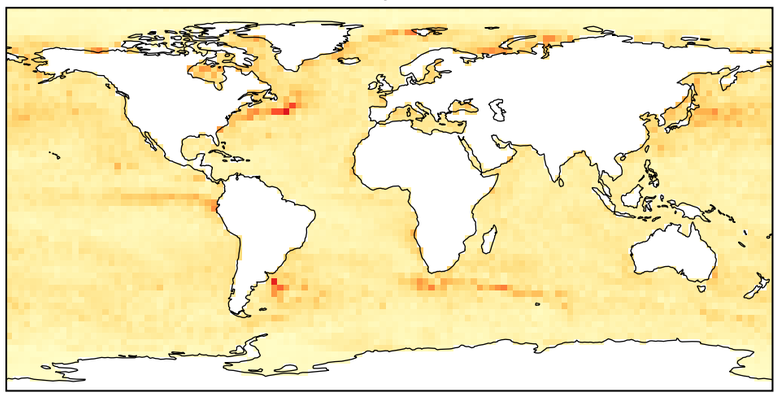
Step 3



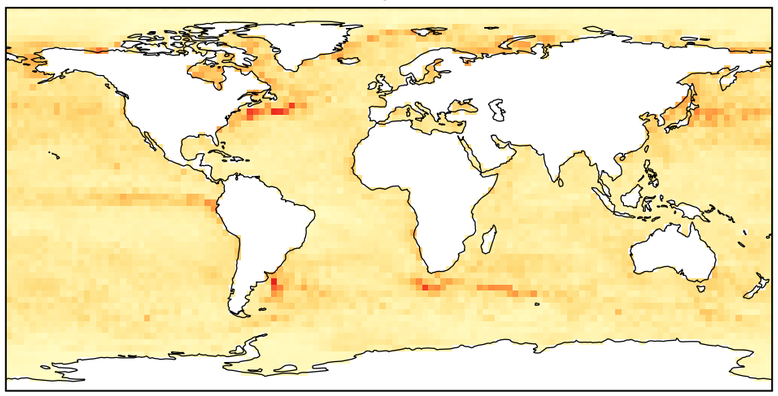
Step 7



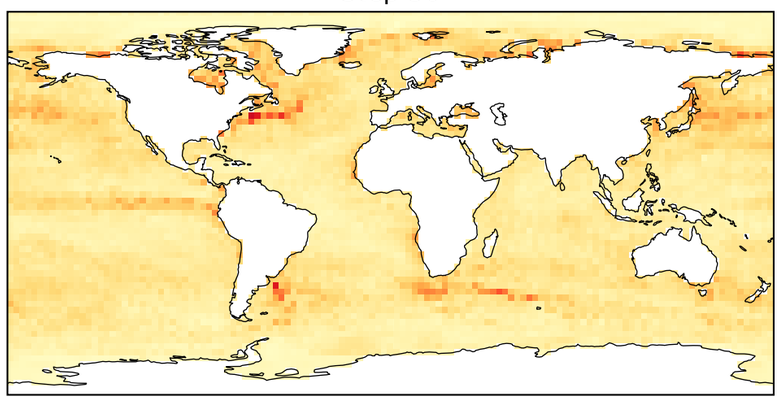
Step 14



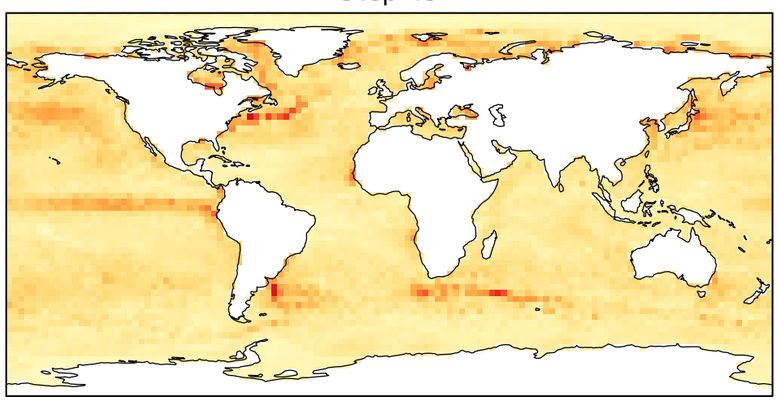
Step 21



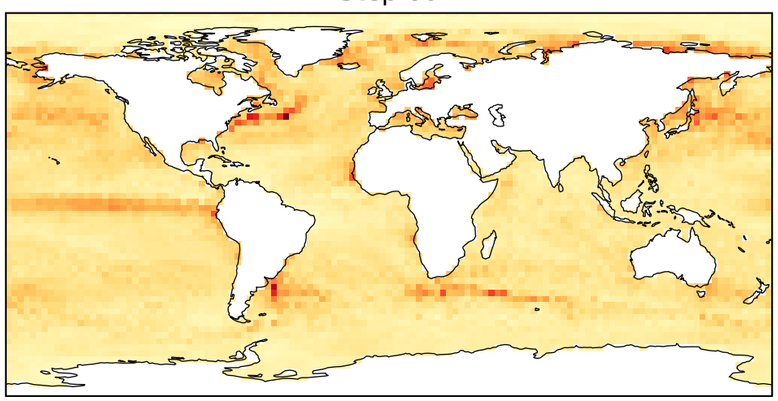
Step 30



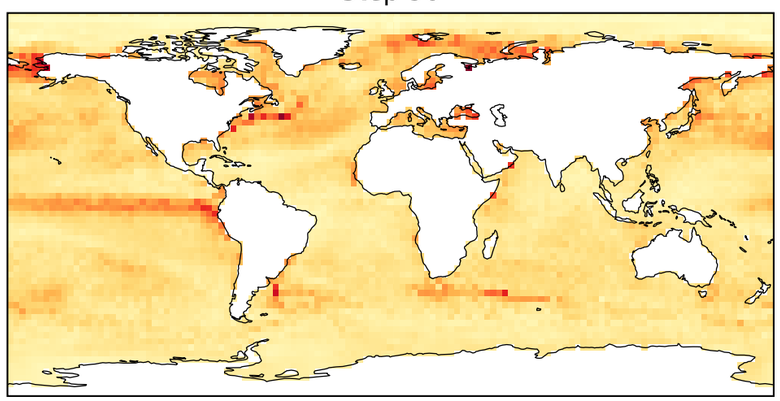
Step 45



Step 60

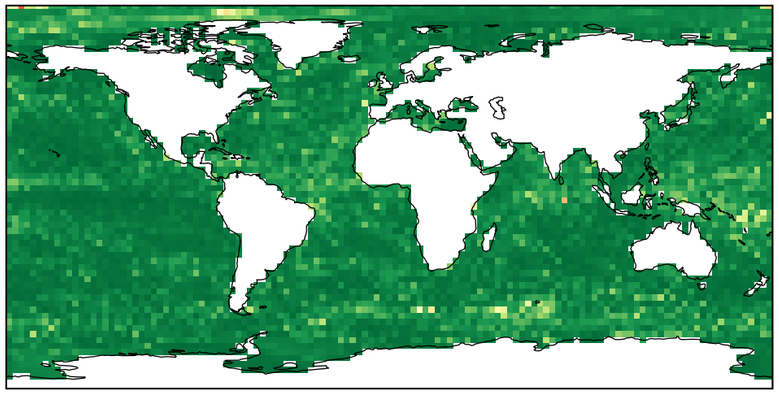


Step 90

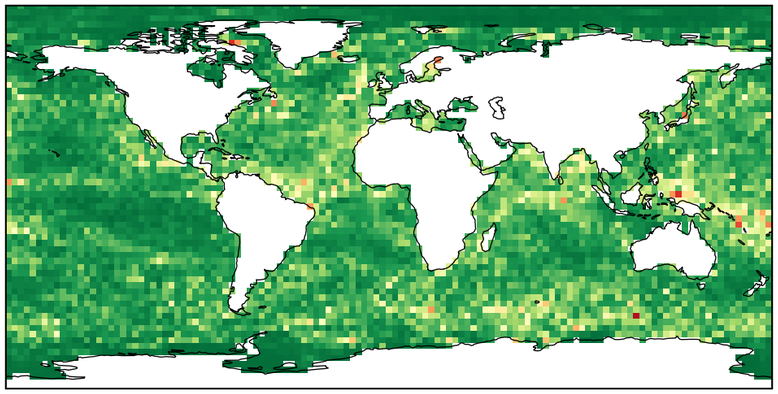


Spatial ACC (ocean only) — Validation

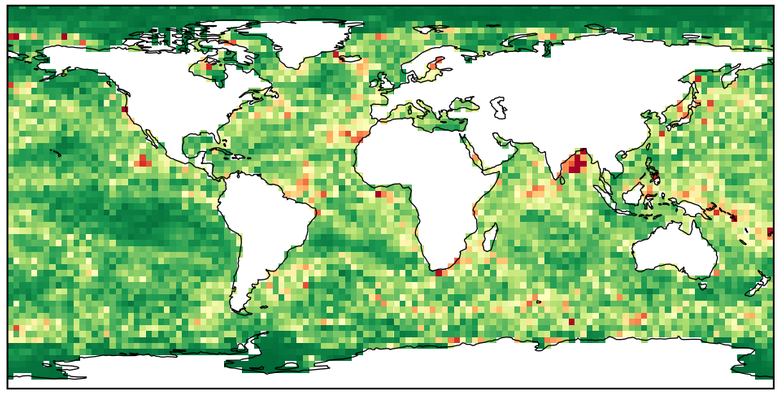
Step 1



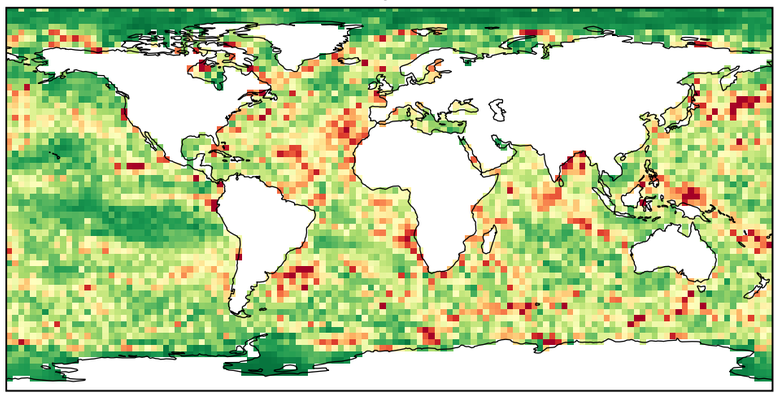
Step 3



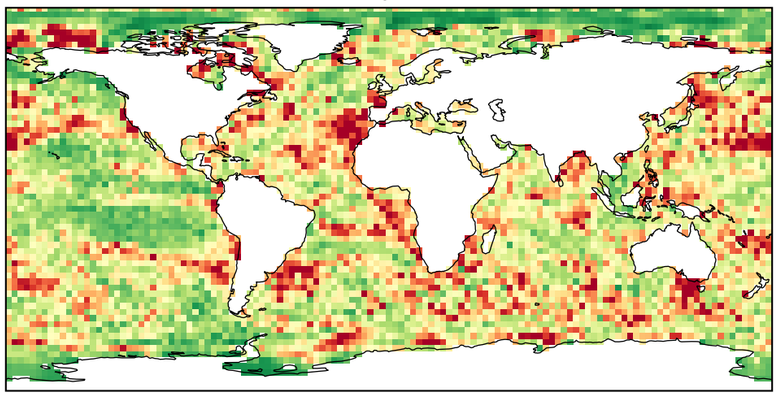
Step 7



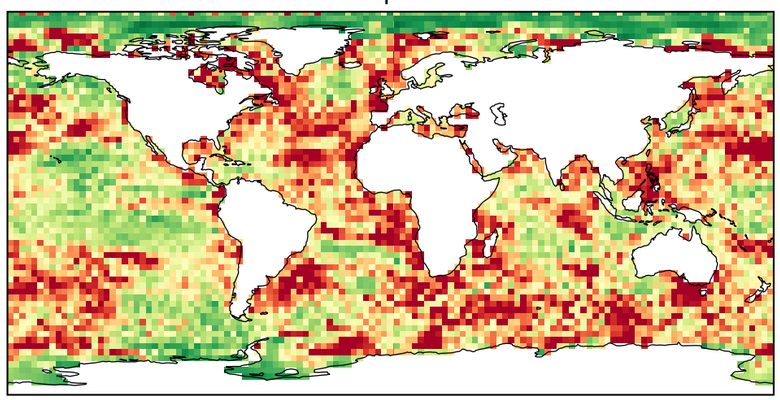
Step 14



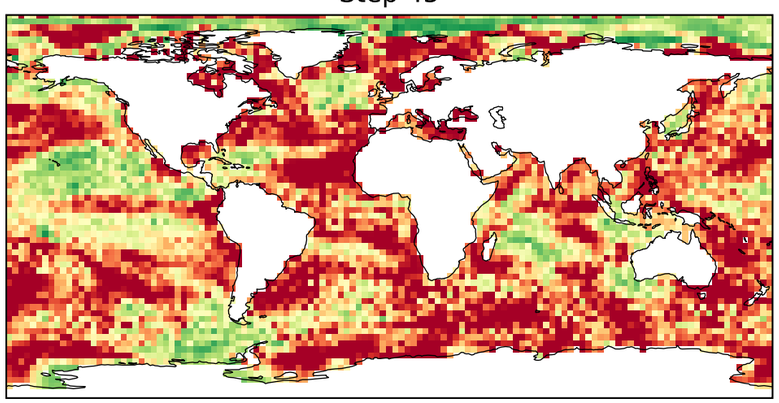
Step 21



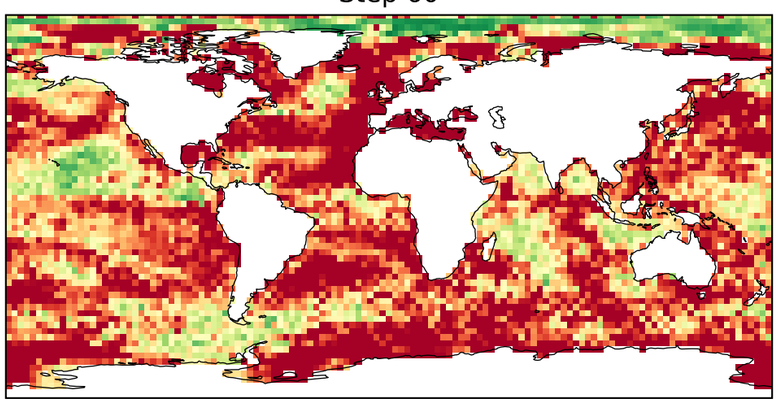
Step 30



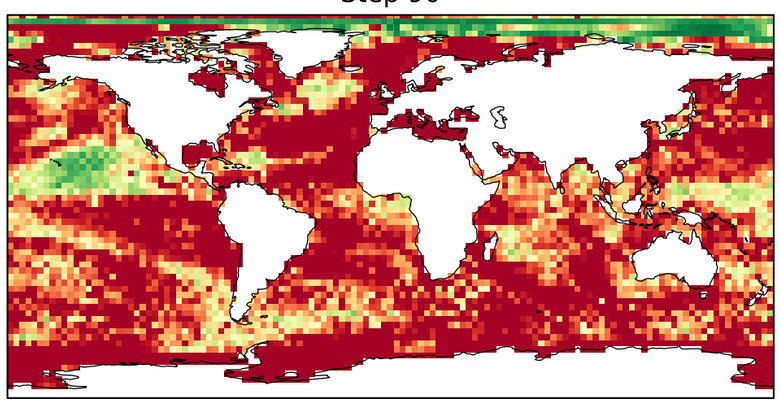
Step 45

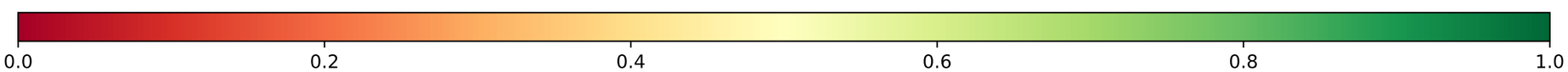


Step 60



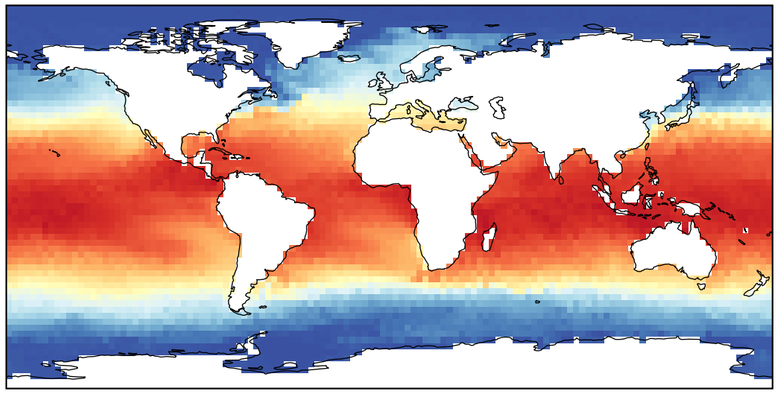
Step 90



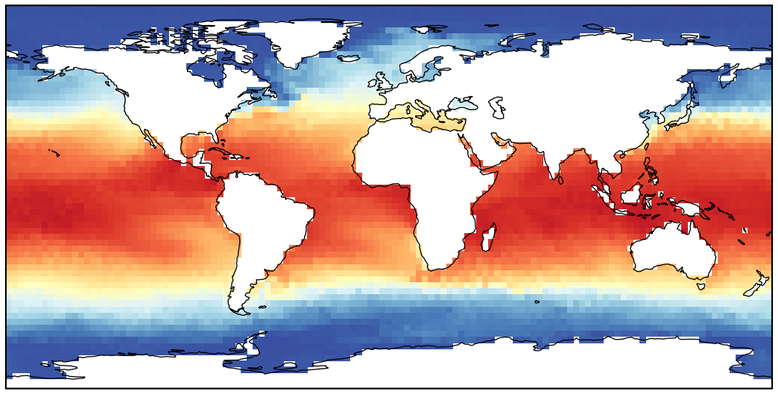


Chain init: 2016-01-11 — Validation (°C)

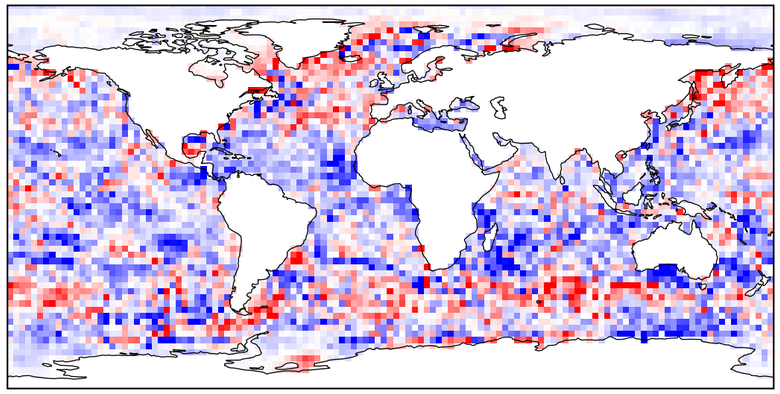
Actual — Step 1 (°C)



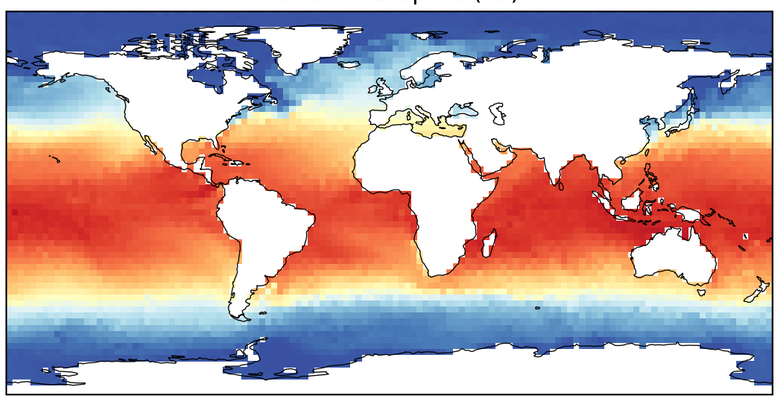
SFNO — Step 1 (°C)



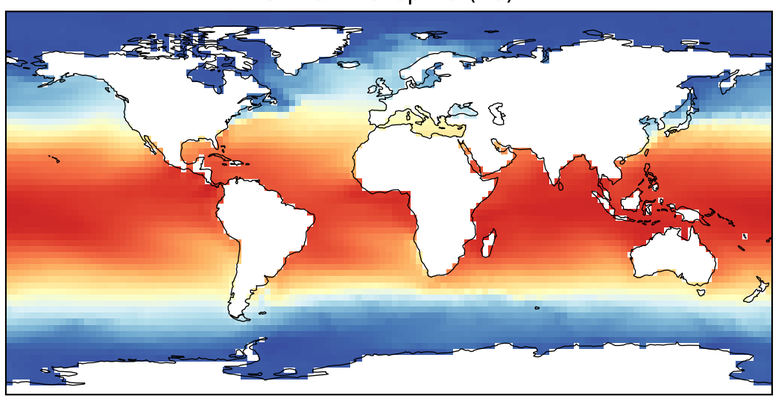
Error — Step 1 (°C)



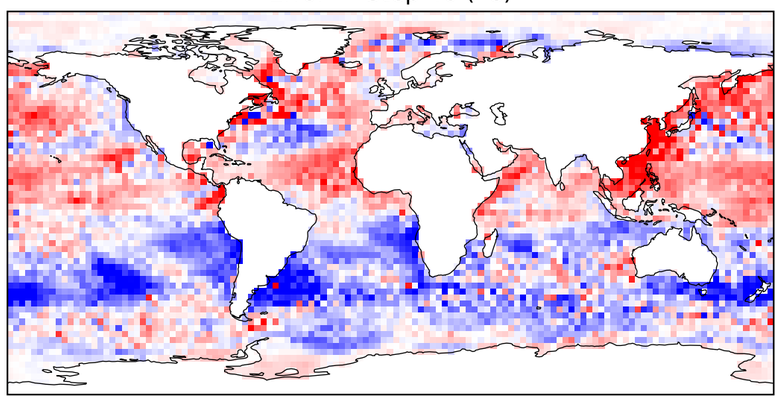
Actual — Step 46 (°C)



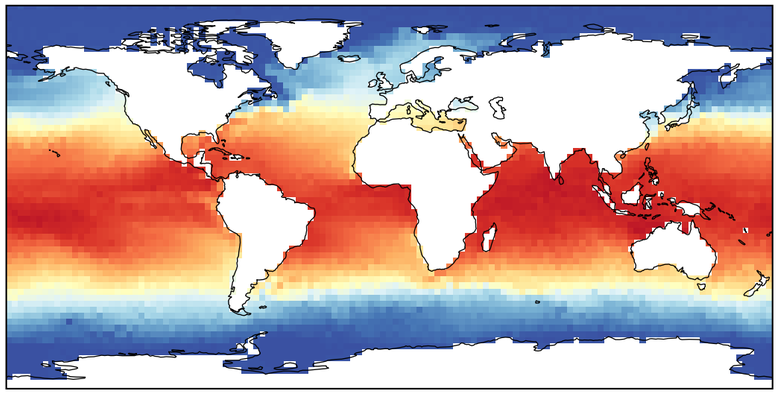
SFNO — Step 46 (°C)



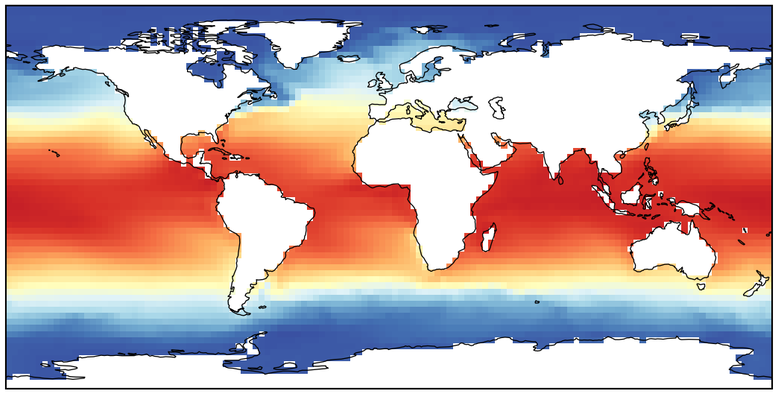
Error — Step 46 (°C)



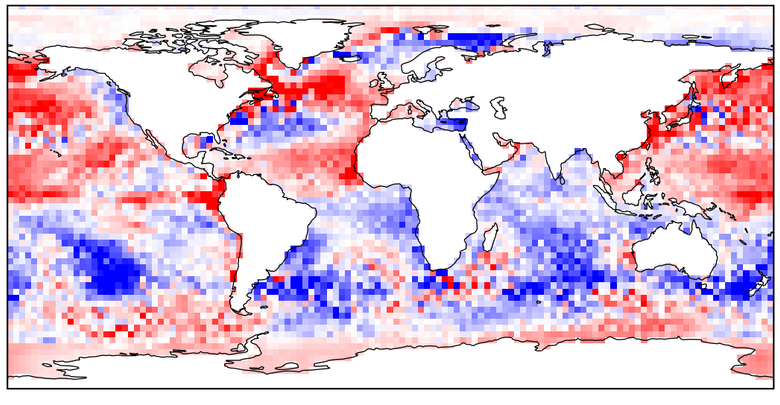
Actual — Step 90 (°C)

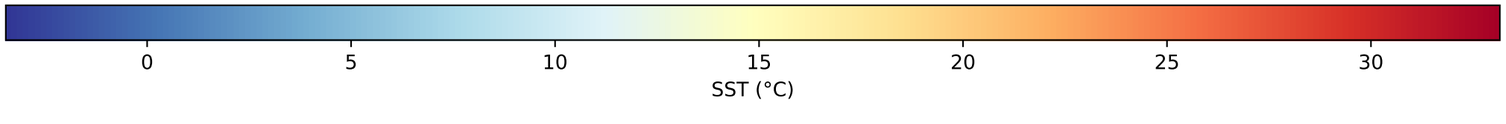


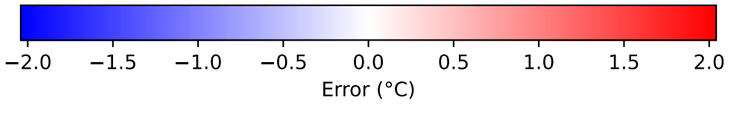
SFNO — Step 90 (°C)



Error — Step 90 (°C)

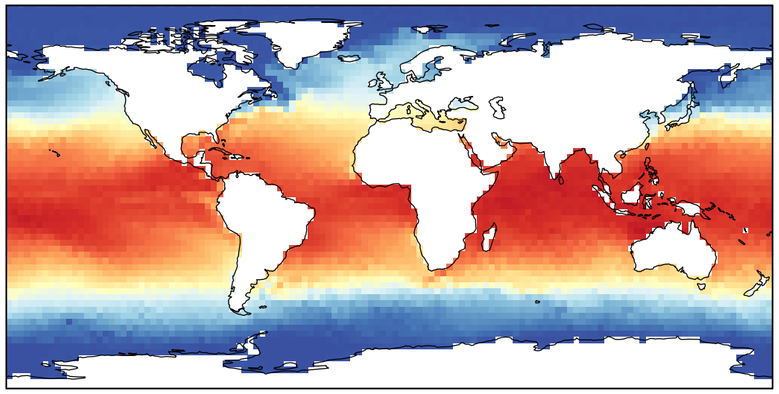




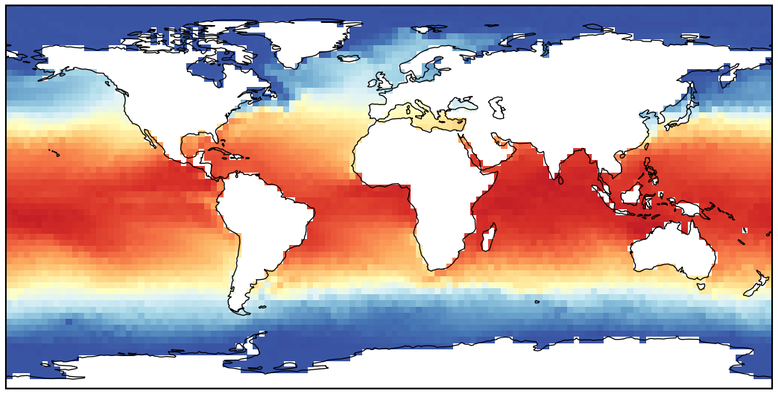


Chain init: 2016-04-10 — Validation (°C)

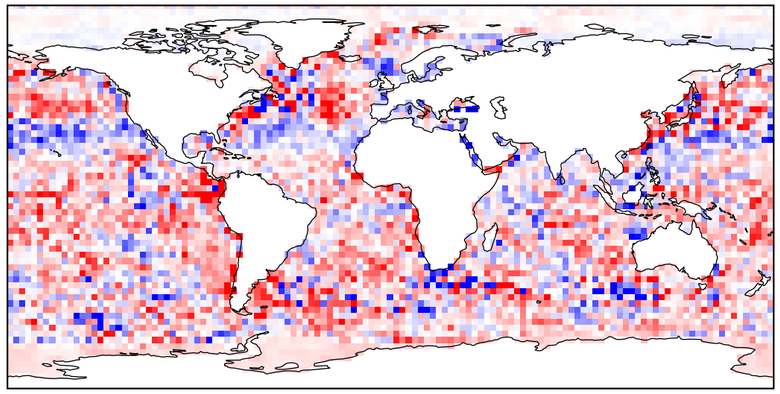
Actual — Step 1 (°C)



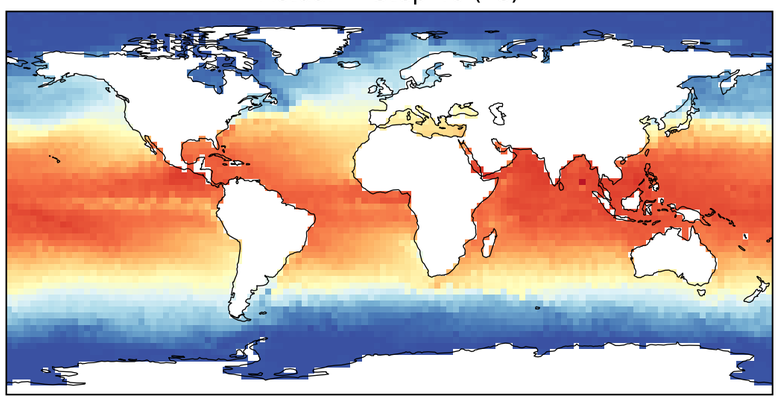
SFNO — Step 1 (°C)



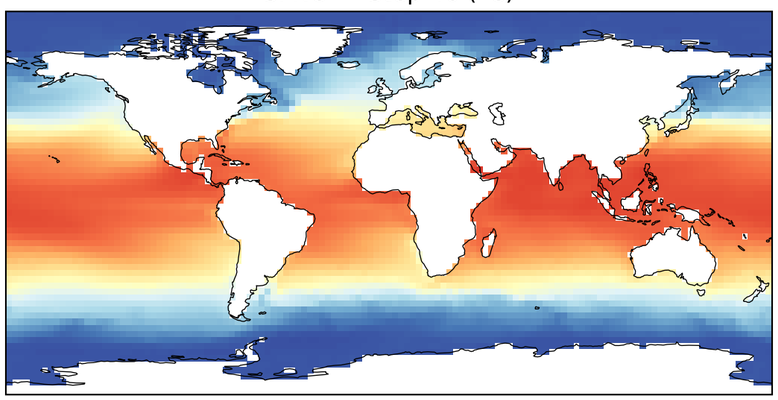
Error — Step 1 (°C)



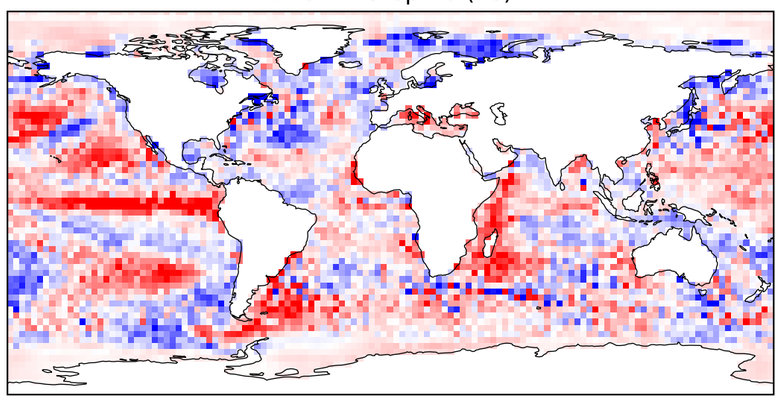
Actual — Step 46 (°C)



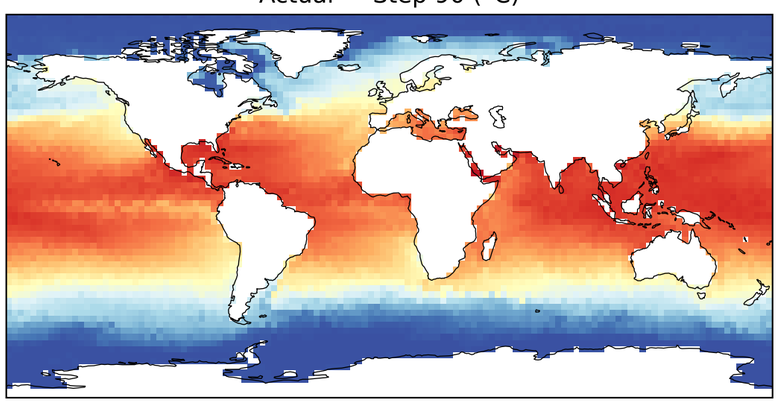
SFNO — Step 46 (°C)



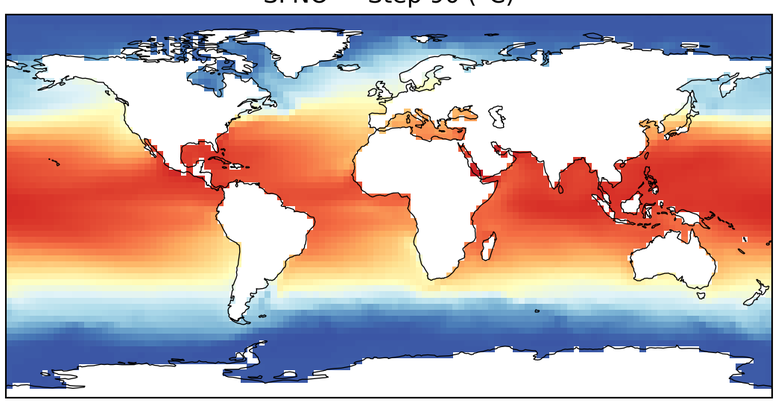
Error — Step 46 (°C)



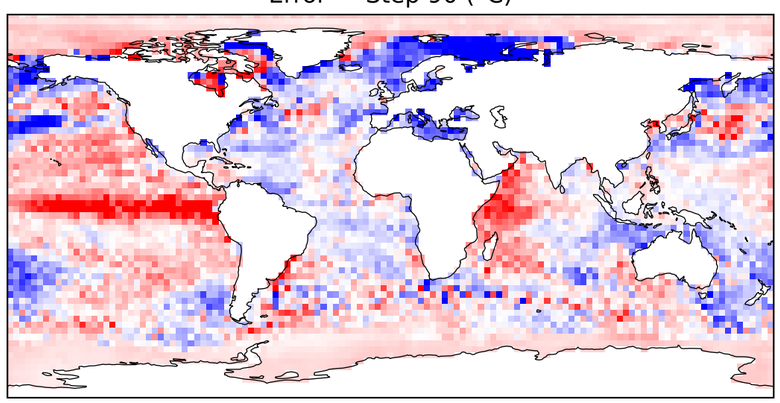
Actual — Step 90 (°C)

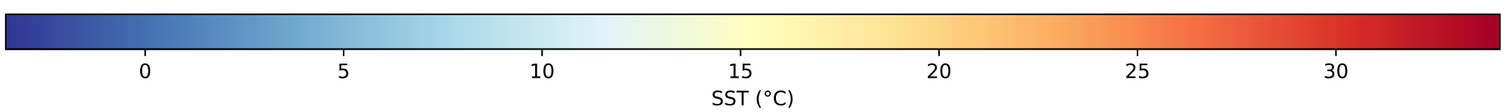


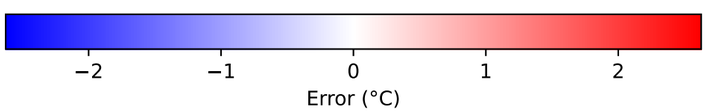
SFNO — Step 90 (°C)



Error — Step 90 (°C)

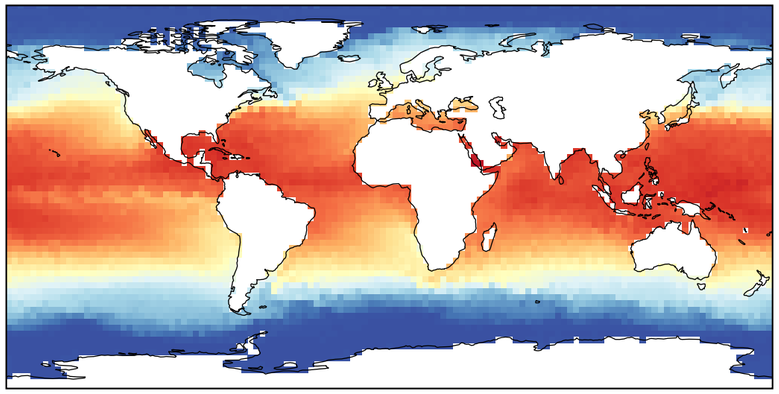




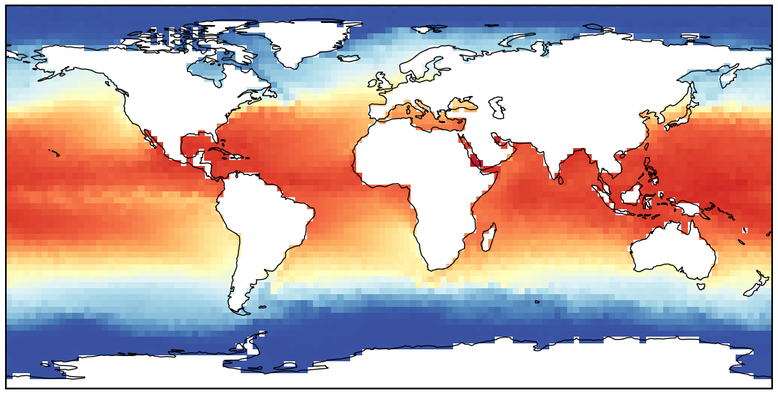


Chain init: 2016-10-07 — Validation (°C)

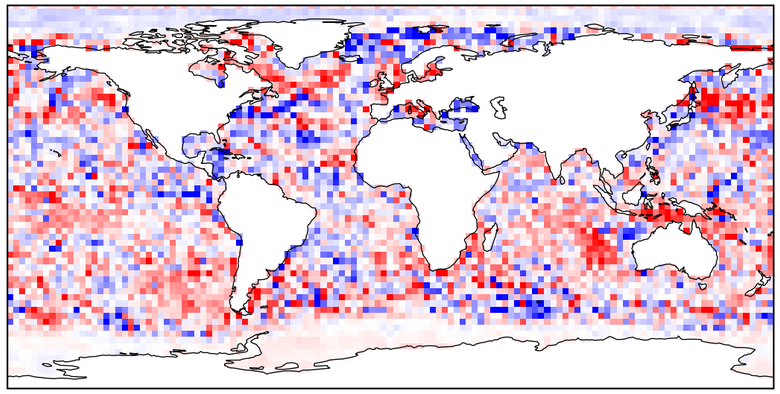
Actual — Step 1 (°C)



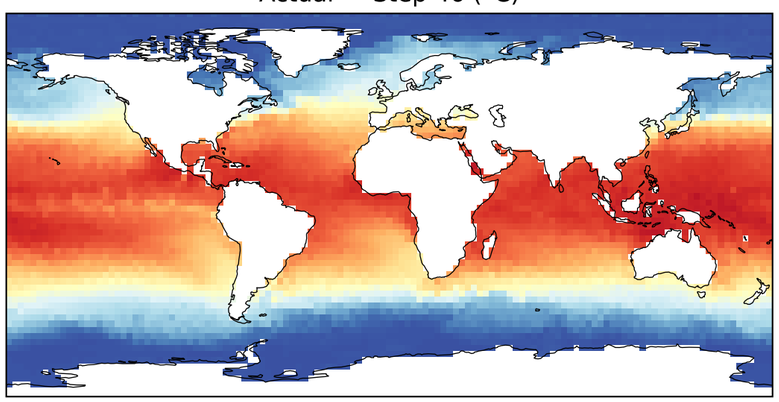
SFNO — Step 1 (°C)



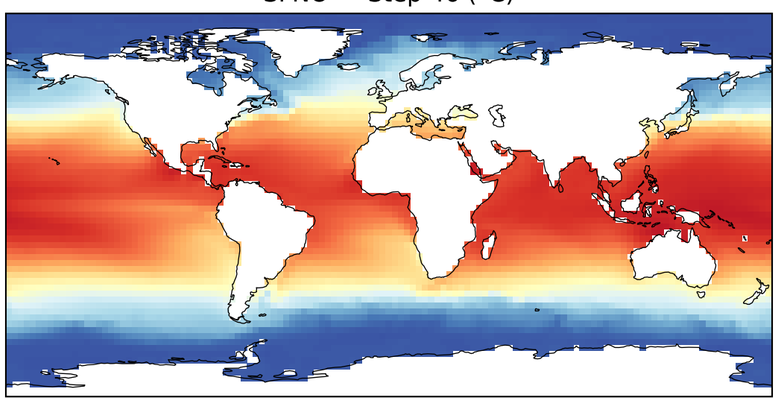
Error — Step 1 (°C)



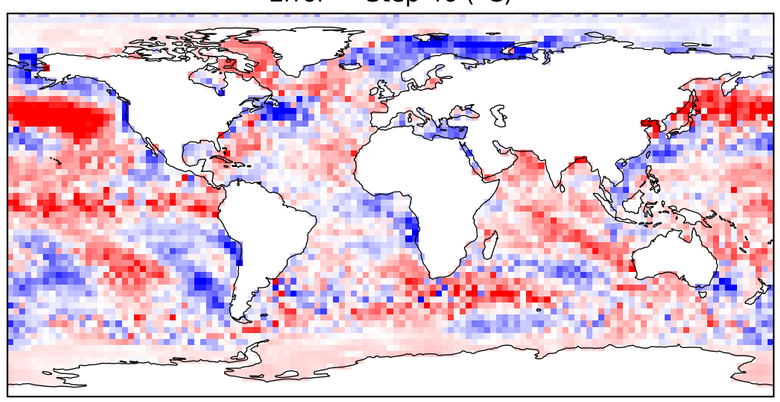
Actual — Step 46 (°C)



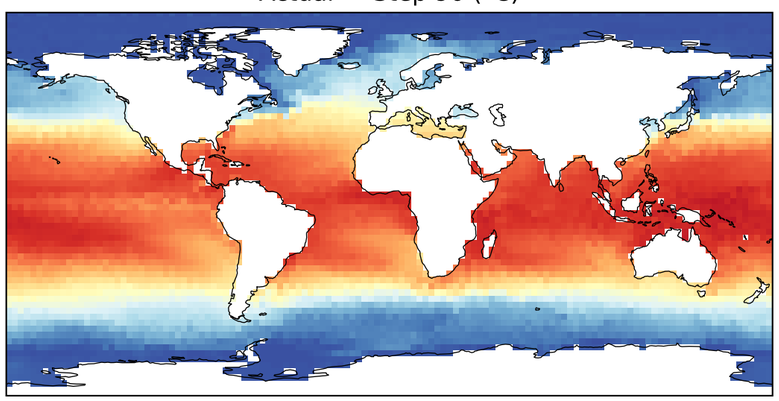
SFNO — Step 46 (°C)



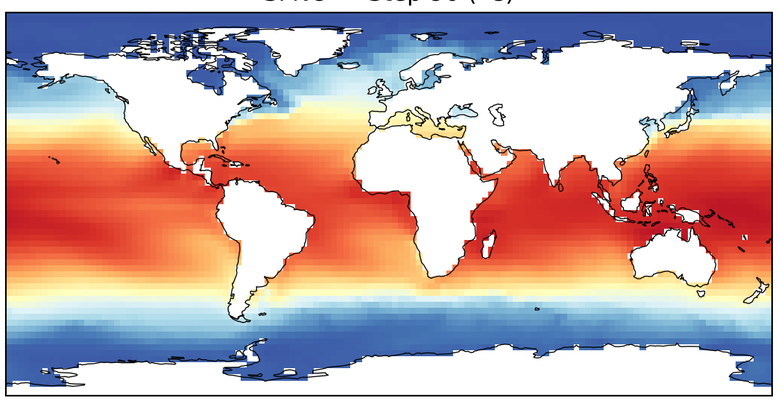
Error — Step 46 (°C)



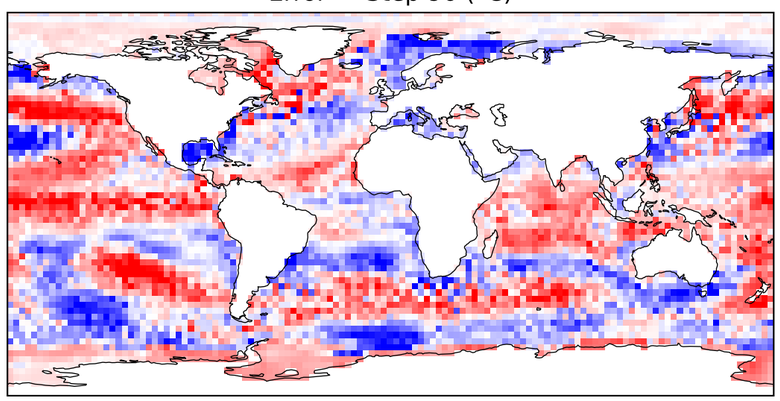
Actual — Step 90 (°C)

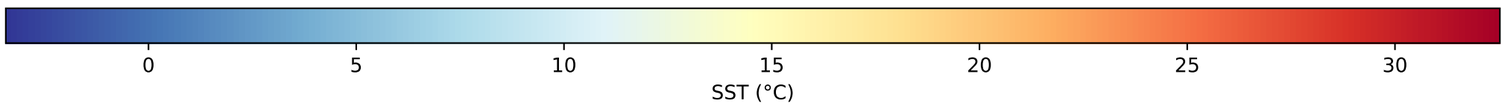


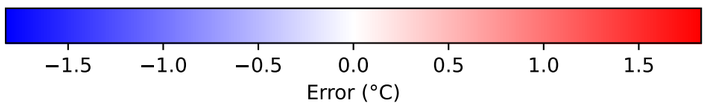
SFNO — Step 90 (°C)



Error — Step 90 (°C)

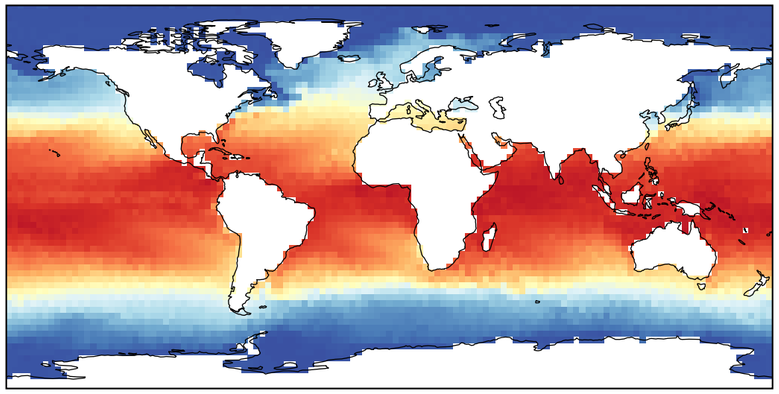




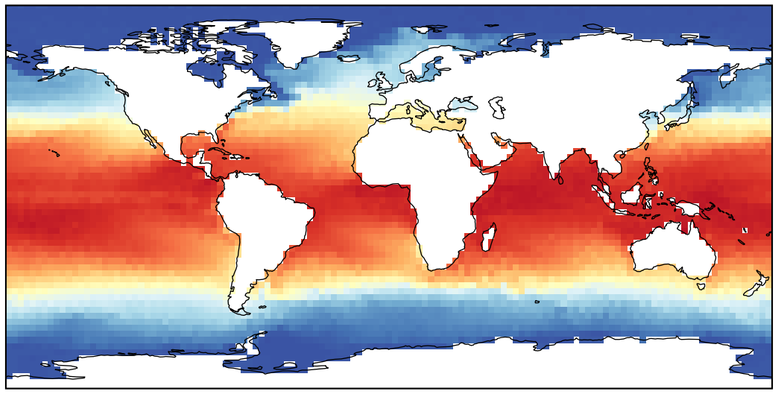


Chain init: 2017-04-05 — Validation (°C)

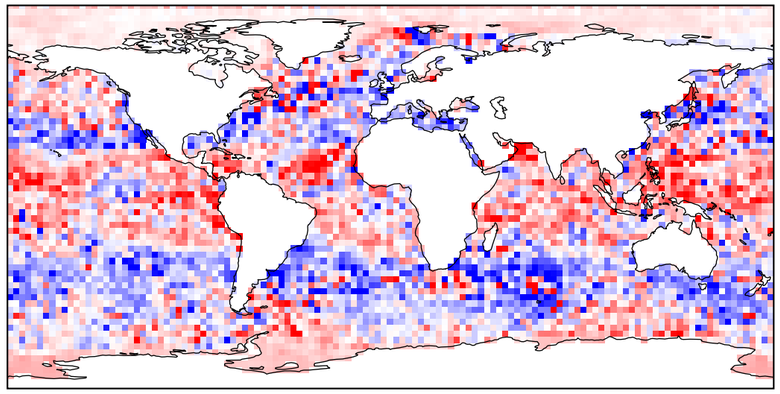
Actual — Step 1 (°C)



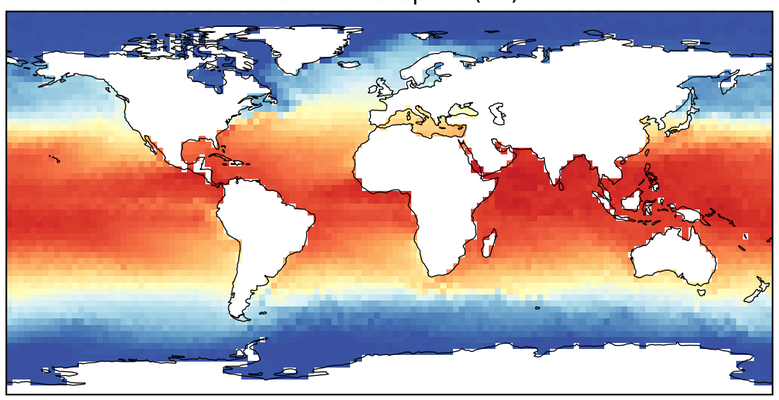
SFNO — Step 1 (°C)



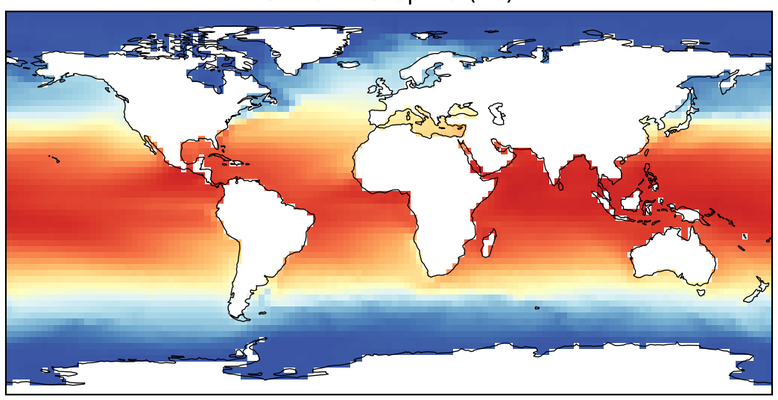
Error — Step 1 (°C)



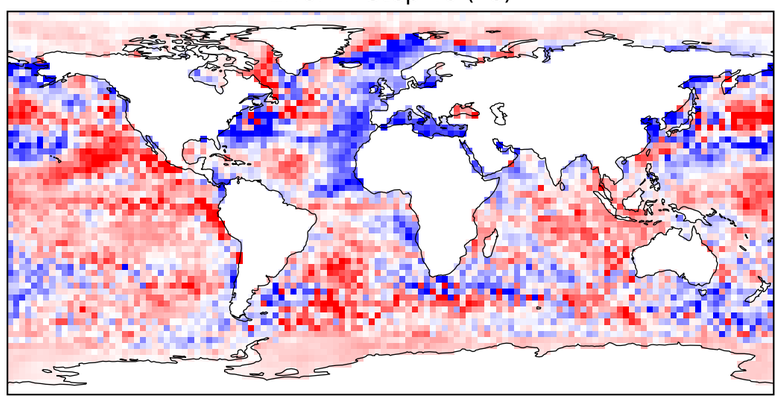
Actual — Step 46 (°C)



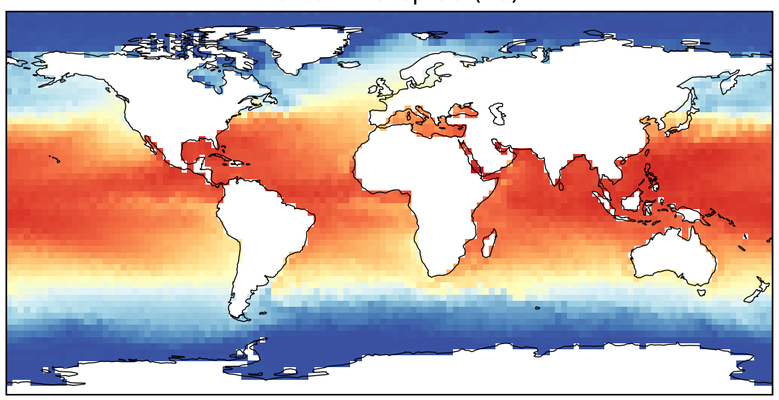
SFNO — Step 46 (°C)



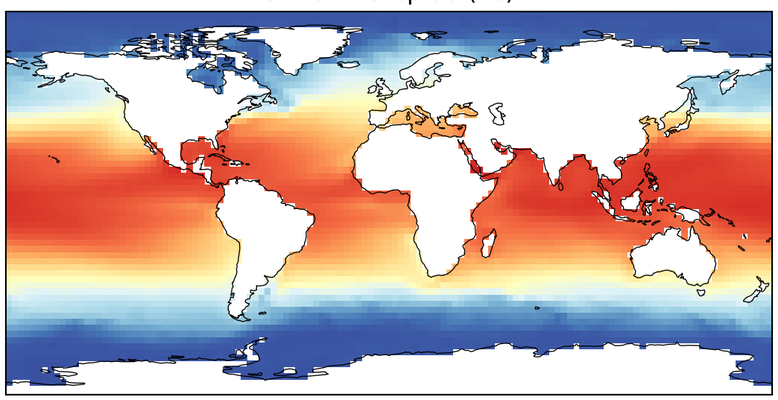
Error — Step 46 (°C)



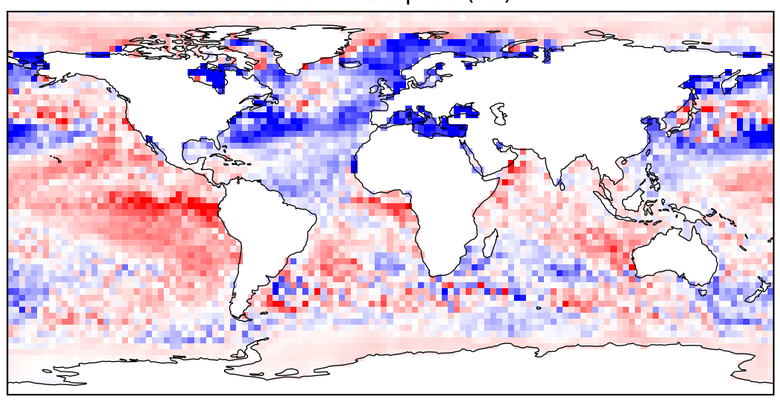
Actual — Step 90 (°C)

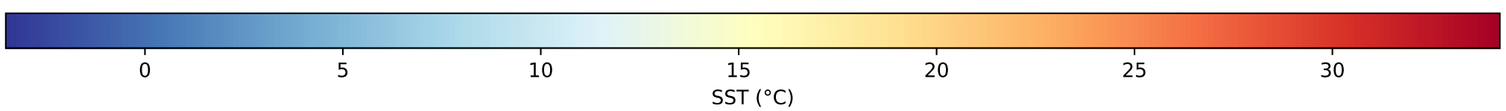


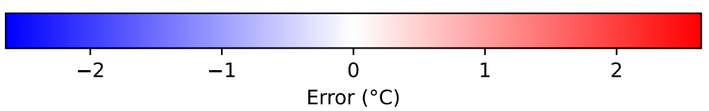
SFNO — Step 90 (°C)



Error — Step 90 (°C)

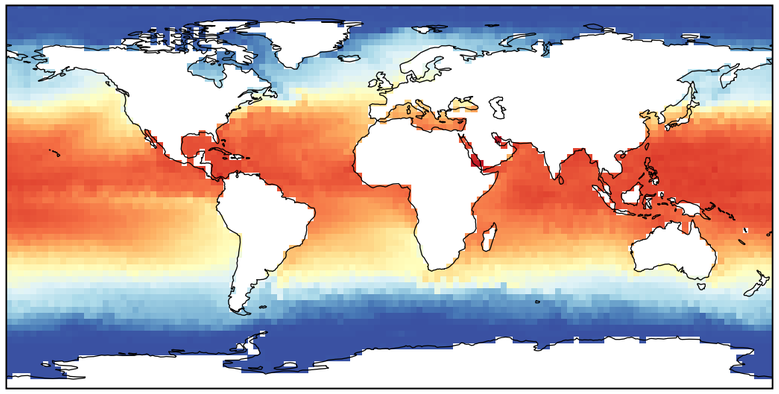




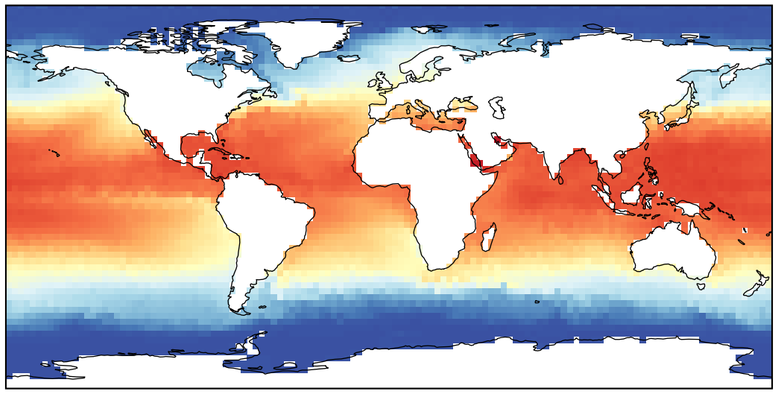


Chain init: 2017-10-02 — Validation (°C)

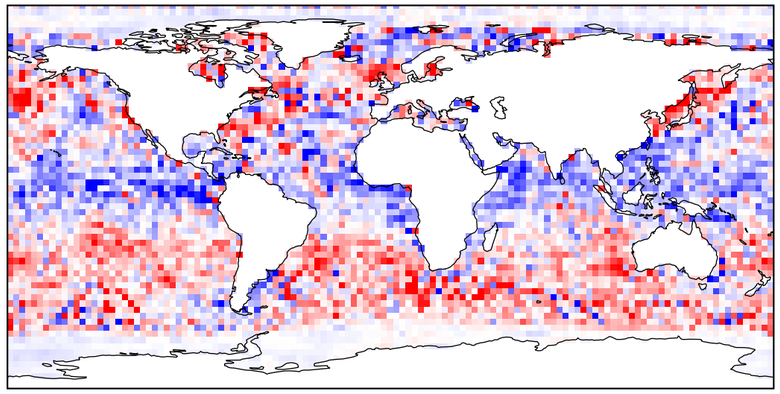
Actual — Step 1 (°C)



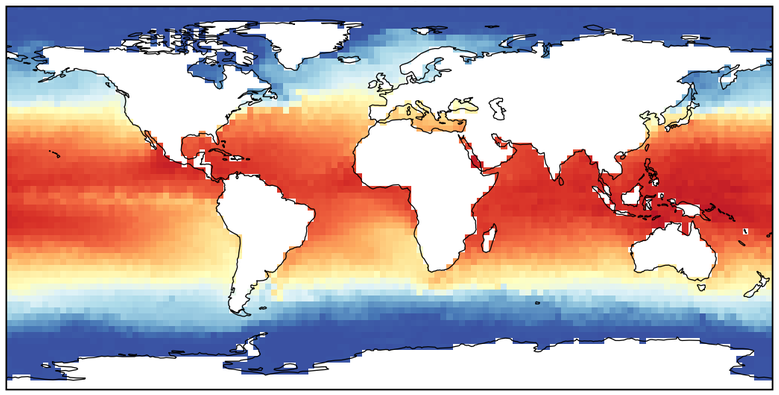
SFNO — Step 1 (°C)



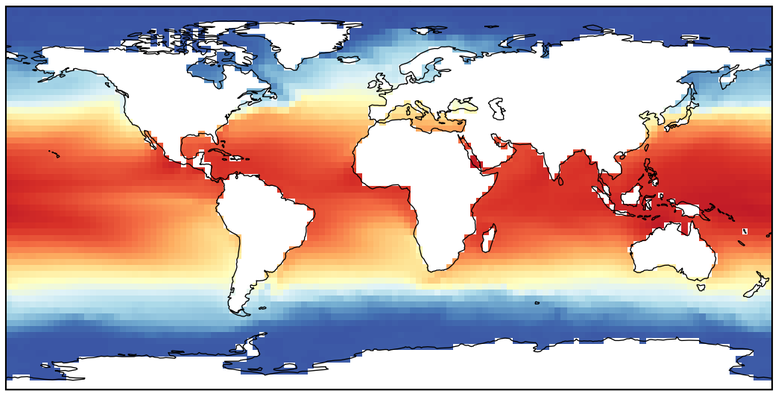
Error — Step 1 (°C)



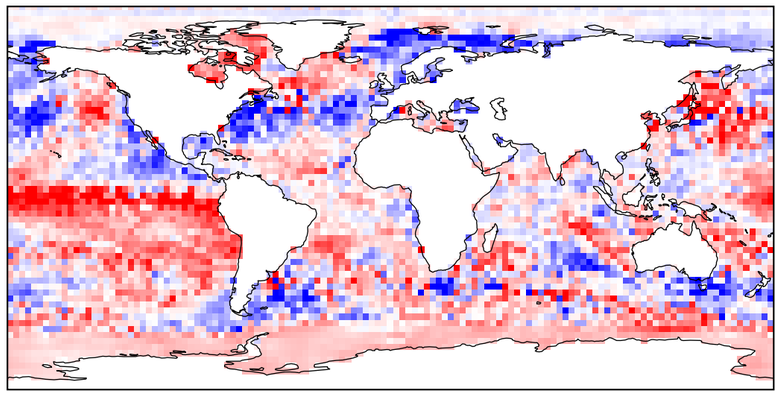
Actual — Step 46 (°C)



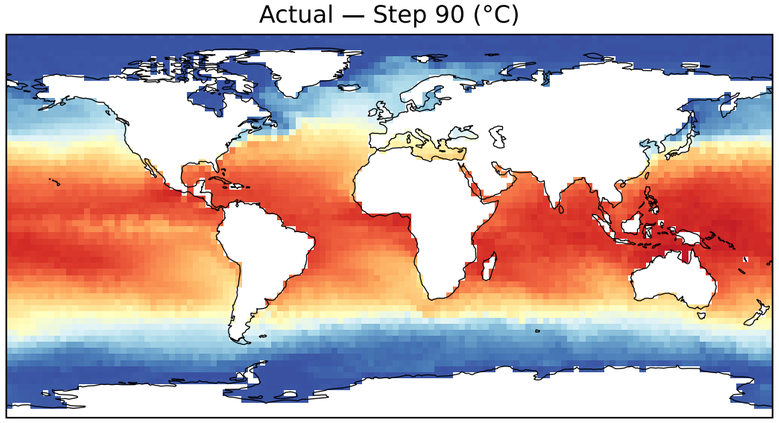
SFNO — Step 46 (°C)



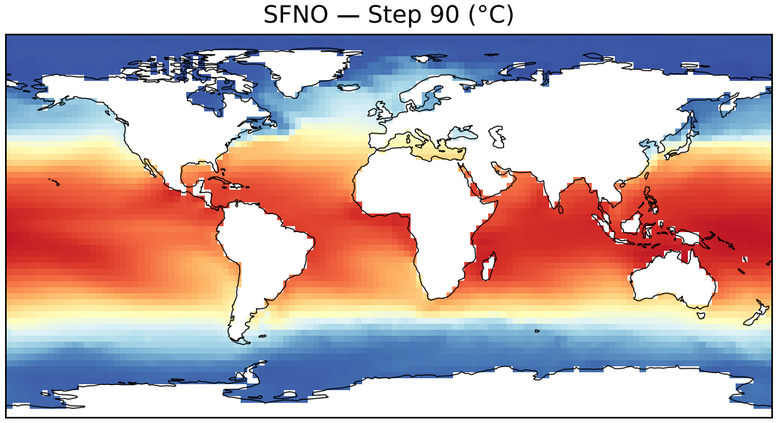
Error — Step 46 (°C)



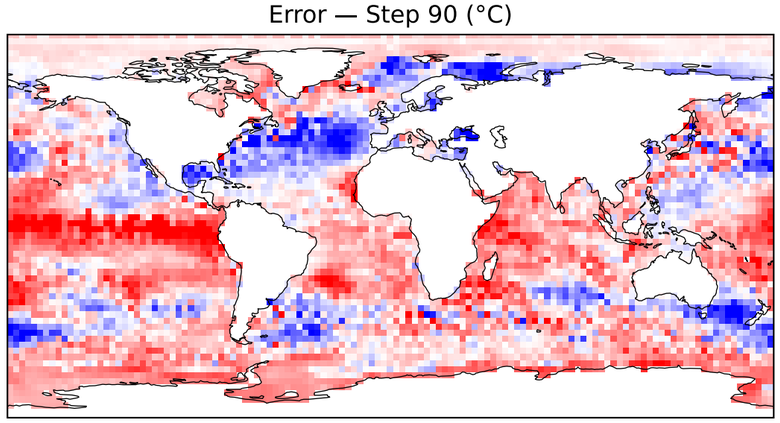
Actual — Step 90 (°C)

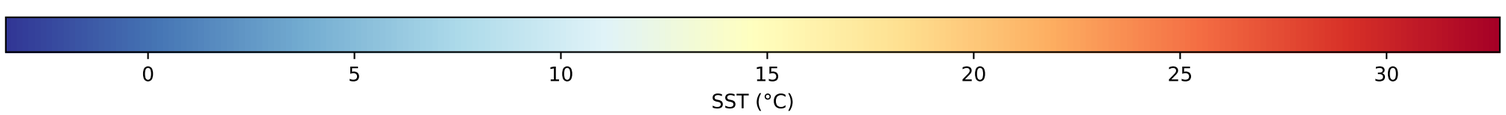


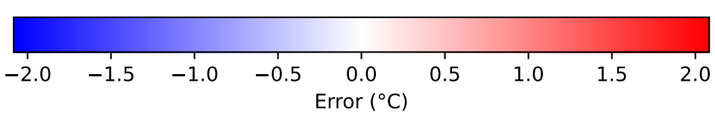
SFNO — Step 90 (°C)



Error — Step 90 (°C)

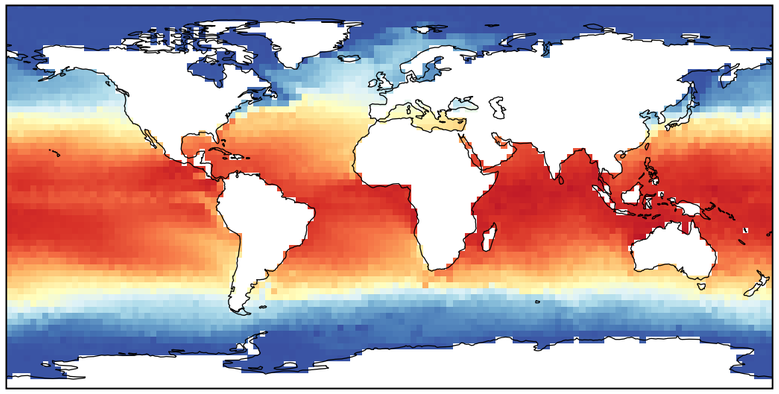




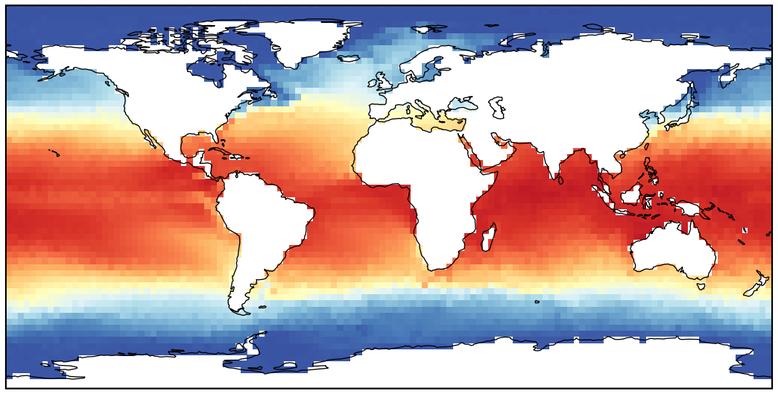


Chain init: 2018-03-31 — Validation (°C)

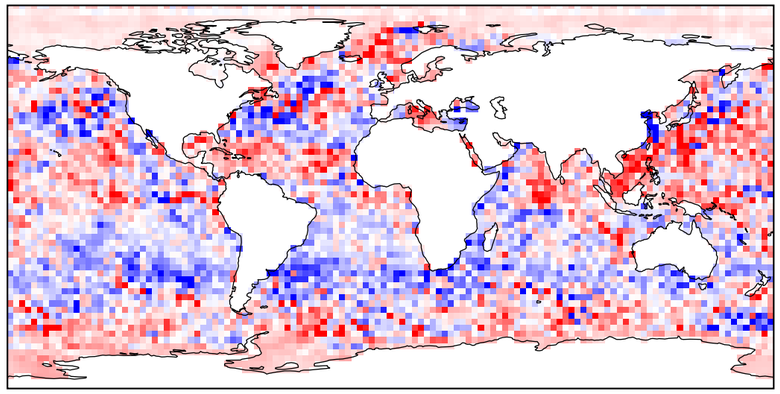
Actual — Step 1 (°C)



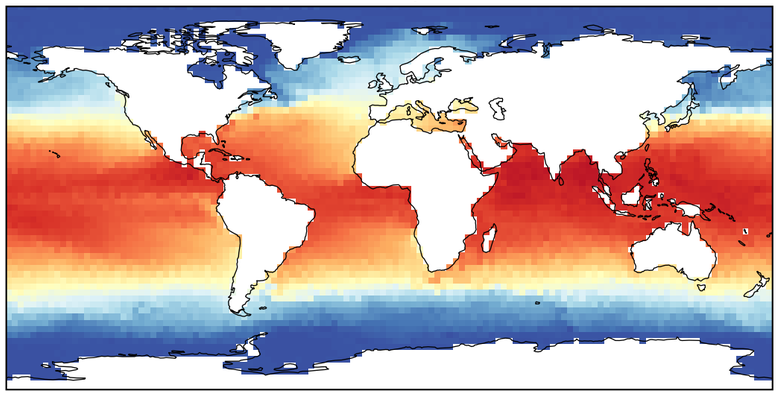
SFNO — Step 1 (°C)



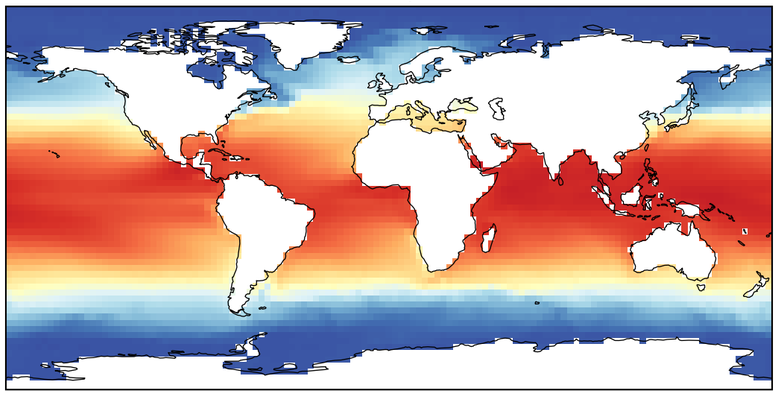
Error — Step 1 (°C)



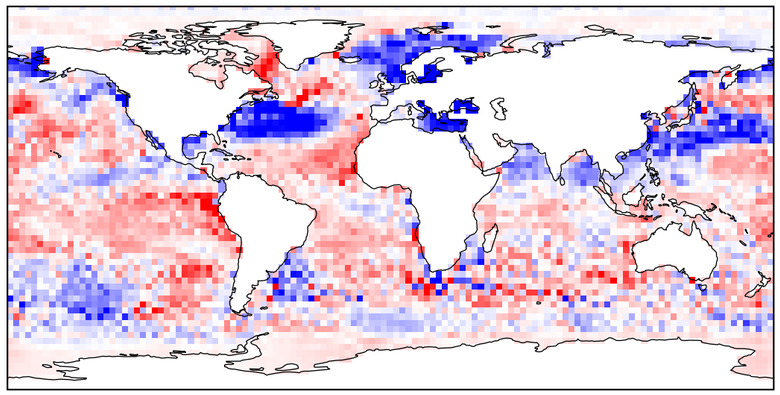
Actual — Step 46 (°C)



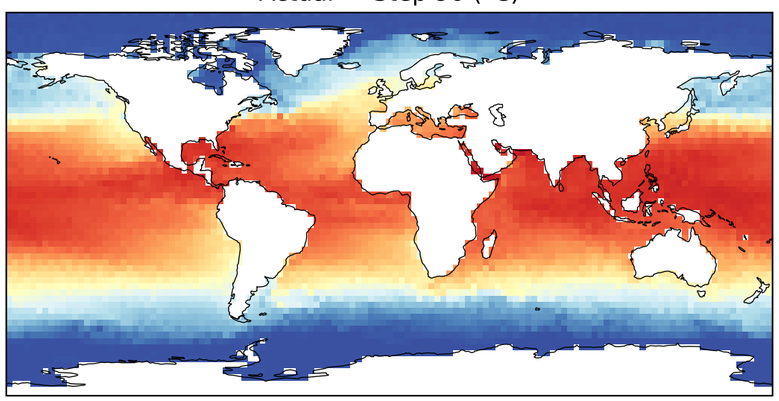
SFNO — Step 46 (°C)



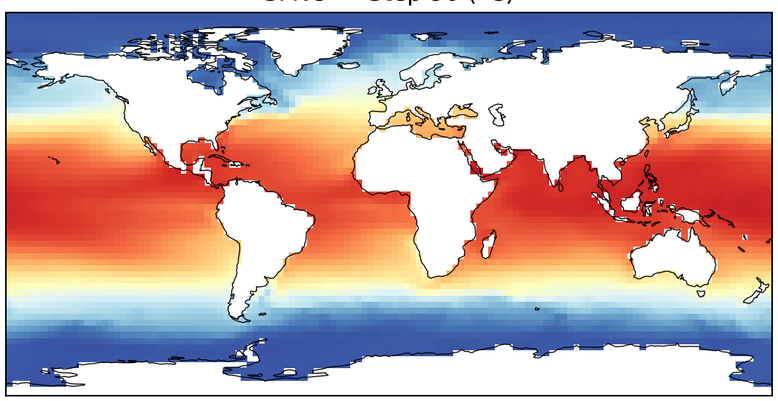
Error — Step 46 (°C)



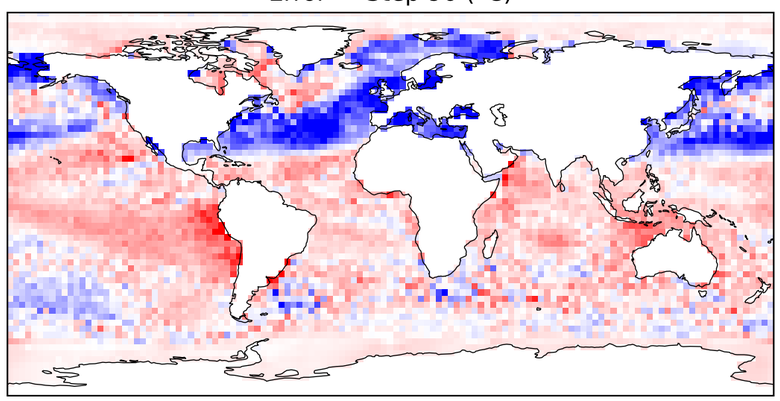
Actual — Step 90 (°C)

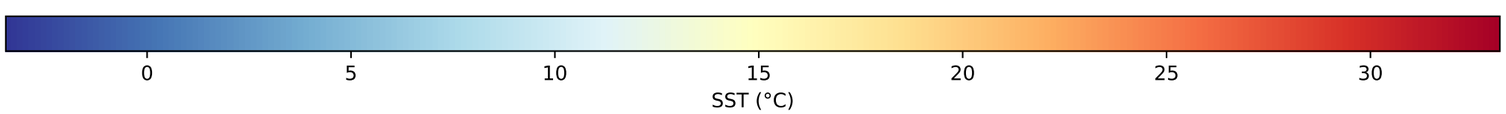


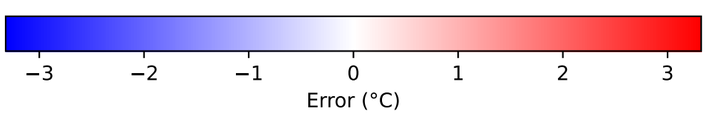
SFNO — Step 90 (°C)



Error — Step 90 (°C)

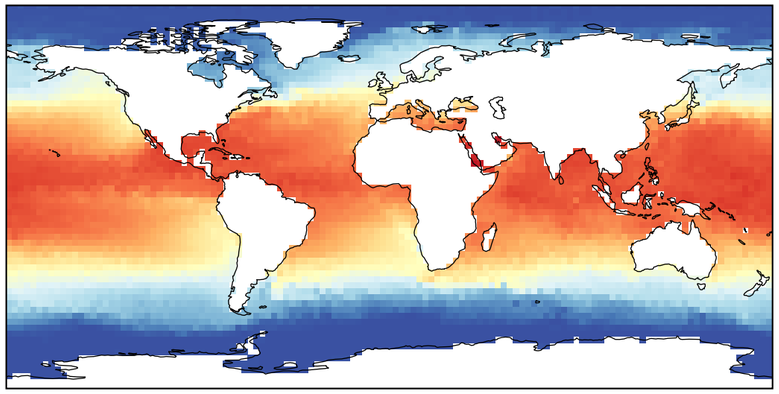




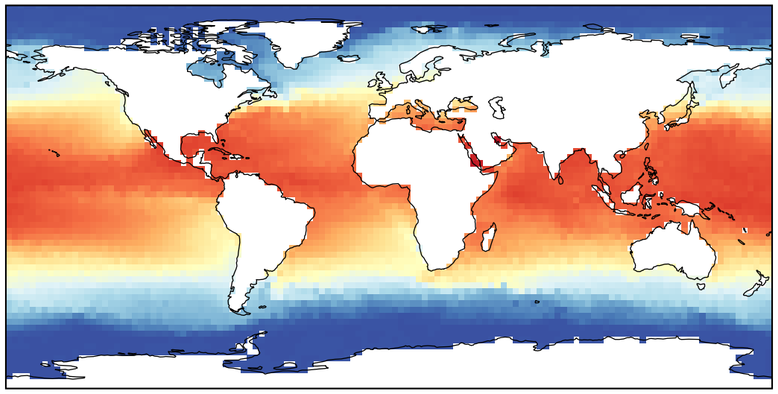


Chain init: 2018-09-27 — Validation (°C)

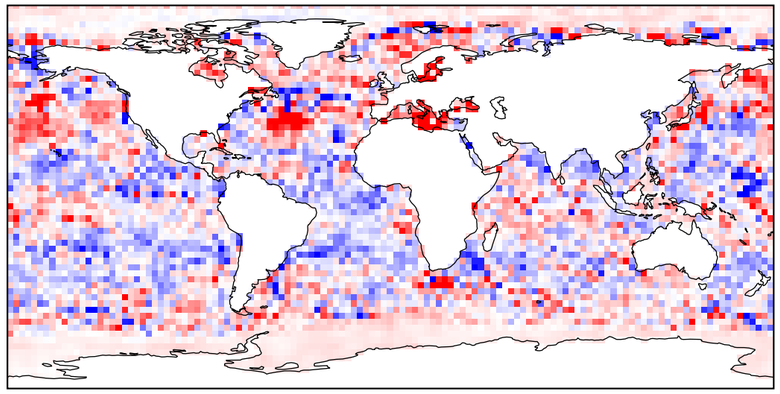
Actual — Step 1 (°C)



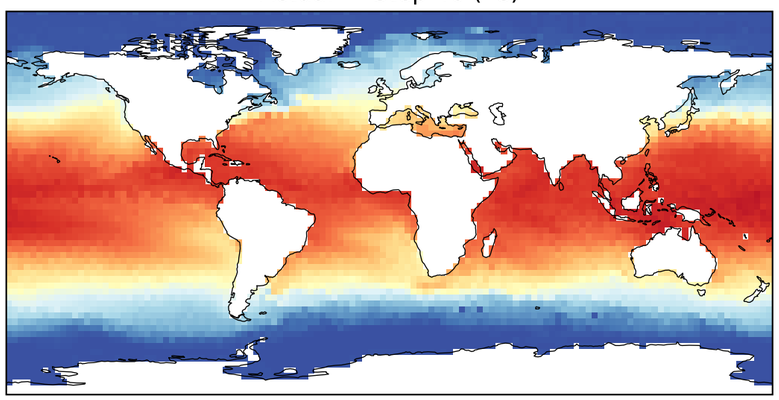
SFNO — Step 1 (°C)



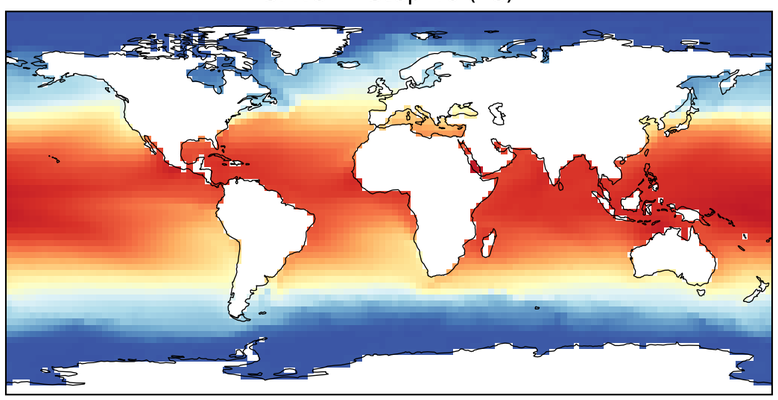
Error — Step 1 (°C)



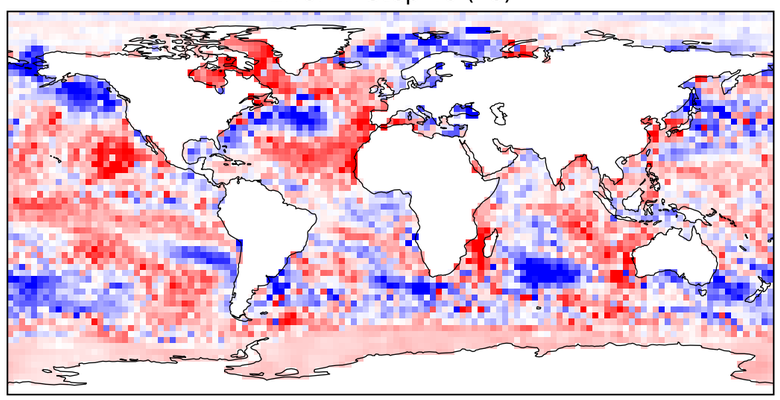
Actual — Step 46 (°C)



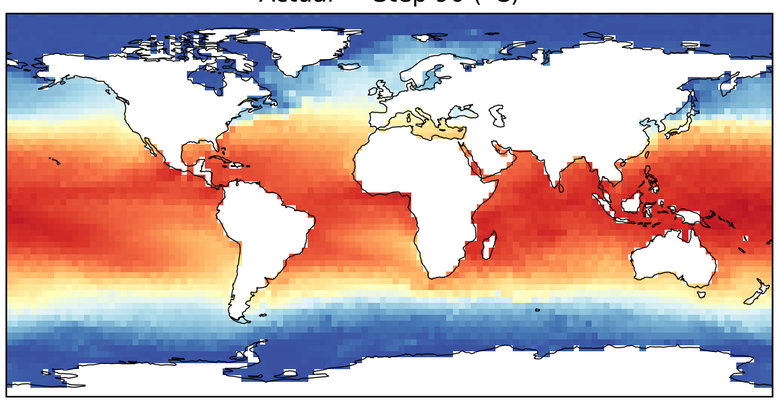
SFNO — Step 46 (°C)



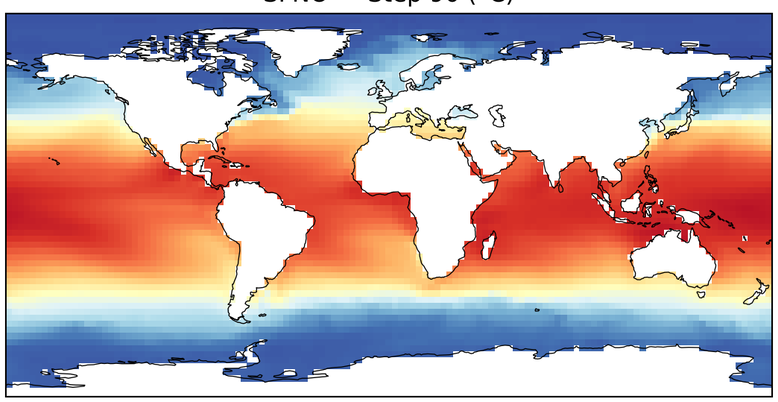
Error — Step 46 (°C)



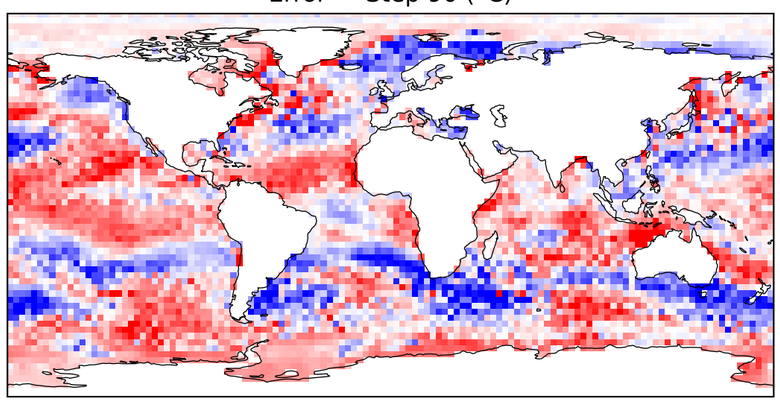
Actual — Step 90 (°C)

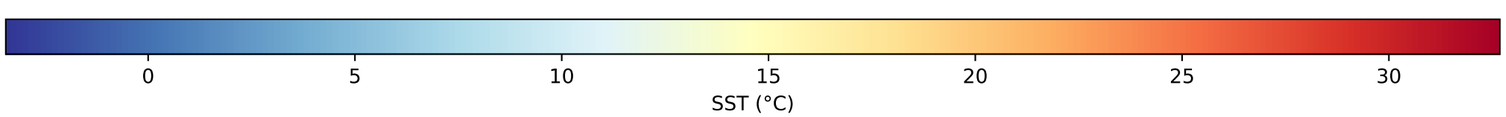


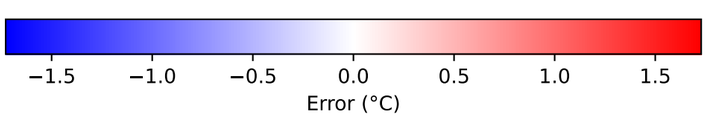
SFNO — Step 90 (°C)



Error — Step 90 (°C)

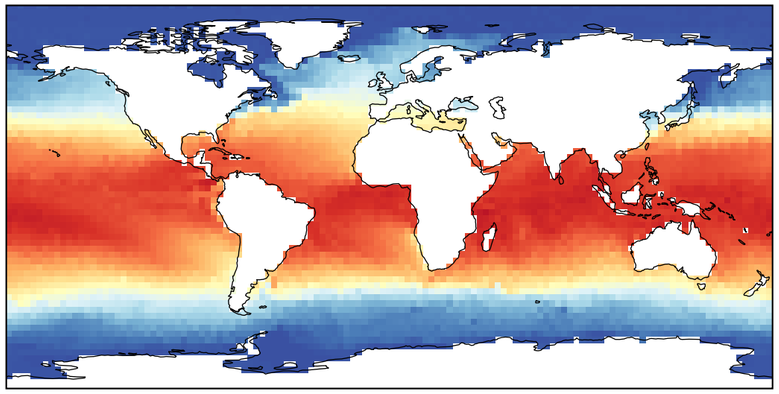




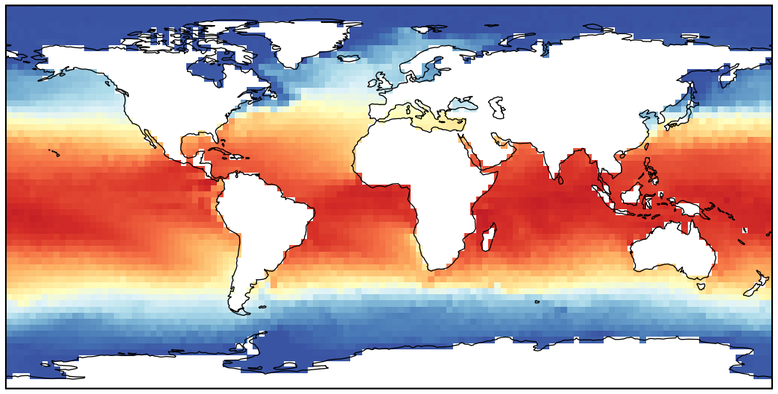


Chain init: 2019-03-26 — Validation (°C)

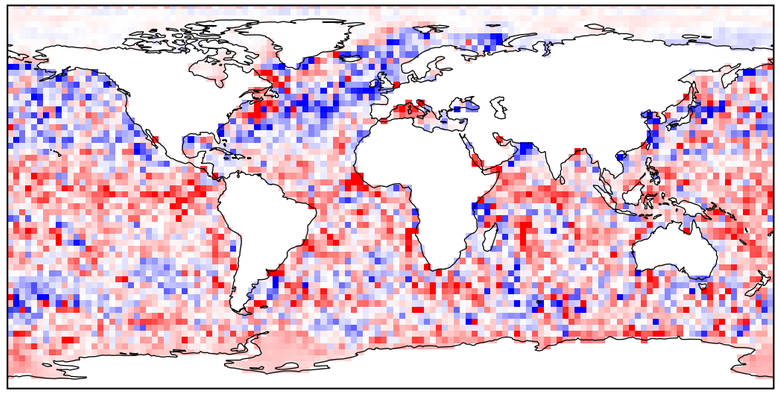
Actual — Step 1 (°C)



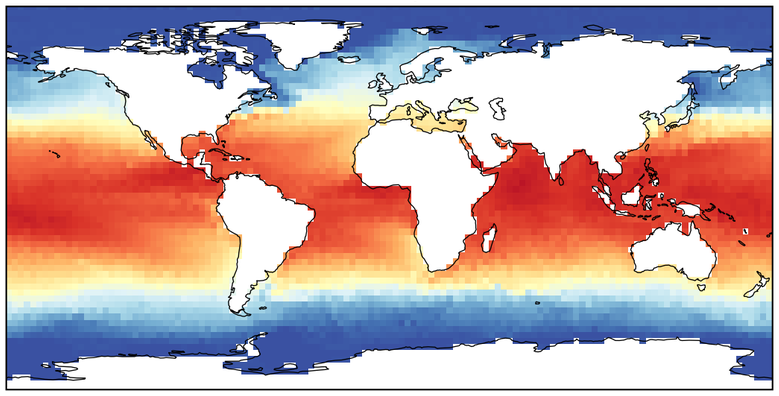
SFNO — Step 1 (°C)



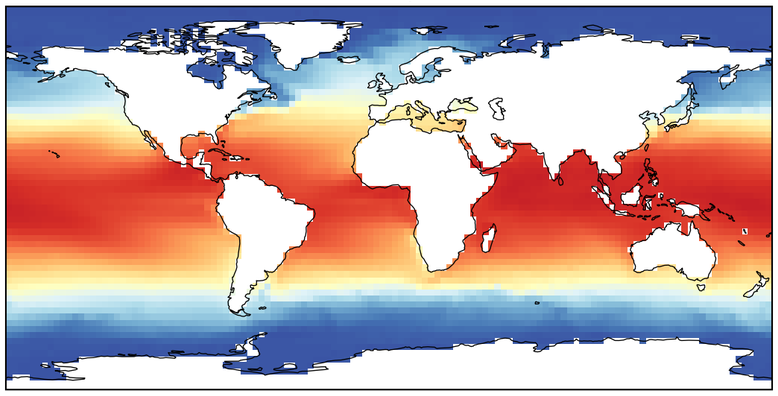
Error — Step 1 (°C)



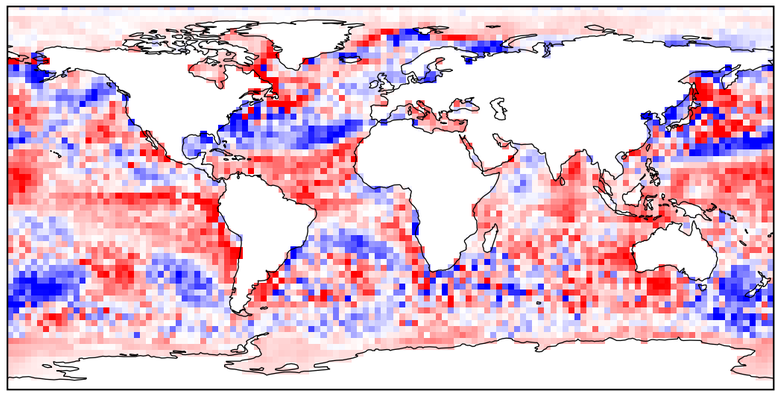
Actual — Step 46 (°C)



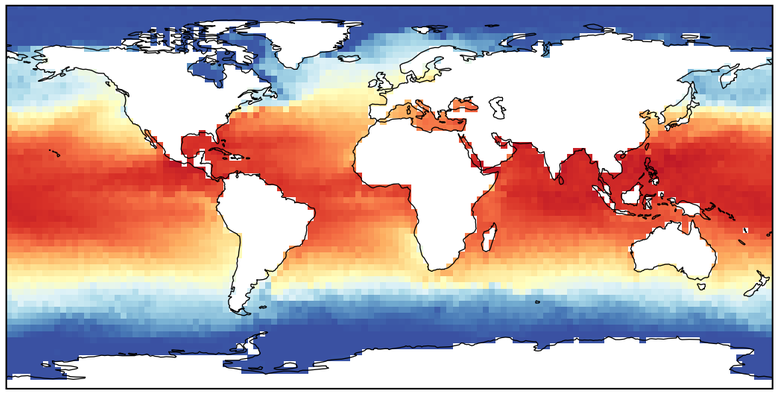
SFNO — Step 46 (°C)



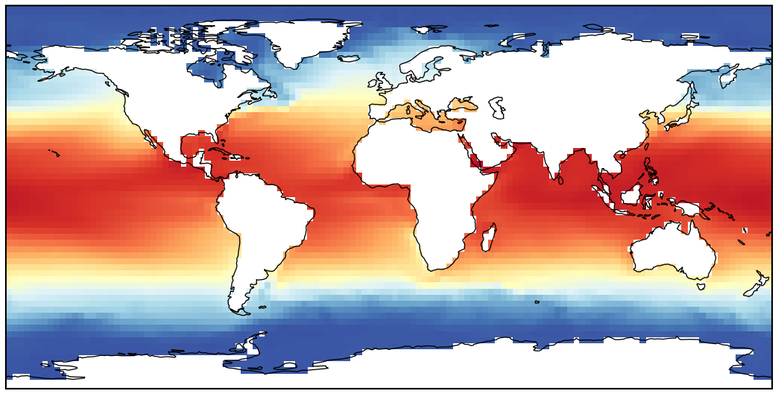
Error — Step 46 (°C)



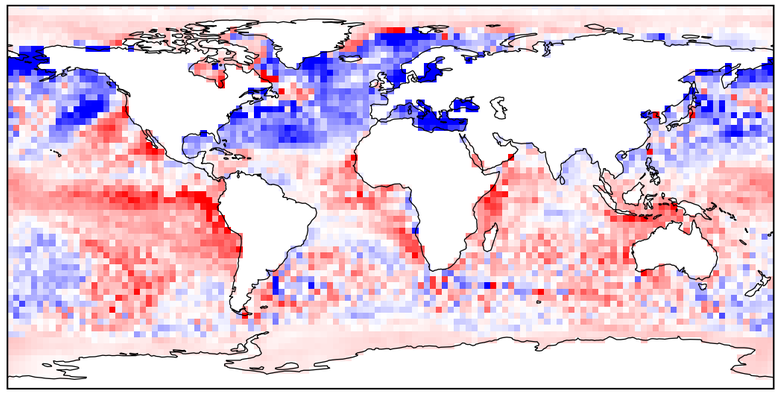
Actual — Step 90 (°C)

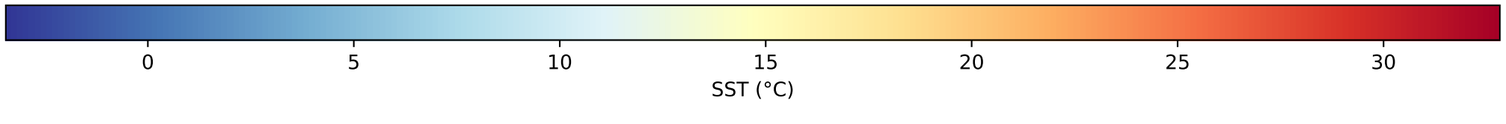


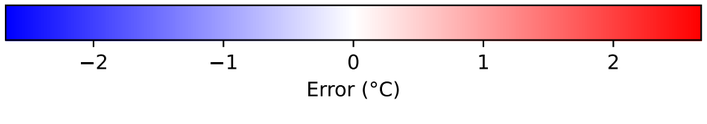
SFNO — Step 90 (°C)



Error — Step 90 (°C)

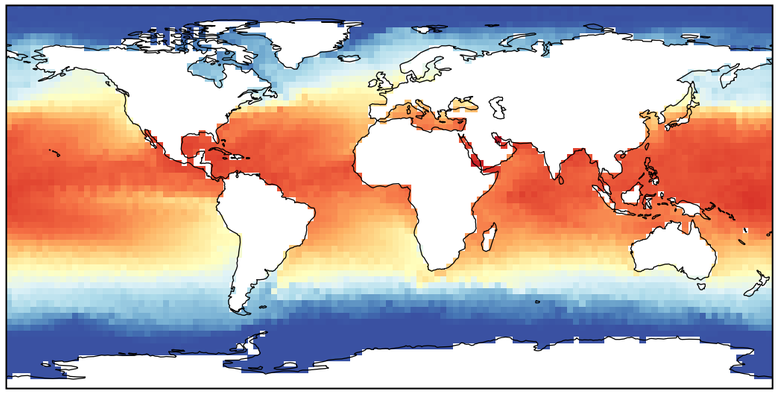




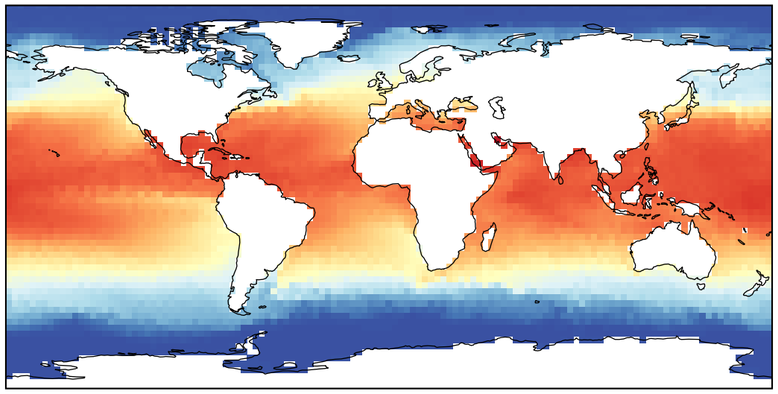


Chain init: 2019-09-22 — Validation (°C)

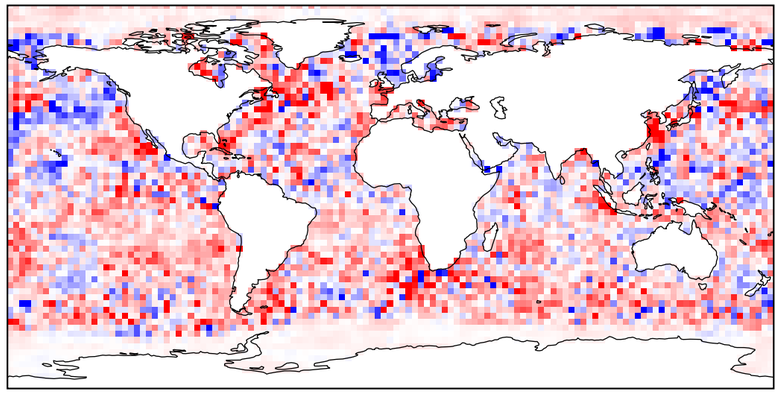
Actual — Step 1 (°C)



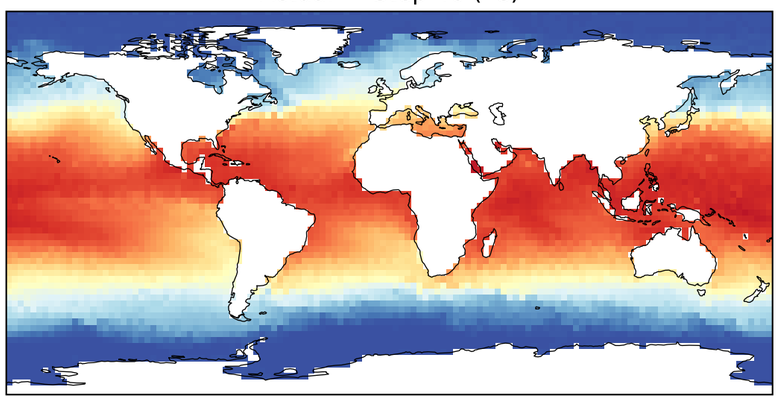
SFNO — Step 1 (°C)



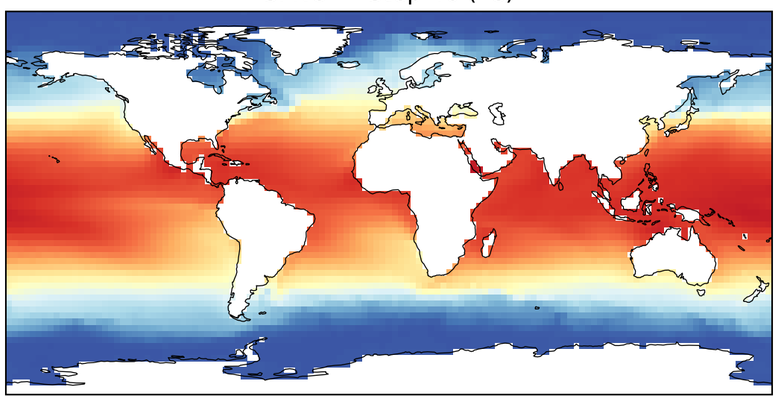
Error — Step 1 (°C)



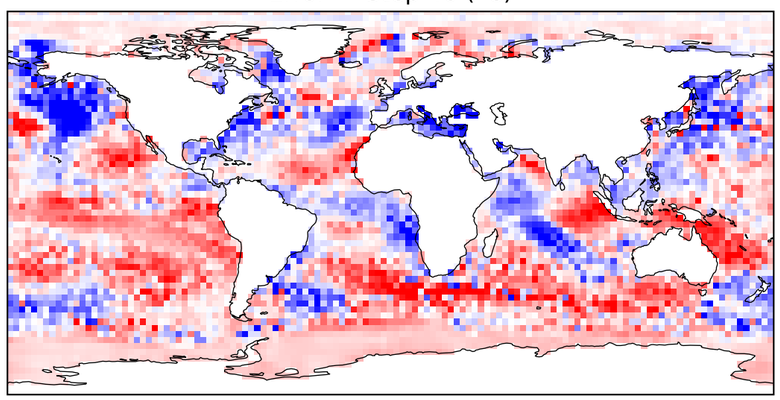
Actual — Step 46 (°C)



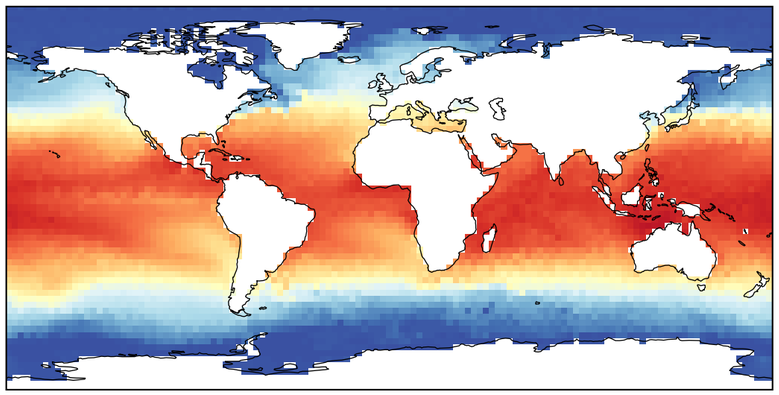
SFNO — Step 46 (°C)



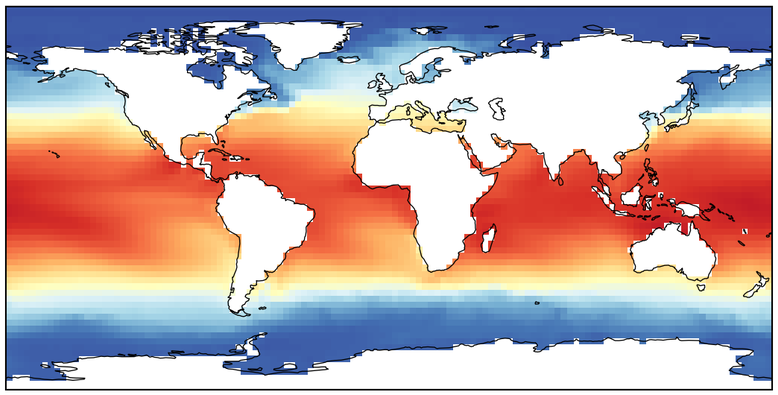
Error — Step 46 (°C)



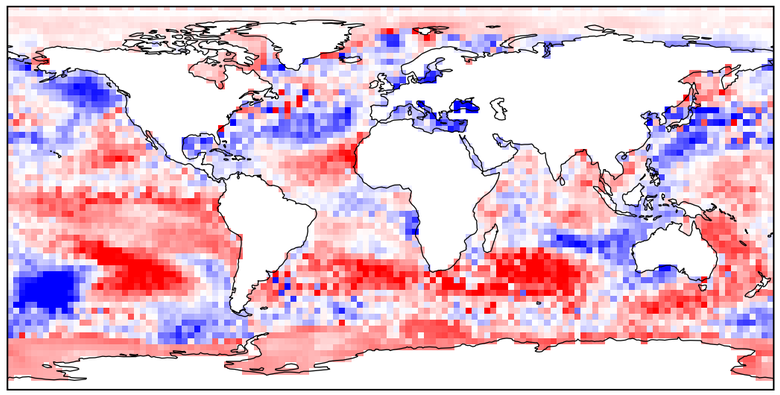
Actual — Step 90 (°C)

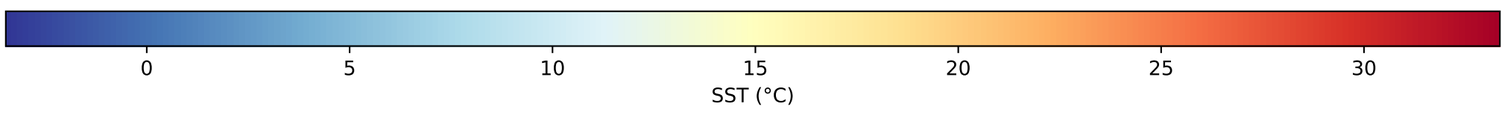


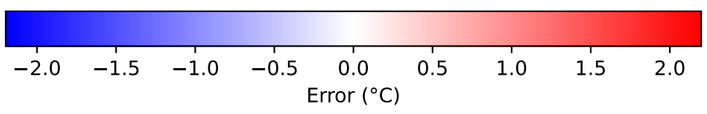
SFNO — Step 90 (°C)



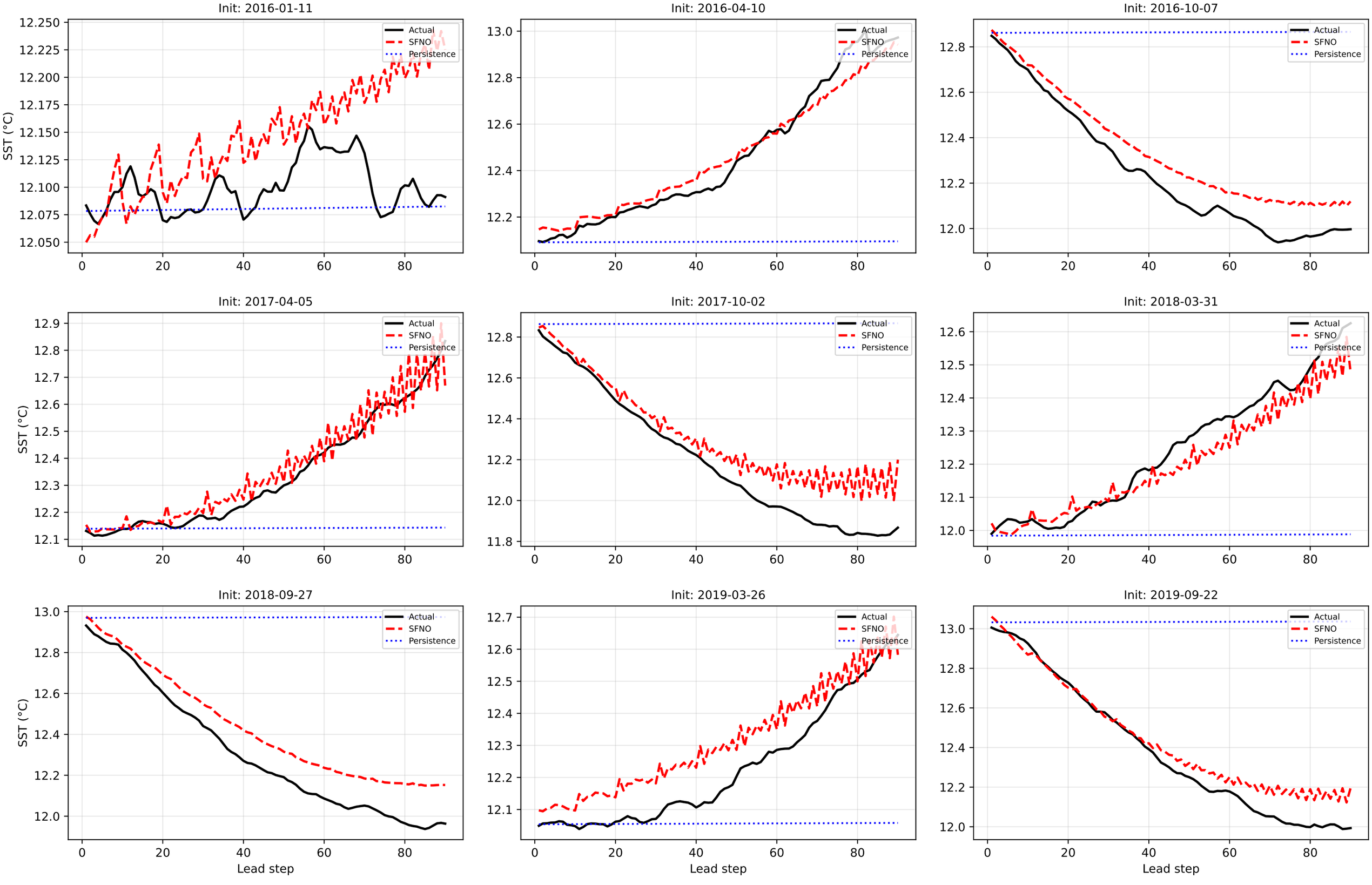
Error — Step 90 (°C)







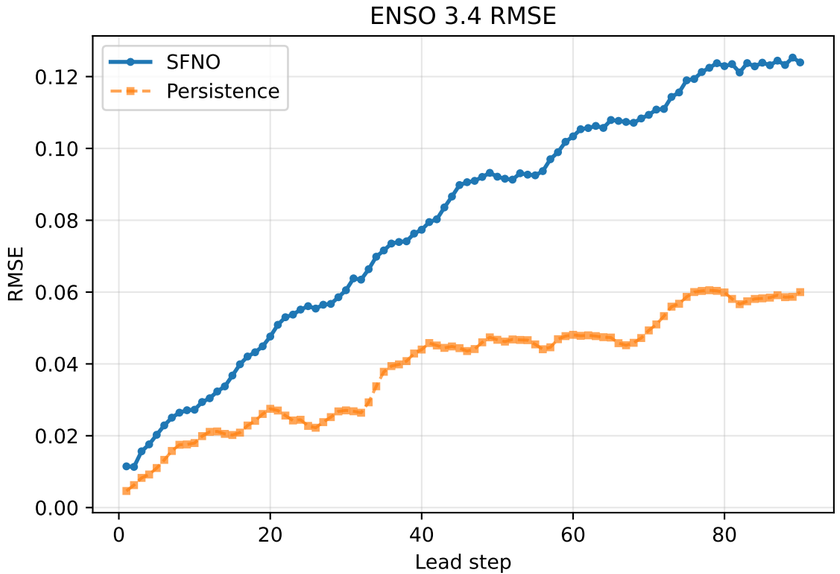
Global-Mean SST — Validation (°C)

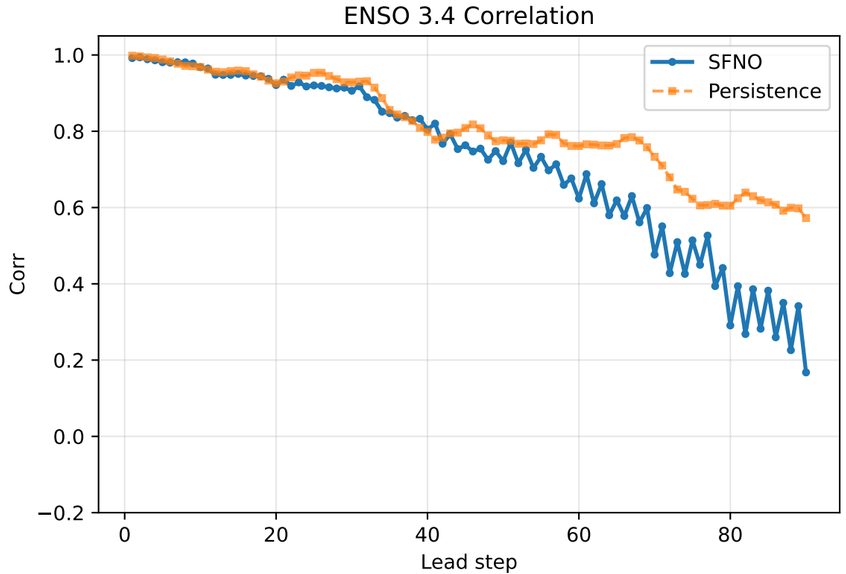


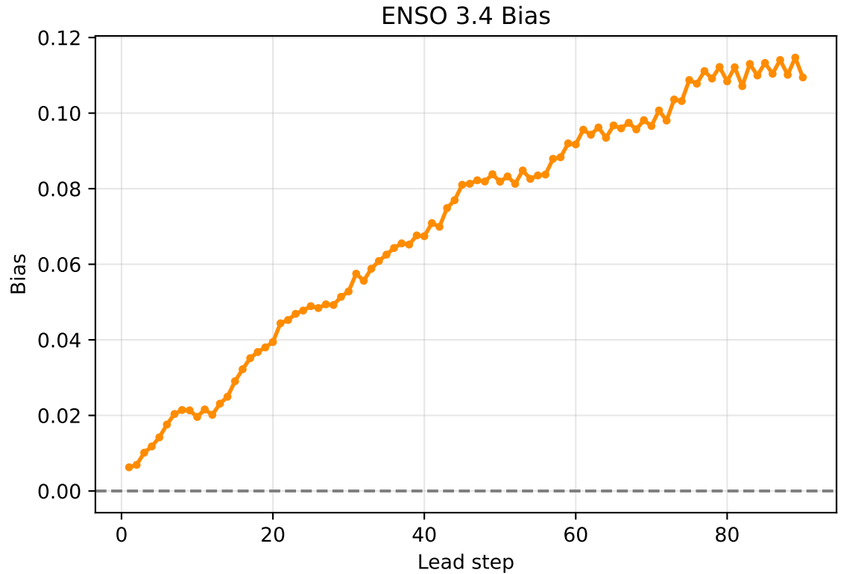
ENSO 3.4 Metrics — Validation

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0115	0.0046	0.9919	0.9981	0.0063
2	0.0113	0.0063	0.9938	0.9964	0.0069
3	0.0157	0.0083	0.9889	0.9938	0.0101
4	0.0176	0.0092	0.9862	0.9920	0.0118
5	0.0203	0.0110	0.9810	0.9881	0.0142
6	0.0229	0.0133	0.9799	0.9833	0.0176
7	0.0250	0.0158	0.9811	0.9771	0.0204
8	0.0264	0.0175	0.9807	0.9714	0.0214
9	0.0271	0.0176	0.9776	0.9705	0.0213
10	0.0272	0.0180	0.9681	0.9684	0.0196
11	0.0294	0.0199	0.9646	0.9615	0.0216
12	0.0304	0.0211	0.9482	0.9562	0.0202
13	0.0323	0.0212	0.9476	0.9548	0.0231
14	0.0337	0.0205	0.9477	0.9576	0.0249
15	0.0368	0.0202	0.9504	0.9592	0.0291
16	0.0399	0.0209	0.9461	0.9576	0.0322
17	0.0421	0.0228	0.9450	0.9494	0.0351
18	0.0432	0.0241	0.9443	0.9431	0.0368
19	0.0449	0.0261	0.9374	0.9328	0.0380
20	0.0476	0.0275	0.9212	0.9257	0.0394
21	0.0509	0.0270	0.9353	0.9303	0.0444
22	0.0530	0.0256	0.9189	0.9410	0.0452
23	0.0537	0.0243	0.9282	0.9466	0.0469
24	0.0551	0.0245	0.9175	0.9458	0.0478
25	0.0561	0.0227	0.9202	0.9531	0.0489
26	0.0554	0.0222	0.9189	0.9533	0.0484
27	0.0565	0.0238	0.9153	0.9448	0.0494
28	0.0567	0.0252	0.9123	0.9366	0.0492
29	0.0586	0.0268	0.9142	0.9284	0.0514
30	0.0605	0.0271	0.9062	0.9280	0.0528
31	0.0638	0.0268	0.9180	0.9302	0.0575
32	0.0635	0.0264	0.8893	0.9312	0.0556
33	0.0664	0.0293	0.8823	0.9144	0.0588
34	0.0698	0.0338	0.8513	0.8869	0.0609
35	0.0716	0.0378	0.8476	0.8557	0.0625
36	0.0735	0.0394	0.8357	0.8439	0.0643
37	0.0739	0.0399	0.8403	0.8370	0.0655
38	0.0741	0.0408	0.8291	0.8272	0.0652
39	0.0763	0.0429	0.8326	0.8091	0.0676
40	0.0773	0.0440	0.8046	0.7981	0.0674
41	0.0795	0.0458	0.8201	0.7782	0.0709
42	0.0803	0.0451	0.7674	0.7835	0.0699
43	0.0836	0.0445	0.7923	0.7945	0.0749
44	0.0866	0.0449	0.7533	0.7965	0.0770
45	0.0898	0.0444	0.7633	0.8085	0.0810
46	0.0906	0.0436	0.7469	0.8178	0.0813
47	0.0910	0.0441	0.7546	0.8079	0.0822
48	0.0921	0.0460	0.7254	0.7888	0.0819
49	0.0932	0.0474	0.7481	0.7738	0.0838
50	0.0921	0.0467	0.7217	0.7763	0.0819
51	0.0916	0.0462	0.7700	0.7751	0.0832
52	0.0913	0.0468	0.7162	0.7672	0.0813
53	0.0931	0.0467	0.7514	0.7680	0.0848
54	0.0927	0.0466	0.7040	0.7665	0.0826
55	0.0925	0.0454	0.7331	0.7763	0.0835
56	0.0937	0.0441	0.6976	0.7919	0.0837
57	0.0970	0.0446	0.7135	0.7902	0.0879
58	0.0989	0.0469	0.6594	0.7685	0.0883
59	0.1018	0.0477	0.6763	0.7616	0.0920
60	0.1034	0.0481	0.6235	0.7614	0.0917
61	0.1053	0.0478	0.6876	0.7659	0.0956
62	0.1057	0.0480	0.6111	0.7649	0.0943
63	0.1063	0.0477	0.6616	0.7628	0.0962
64	0.1057	0.0474	0.5798	0.7625	0.0935
65	0.1079	0.0473	0.6188	0.7666	0.0967
66	0.1077	0.0458	0.5780	0.7823	0.0960
67	0.1074	0.0452	0.6302	0.7845	0.0974
68	0.1071	0.0459	0.5610	0.7758	0.0957
69	0.1083	0.0472	0.5989	0.7581	0.0981
70	0.1093	0.0493	0.4766	0.7330	0.0966
71	0.1108	0.0510	0.5508	0.7100	0.1007
72	0.1110	0.0533	0.4280	0.6791	0.0980
73	0.1143	0.0559	0.5094	0.6473	0.1036
74	0.1156	0.0567	0.4260	0.6412	0.1032
75	0.1190	0.0587	0.5139	0.6229	0.1088
76	0.1193	0.0600	0.4498	0.6055	0.1078
77	0.1213	0.0603	0.5265	0.6065	0.1111
78	0.1225	0.0605	0.3940	0.6097	0.1091
79	0.1237	0.0603	0.4415	0.6050	0.1122
80	0.1229	0.0599	0.2909	0.6048	0.1084
81	0.1235	0.0581	0.3939	0.6243	0.1121
82	0.1211	0.0566	0.2683	0.6391	0.1071
83	0.1238	0.0574	0.3859	0.6298	0.1130
84	0.1229	0.0581	0.2822	0.6190	0.1100
85	0.1239	0.0583	0.3821	0.6136	0.1133
86	0.1232	0.0585	0.2598	0.6072	0.1104
87	0.1244	0.0591	0.3501	0.5917	0.1141
88	0.1232	0.0586	0.2260	0.5994	0.1101
89	0.1253	0.0587	0.3417	0.5977	0.1147
90	0.1239	0.0600	0.1677	0.5725	0.1094

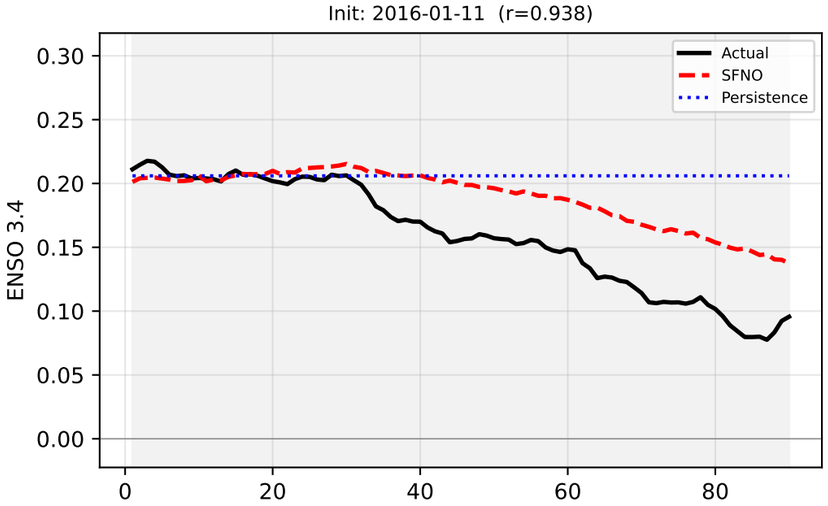
ENSO 3.4 Forecast Skill — Validation

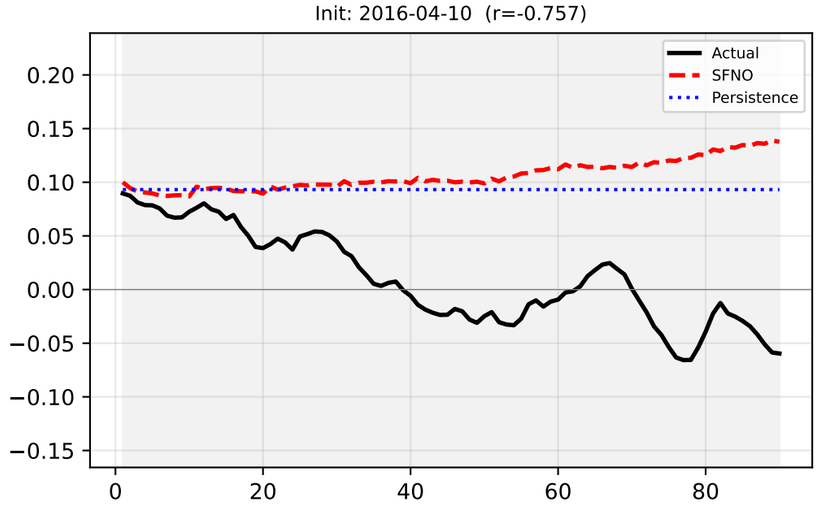


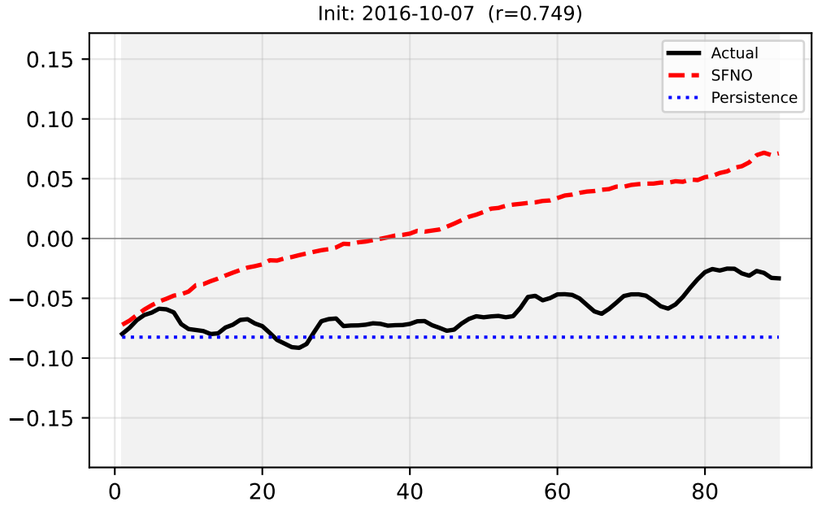


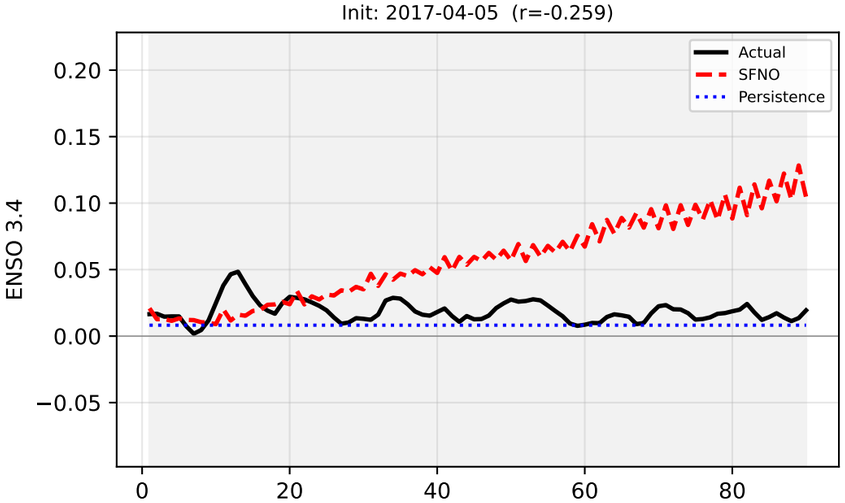


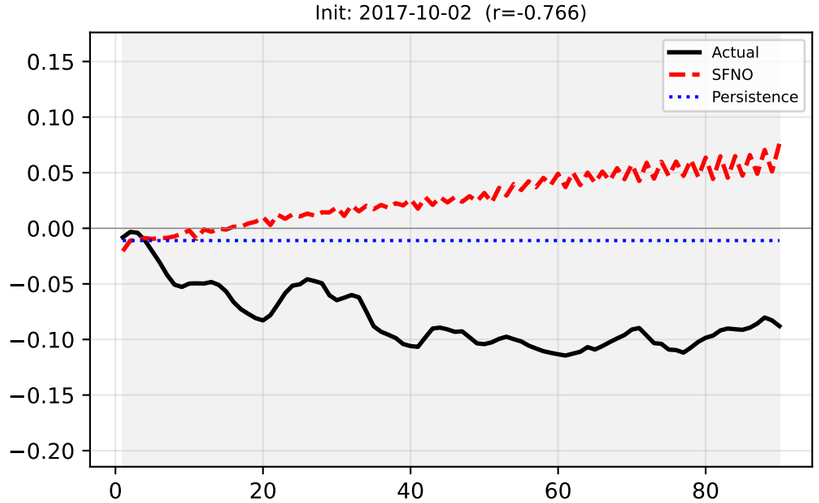
ENSO 3.4 Index — Validation

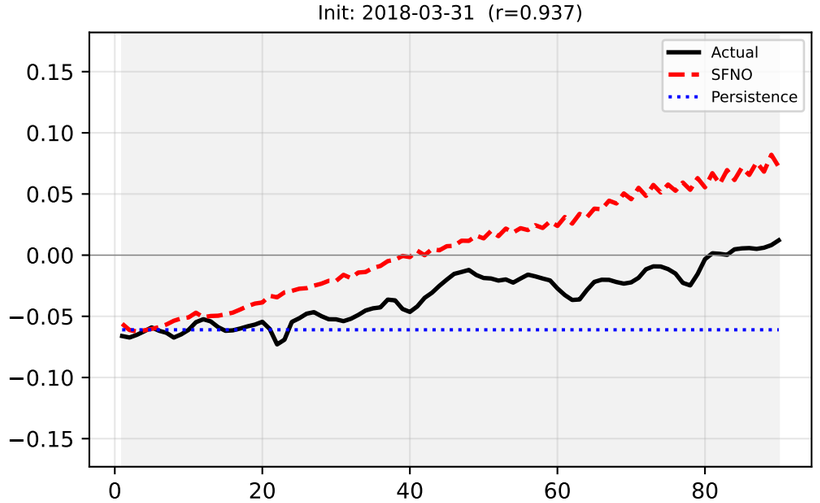


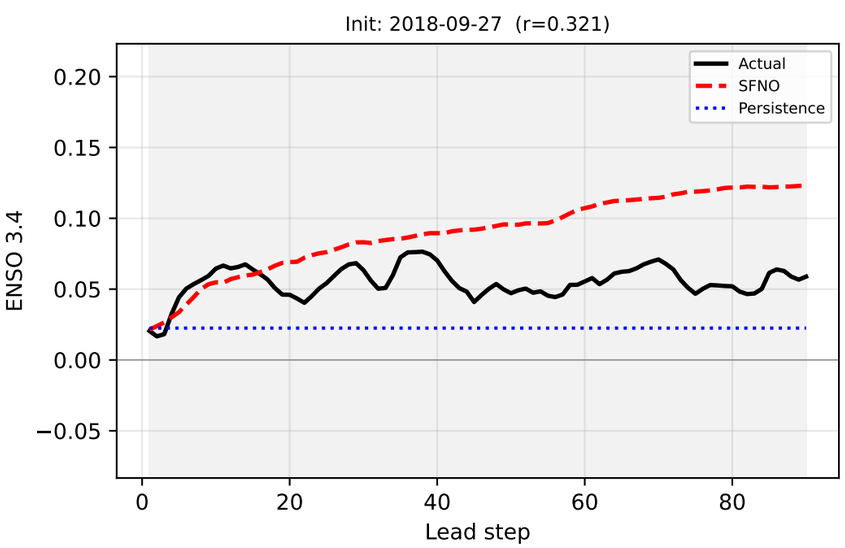


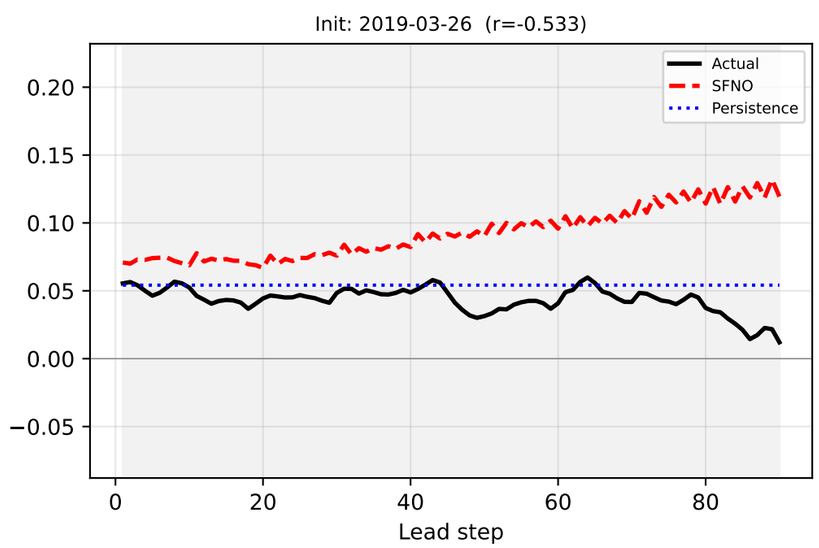


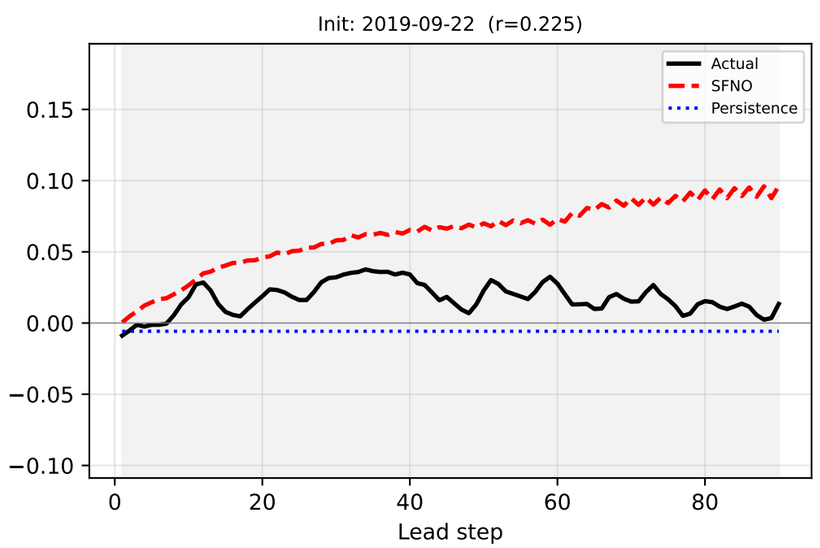




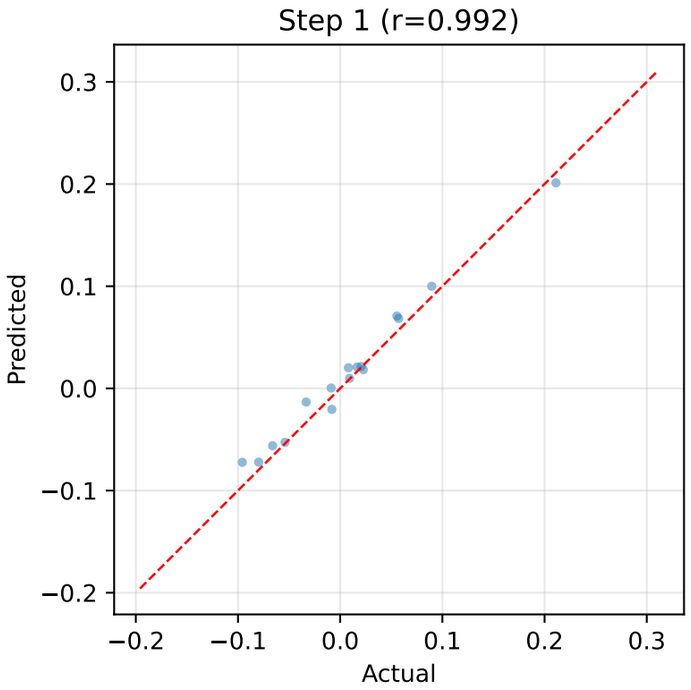


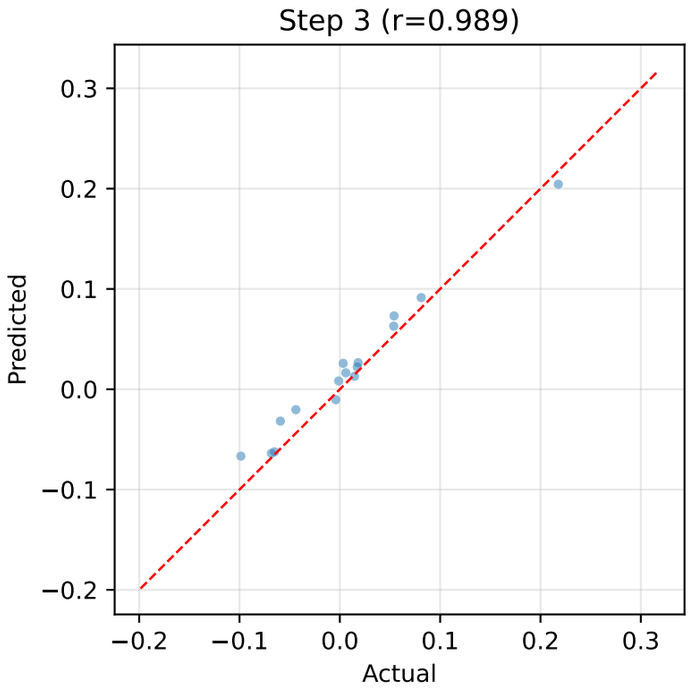


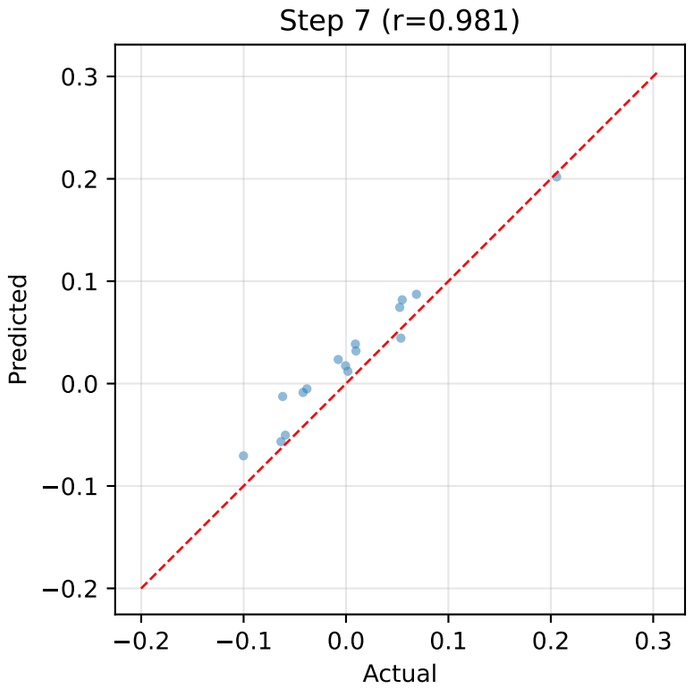


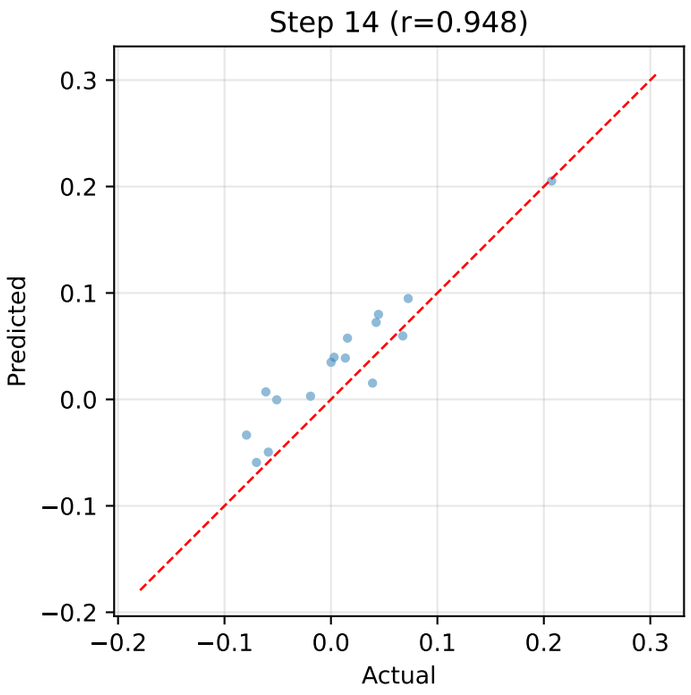


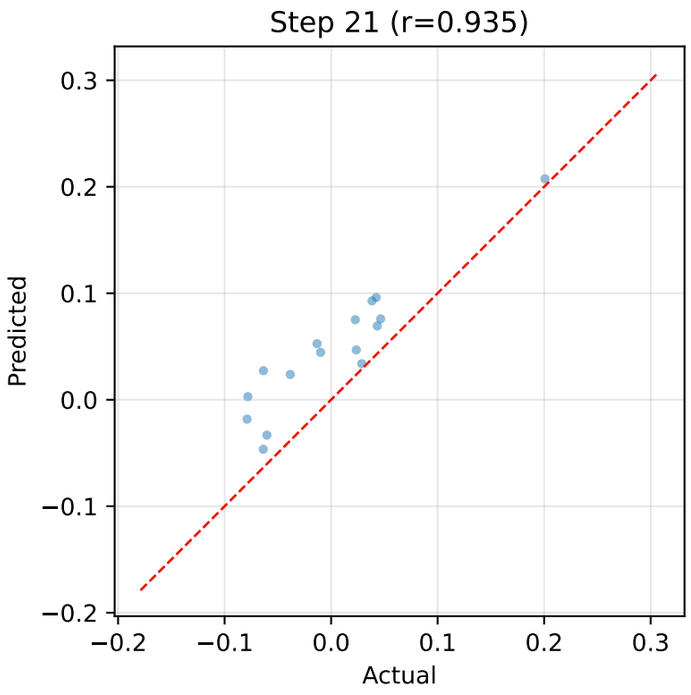
ENSO 3.4 Scatter — Validation

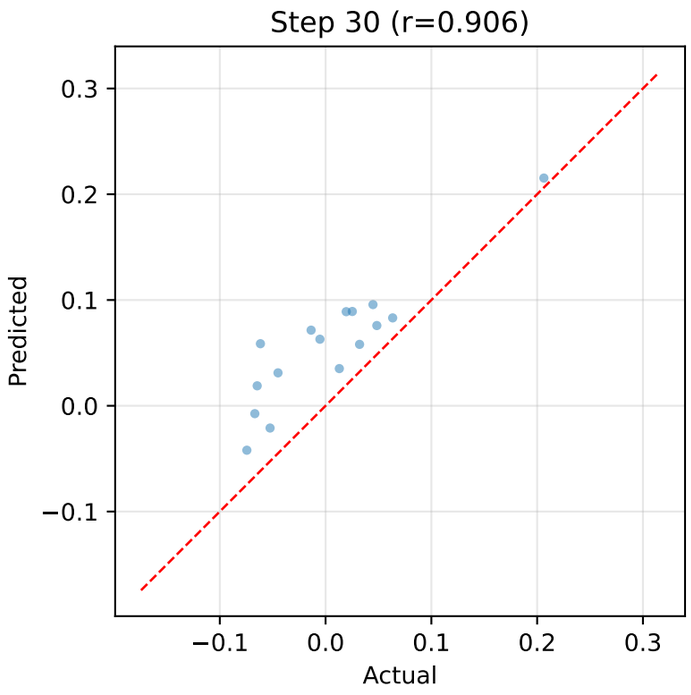


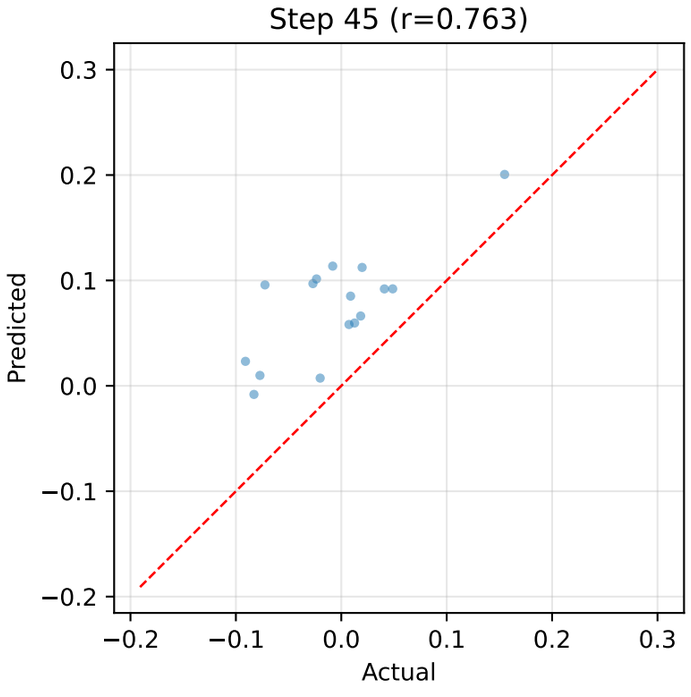


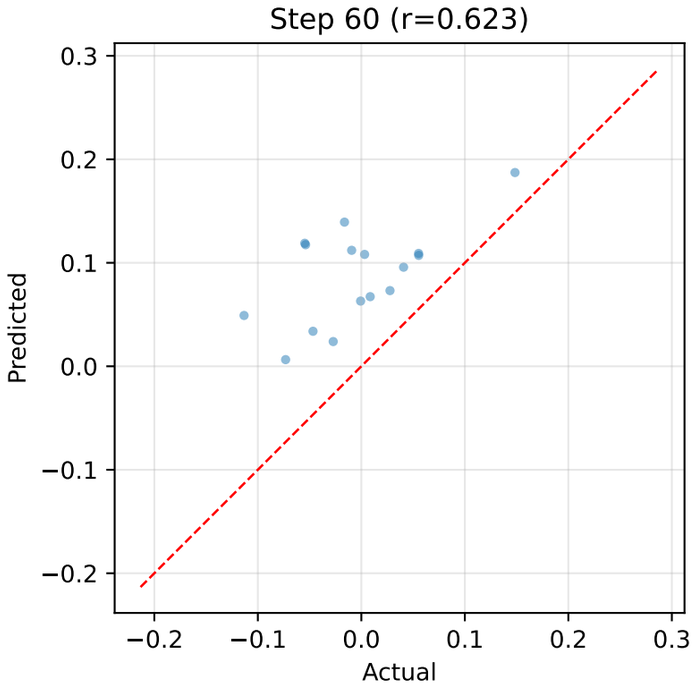


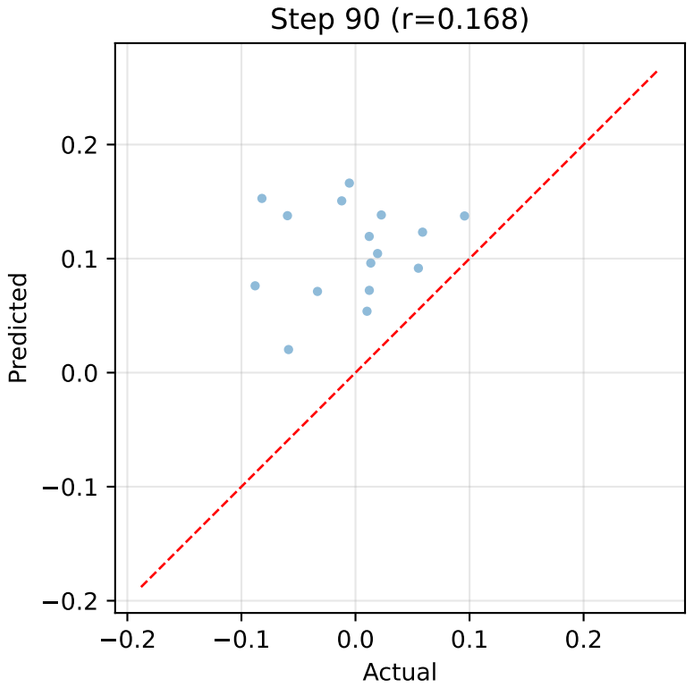








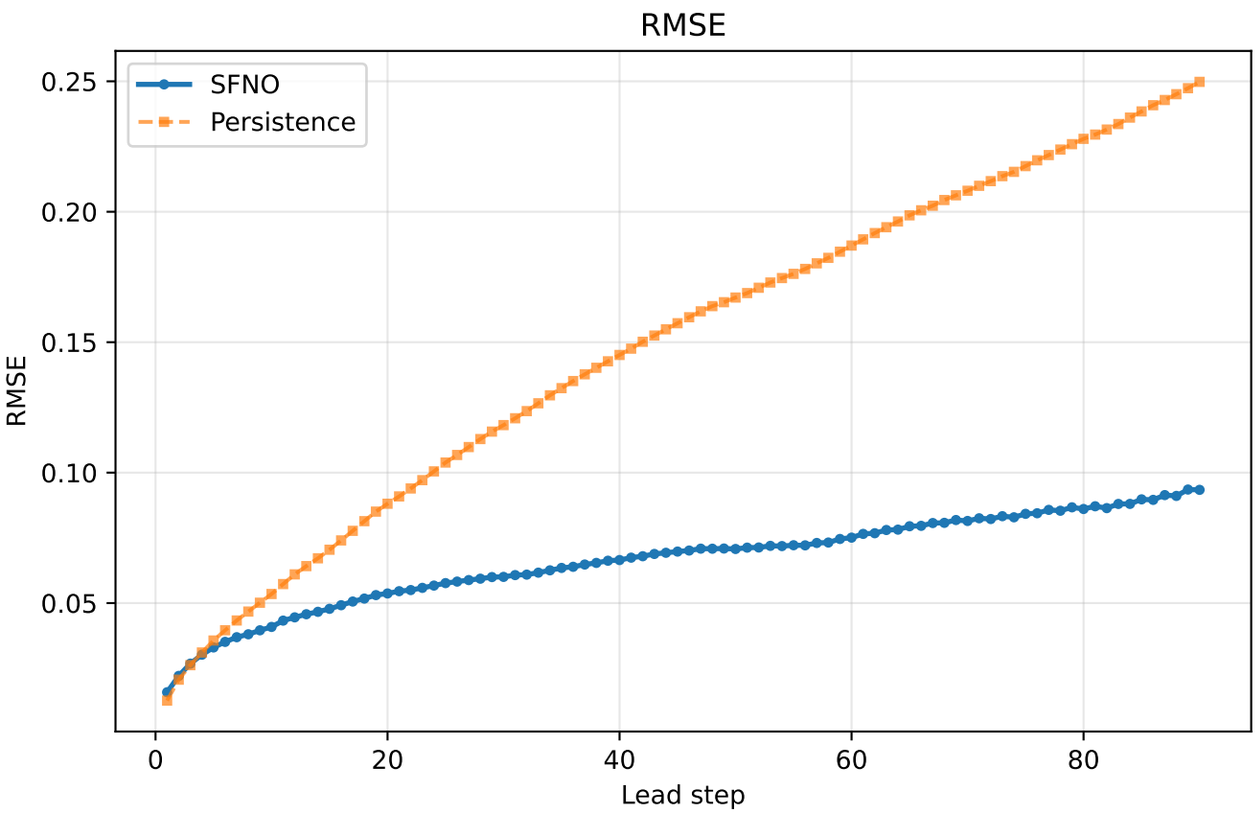


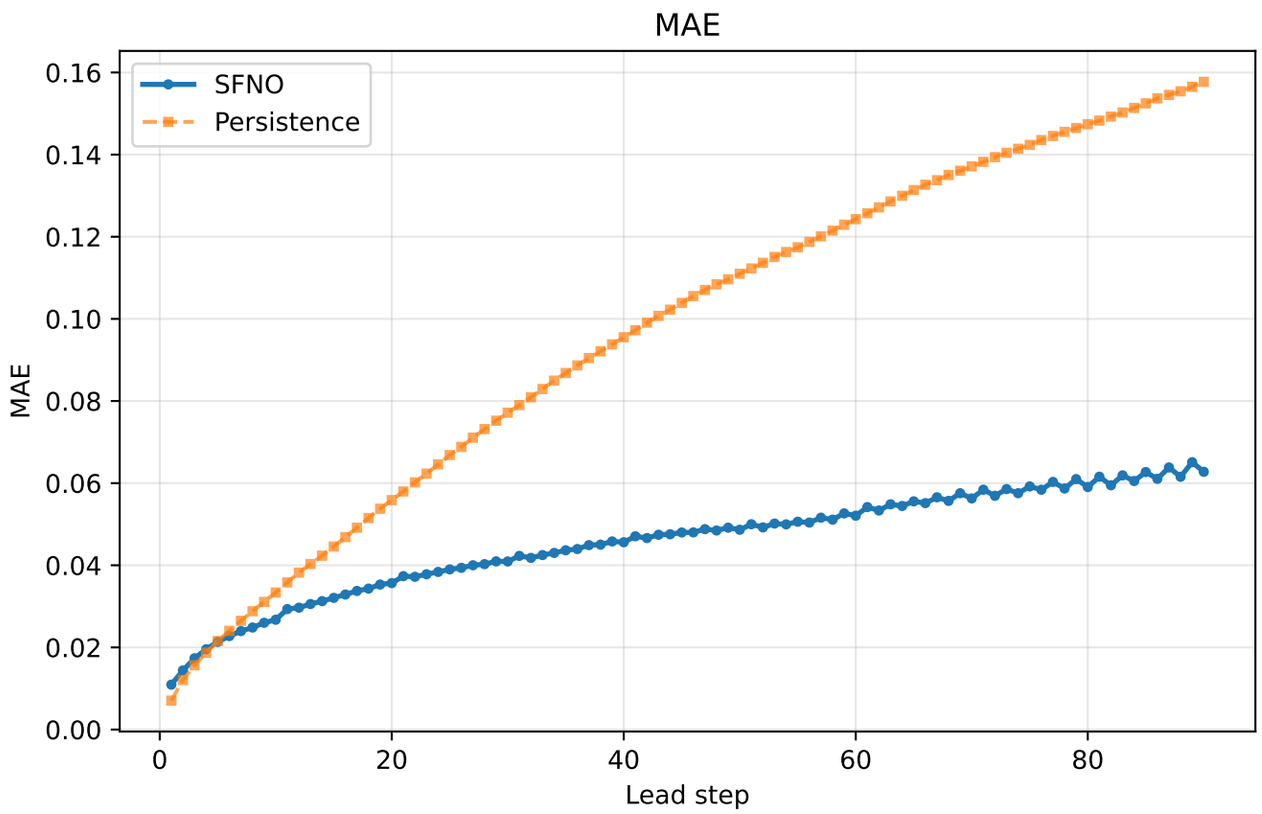


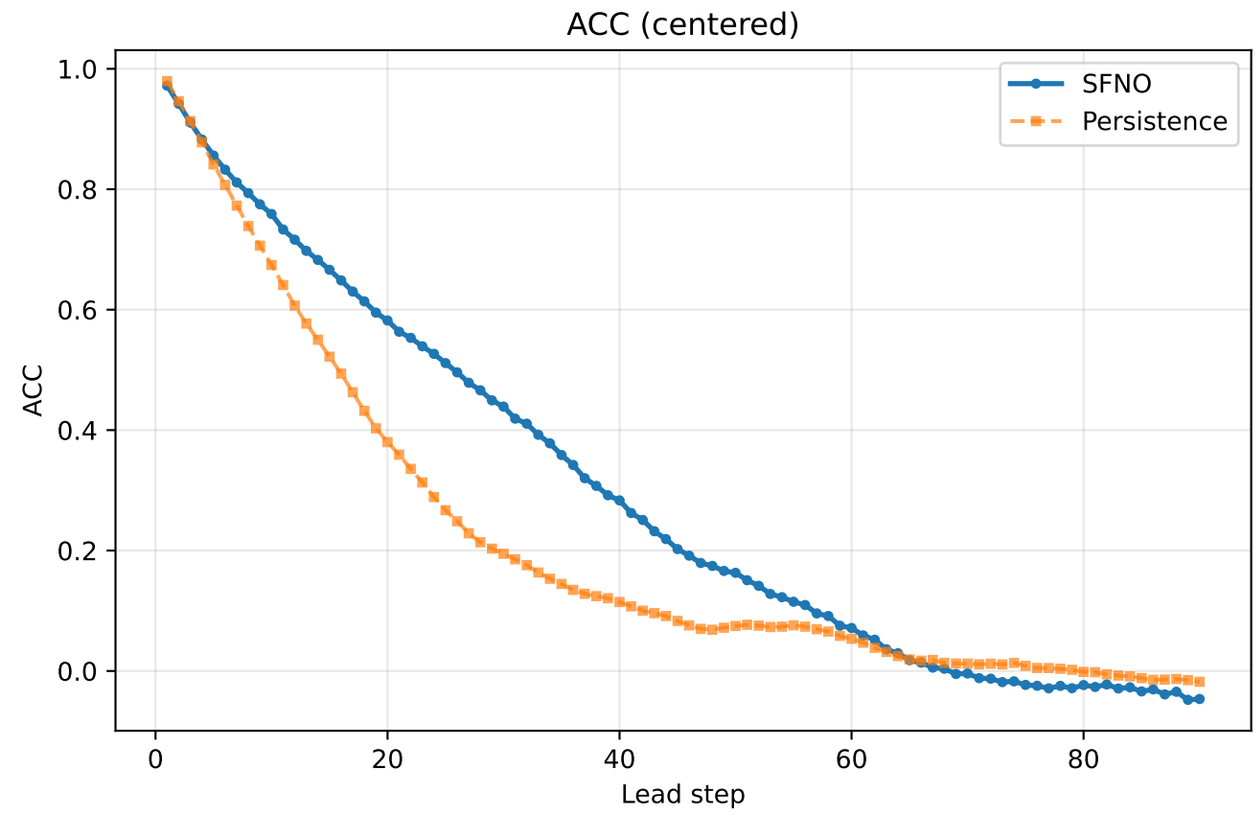
Metrics by Lead Step — Test

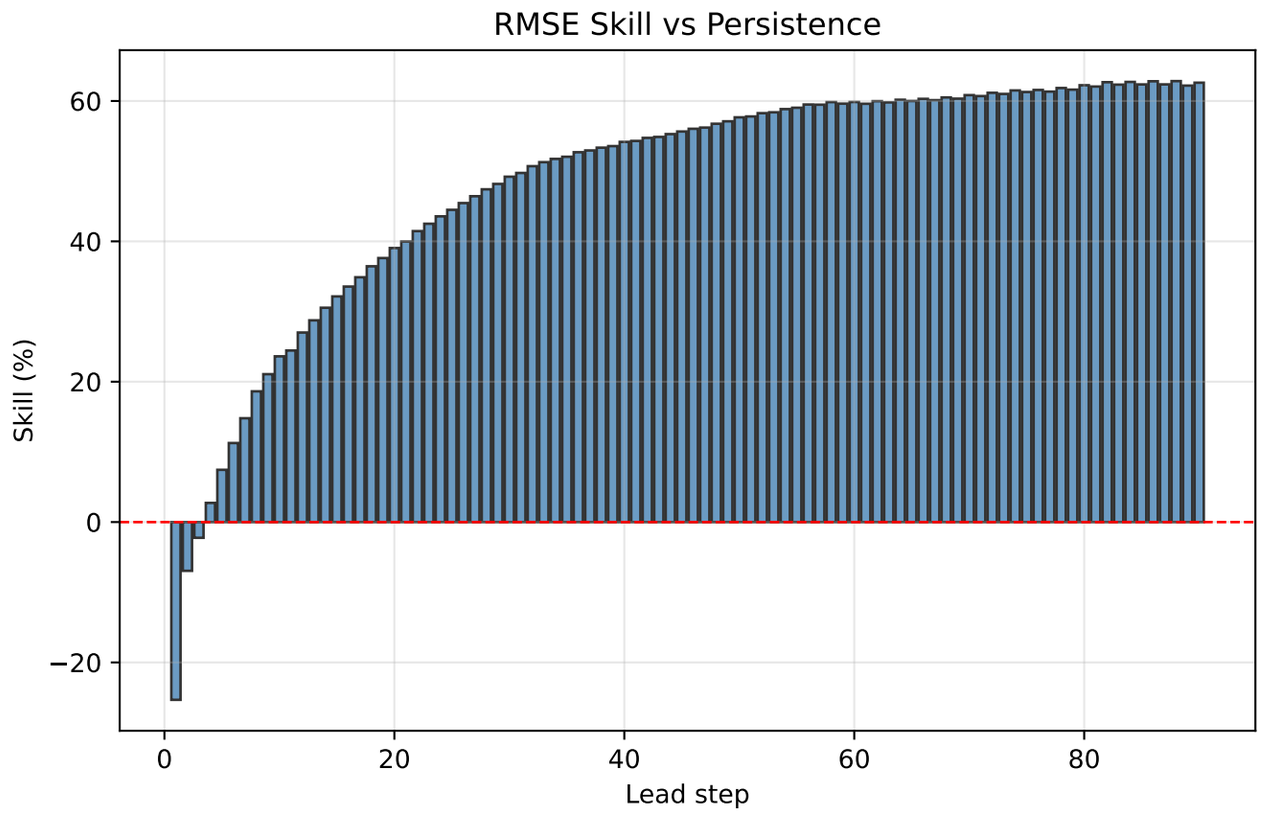
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0158	0.0126	-25.3%	0.0109	0.0024	0.9720	0.9796	0.9998
2	0.0221	0.0206	-6.9%	0.0144	0.0011	0.9419	0.9462	0.9995
3	0.0268	0.0262	-2.2%	0.0173	0.0008	0.9107	0.9132	0.9993
4	0.0302	0.0311	2.7%	0.0195	0.0005	0.8825	0.8780	0.9991
5	0.0330	0.0357	7.4%	0.0214	0.0006	0.8559	0.8411	0.9989
6	0.0351	0.0396	11.3%	0.0228	0.0007	0.8325	0.8072	0.9988
7	0.0369	0.0433	14.8%	0.0240	0.0005	0.8113	0.7728	0.9987
8	0.0380	0.0467	18.6%	0.0248	0.0002	0.7936	0.7389	0.9986
9	0.0396	0.0501	21.1%	0.0260	0.0002	0.7750	0.7063	0.9985
10	0.0409	0.0535	23.6%	0.0267	0.0000	0.7589	0.6743	0.9984
11	0.0433	0.0573	24.4%	0.0293	0.0027	0.7331	0.6410	0.9982
12	0.0445	0.0610	27.0%	0.0297	0.0005	0.7163	0.6068	0.9981
13	0.0457	0.0642	28.7%	0.0306	0.0001	0.6977	0.5769	0.9980
14	0.0467	0.0672	30.5%	0.0313	-0.0002	0.6826	0.5500	0.9979
15	0.0478	0.0705	32.2%	0.0321	-0.0003	0.6662	0.5219	0.9978
16	0.0492	0.0740	33.6%	0.0329	-0.0007	0.6486	0.4938	0.9977
17	0.0506	0.0777	34.9%	0.0338	-0.0009	0.6301	0.4629	0.9975
18	0.0517	0.0814	36.4%	0.0343	-0.0012	0.6138	0.4320	0.9974
19	0.0531	0.0851	37.6%	0.0353	-0.0012	0.5951	0.4030	0.9973
20	0.0537	0.0881	39.0%	0.0357	-0.0014	0.5820	0.3801	0.9972
21	0.0546	0.0909	40.0%	0.0374	0.0017	0.5633	0.3592	0.9971
22	0.0550	0.0939	41.5%	0.0372	-0.0007	0.5532	0.3356	0.9971
23	0.0558	0.0971	42.5%	0.0378	-0.0003	0.5392	0.3131	0.9970
24	0.0567	0.1005	43.6%	0.0384	-0.0007	0.5267	0.2889	0.9969
25	0.0577	0.1039	44.5%	0.0390	-0.0007	0.5113	0.2670	0.9968
26	0.0582	0.1068	45.5%	0.0394	-0.0011	0.4959	0.2487	0.9967
27	0.0588	0.1098	46.4%	0.0400	-0.0010	0.4787	0.2287	0.9967
28	0.0594	0.1129	47.4%	0.0403	-0.0018	0.4661	0.2138	0.9966
29	0.0600	0.1157	48.2%	0.0410	-0.0012	0.4494	0.2032	0.9965
30	0.0600	0.1182	49.2%	0.0409	-0.0018	0.4391	0.1947	0.9965
31	0.0607	0.1208	49.7%	0.0423	0.0015	0.4190	0.1854	0.9964
32	0.0609	0.1236	50.7%	0.0418	-0.0013	0.4107	0.1758	0.9964
33	0.0617	0.1266	51.3%	0.0425	-0.0006	0.3923	0.1635	0.9963
34	0.0626	0.1296	51.7%	0.0430	-0.0012	0.3780	0.1532	0.9962
35	0.0635	0.1324	52.1%	0.0437	-0.0011	0.3586	0.1445	0.9961
36	0.0639	0.1351	52.7%	0.0440	-0.0016	0.3421	0.1347	0.9961
37	0.0648	0.1377	53.0%	0.0449	-0.0012	0.3201	0.1282	0.9959
38	0.0654	0.1402	53.3%	0.0450	-0.0022	0.3074	0.1243	0.9959
39	0.0662	0.1427	53.6%	0.0458	-0.0017	0.2919	0.1209	0.9958
40	0.0665	0.1451	54.2%	0.0456	-0.0031	0.2833	0.1145	0.9957
41	0.0674	0.1475	54.3%	0.0471	-0.0001	0.2624	0.1073	0.9956
42	0.0680	0.1502	54.7%	0.0466	-0.0033	0.2508	0.1002	0.9955
43	0.0688	0.1526	54.9%	0.0474	-0.0020	0.2320	0.0960	0.9954
44	0.0693	0.1549	55.3%	0.0476	-0.0027	0.2190	0.0912	0.9954
45	0.0697	0.1573	55.7%	0.0480	-0.0022	0.2024	0.0831	0.9953
46	0.0701	0.1596	56.0%	0.0480	-0.0029	0.1915	0.0757	0.9952
47	0.0709	0.1618	56.2%	0.0488	-0.0019	0.1793	0.0700	0.9952
48	0.0708	0.1638	56.8%	0.0485	-0.0031	0.1745	0.0685	0.9952
49	0.0709	0.1653	57.1%	0.0492	-0.0015	0.1662	0.0718	0.9951
50	0.0707	0.1671	57.7%	0.0487	-0.0030	0.1631	0.0747	0.9952
51	0.0713	0.1688	57.8%	0.0500	0.0007	0.1507	0.0768	0.9951
52	0.0713	0.1709	58.3%	0.0492	-0.0028	0.1414	0.0753	0.9951
53	0.0719	0.1729	58.4%	0.0502	-0.0009	0.1279	0.0730	0.9950
54	0.0718	0.1746	58.8%	0.0500	-0.0019	0.1226	0.0736	0.9950
55	0.0722	0.1762	59.0%	0.0506	-0.0009	0.1150	0.0758	0.9950
56	0.0721	0.1781	59.5%	0.0504	-0.0019	0.1095	0.0736	0.9950
57	0.0730	0.1802	59.5%	0.0516	-0.0005	0.0956	0.0694	0.9948
58	0.0732	0.1823	59.8%	0.0511	-0.0025	0.0913	0.0654	0.9948
59	0.0746	0.1847	59.6%	0.0526	-0.0006	0.0752	0.0584	0.9946
60	0.0751	0.1870	59.9%	0.0521	-0.0032	0.0714	0.0535	0.9946
61	0.0765	0.1894	59.6%	0.0542	0.0005	0.0591	0.0470	0.9943
62	0.0768	0.1918	60.0%	0.0533	-0.0037	0.0519	0.0385	0.9943
63	0.0780	0.1941	59.8%	0.0548	-0.0015	0.0361	0.0316	0.9941
64	0.0782	0.1963	60.2%	0.0544	-0.0033	0.0295	0.0248	0.9941
65	0.0795	0.1986	60.0%	0.0556	-0.0021	0.0179	0.0192	0.9939
66	0.0796	0.2006	60.3%	0.0551	-0.0038	0.0138	0.0171	0.9939
67	0.0807	0.2023	60.1%	0.0565	-0.0018	0.0056	0.0183	0.9937
68	0.0808	0.2045	60.5%	0.0557	-0.0044	0.0037	0.0135	0.9937
69	0.0818	0.2063	60.3%	0.0575	-0.0017	-0.0050	0.0125	0.9935
70	0.0815	0.2080	60.8%	0.0563	-0.0049	-0.0042	0.0122	0.9936
71	0.0825	0.2099	60.7%	0.0584	-0.0003	-0.0119	0.0110	0.9934
72	0.0822	0.2117	61.2%	0.0569	-0.0047	-0.0127	0.0120	0.9935
73	0.0833	0.2136	61.0%	0.0586	-0.0016	-0.0185	0.0109	0.9933
74	0.0829	0.2153	61.5%	0.0575	-0.0040	-0.0172	0.0133	0.9934
75	0.0842	0.2175	61.3%	0.0592	-0.0019	-0.0231	0.0086	0.9931
76	0.0844	0.2197	61.6%	0.0584	-0.0043	-0.0245	0.0050	0.9931
77	0.0857	0.2217	61.3%	0.0603	-0.0012	-0.0288	0.0049	0.9929
78	0.0854	0.2238	61.8%	0.0587	-0.0045	-0.0247	0.0037	0.9930
79	0.0867	0.2259	61.6%	0.0610	-0.0005	-0.0286	0.0019	0.9927
80	0.0861	0.2279	62.2%	0.0591	-0.0045	-0.0234	-0.0015	0.9928
81	0.0871	0.2296	62.1%	0.0616	0.0009	-0.0265	-0.0020	0.9927
82	0.0864	0.2315	62.7%	0.0595	-0.0041	-0.0224	-0.0056	0.9928
83	0.0880	0.2336	62.3%	0.0619	0.0001	-0.0291	-0.0076	0.9925
84	0.0880	0.2361	62.7%	0.0605	-0.0035	-0.0271	-0.0090	0.9925
85	0.0898	0.2385	62.4%	0.0627	-0.0008	-0.0339	-0.0119	0.9922
86	0.0896	0.2408	62.8%	0.0610	-0.0043	-0.0306	-0.0147	0.9922
87	0.0914	0.2428	62.4%	0.0638	-0.0004	-0.0389	-0.0142	0.9919
88	0.0911	0.2451	62.8%	0.0616	-0.0053	-0.0345	-0.0133	0.9920
89	0.0935	0.2474	62.2%	0.0651	-0.0003	-0.0479	-0.0151	0.9916
90	0.0934	0.2498	62.6%	0.0628	-0.0058	-0.0465	-0.0180	0.9916

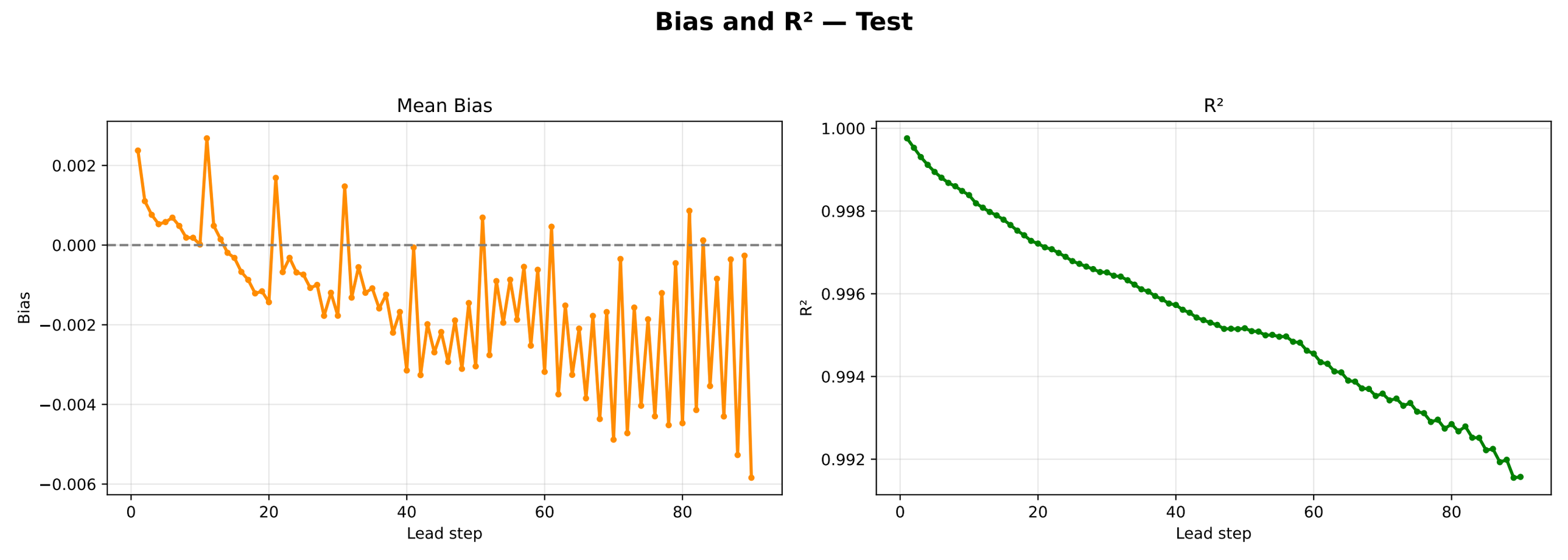
Forecast Skill vs Lead Step — Test





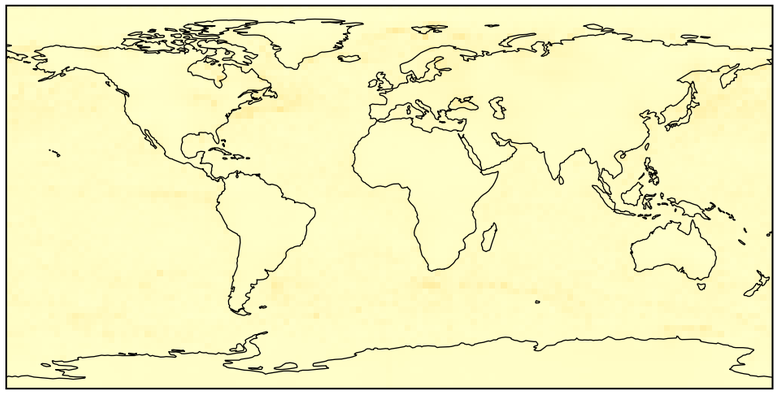




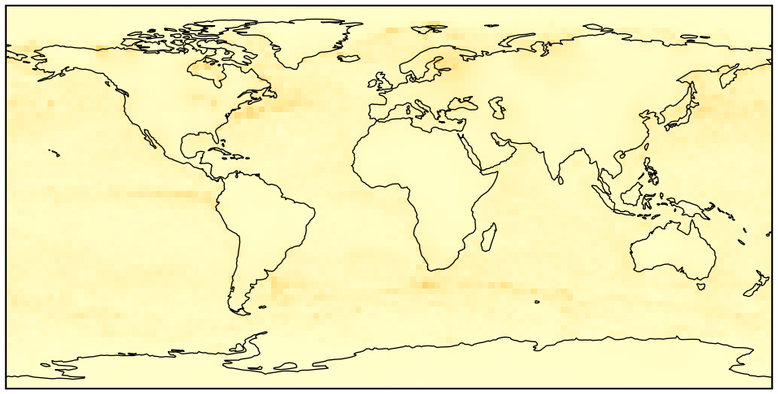

Bias and R² — Test
Mean Bias
Bias
0.002
0.000
−0.002
−0.004
−0.006
0
20
40
60
80
Lead step
R²
R²
1.000
0.998
0.996
0.994
0.992
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Test

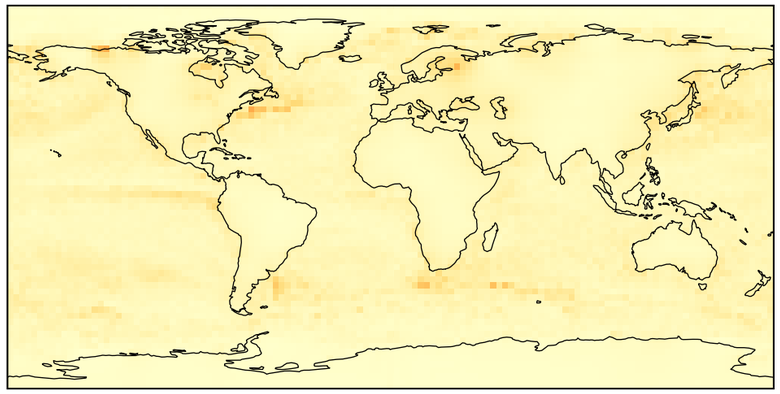
Step 1



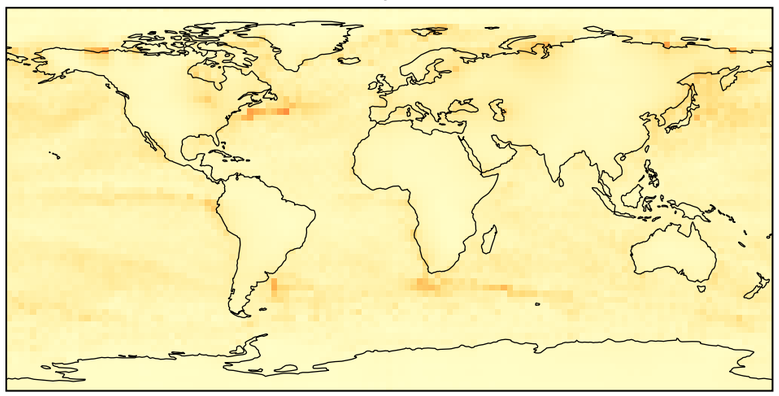
Step 3



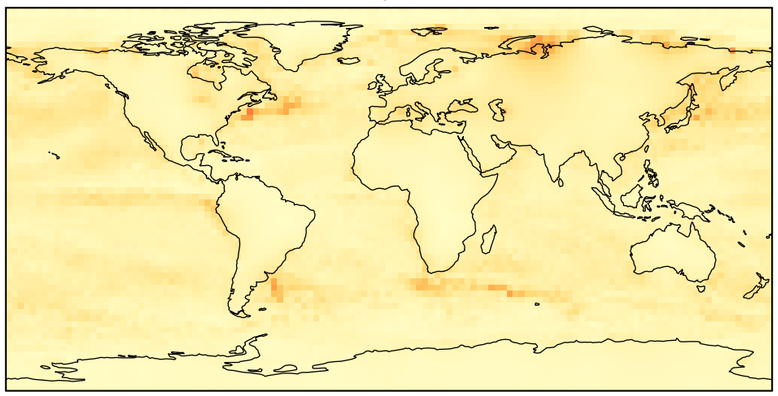
Step 7



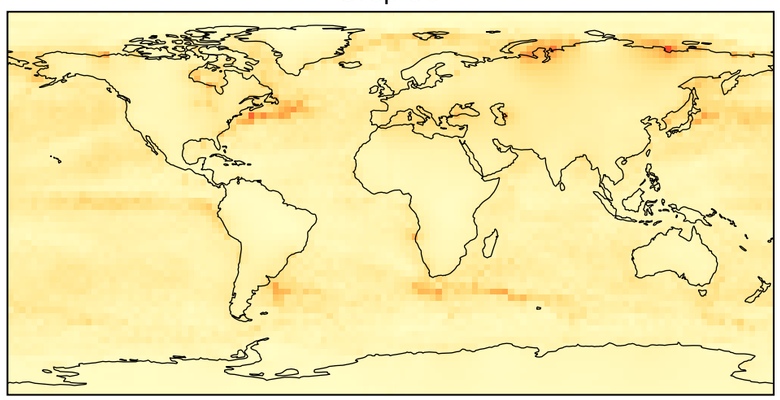
Step 14



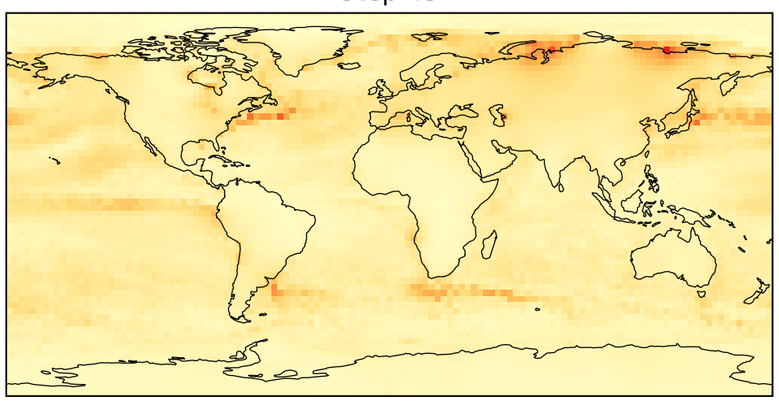
Step 21



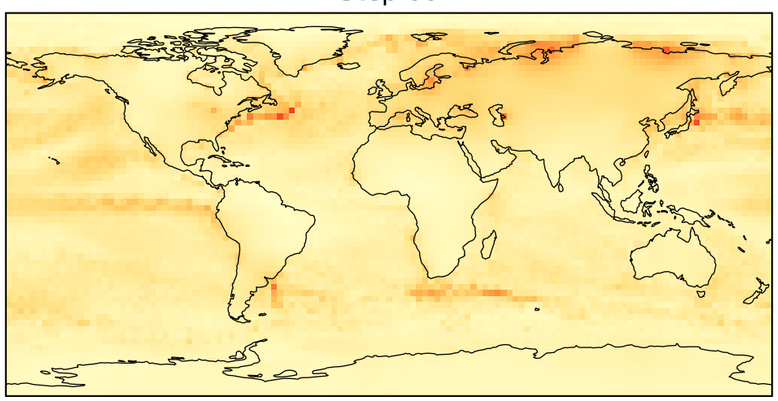
Step 30



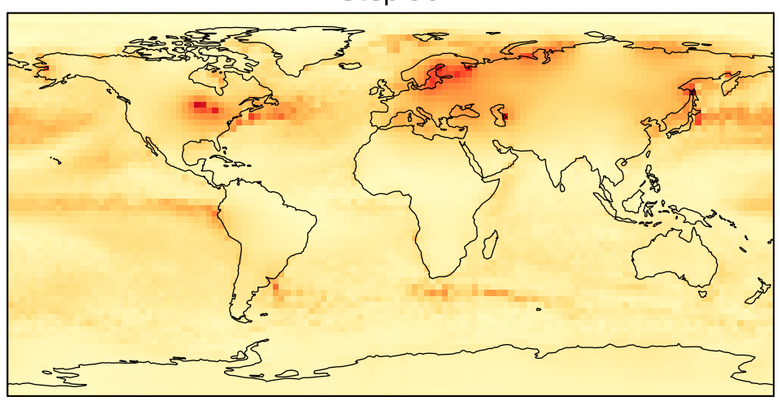
Step 45



Step 60

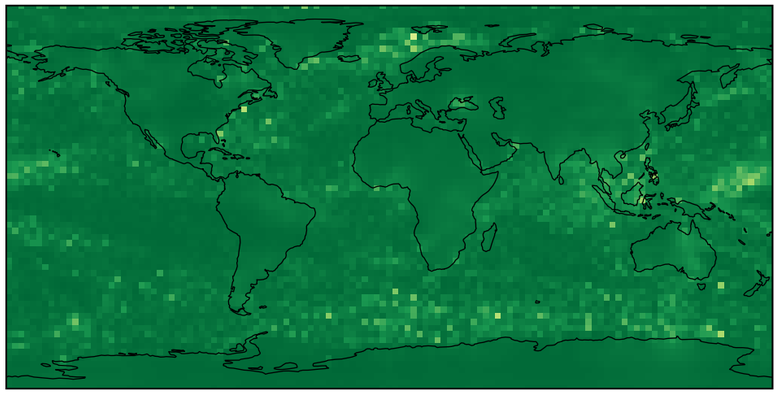


Step 90

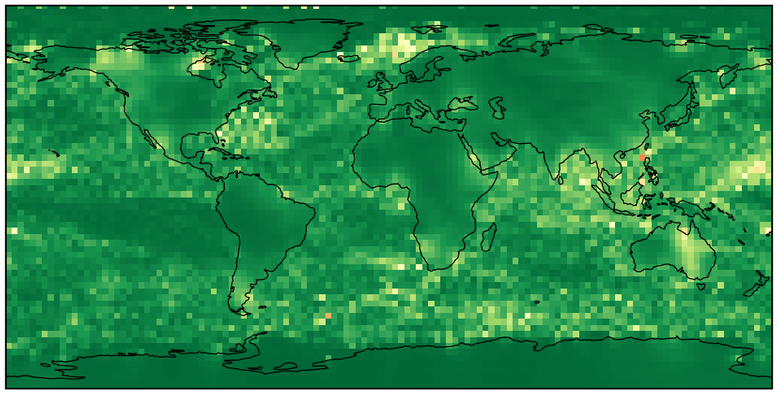


Spatial ACC (all grid) — Test

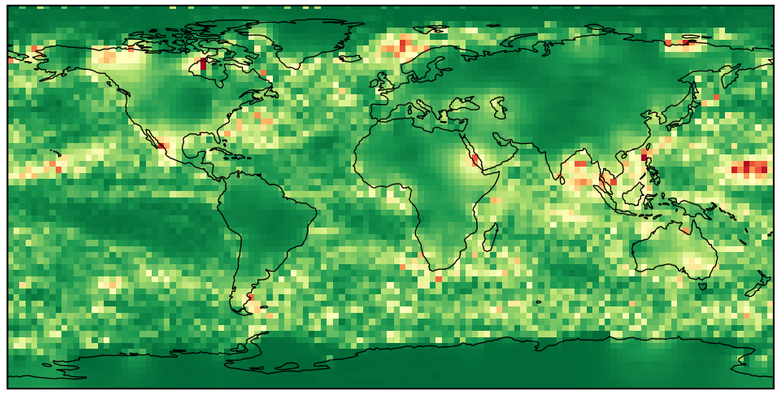
Step 1



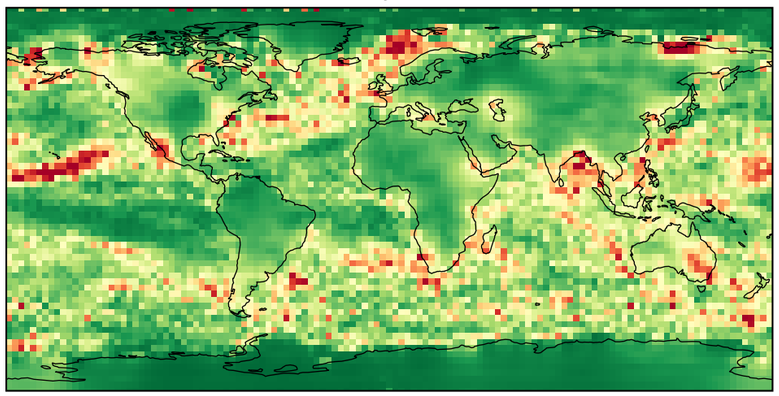
Step 3



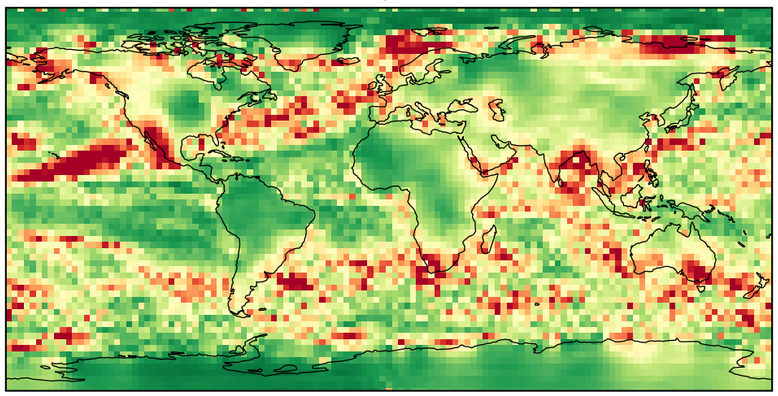
Step 7



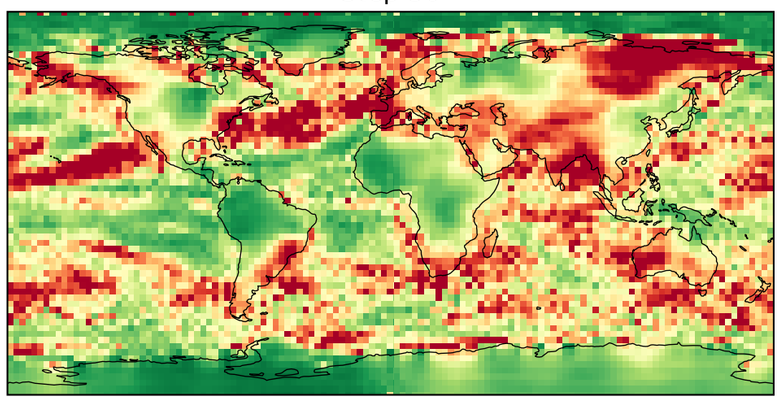
Step 14



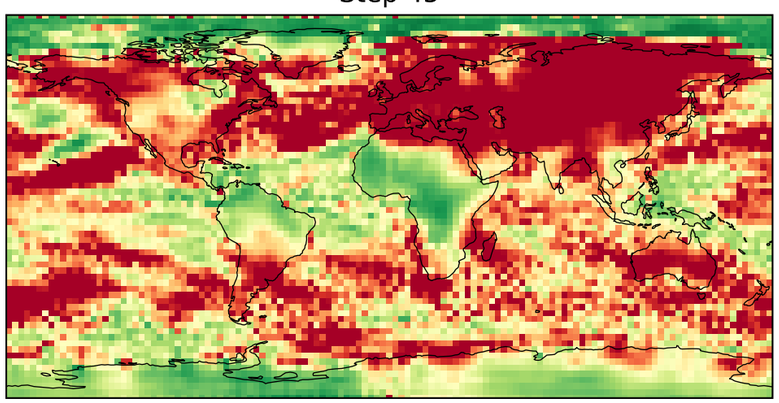
Step 21



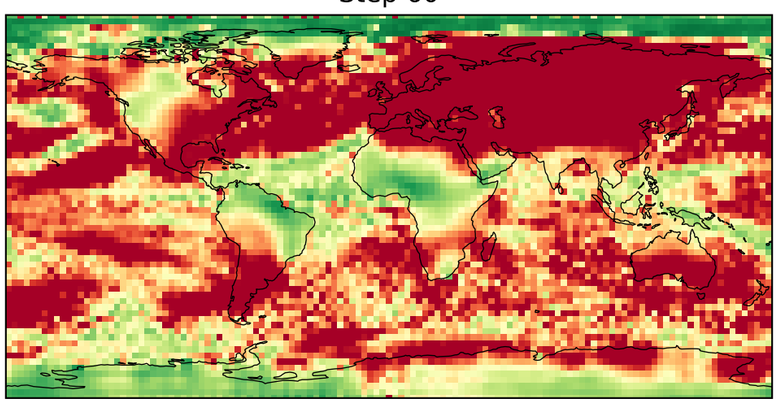
Step 30



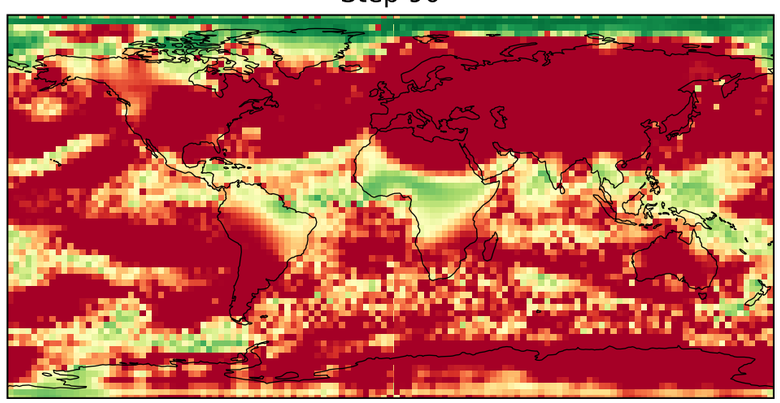
Step 45



Step 60

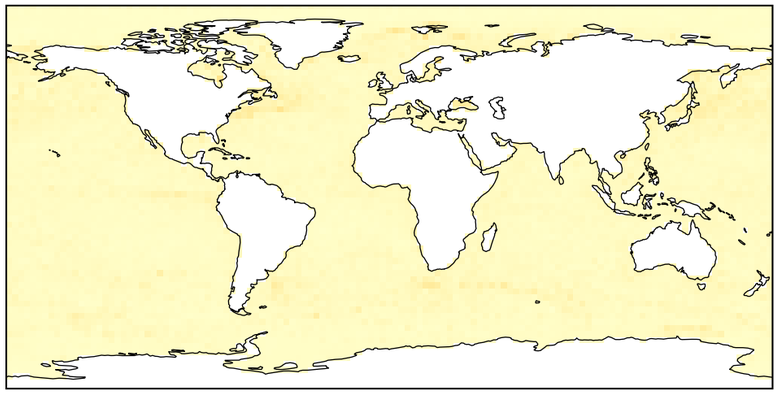


Step 90

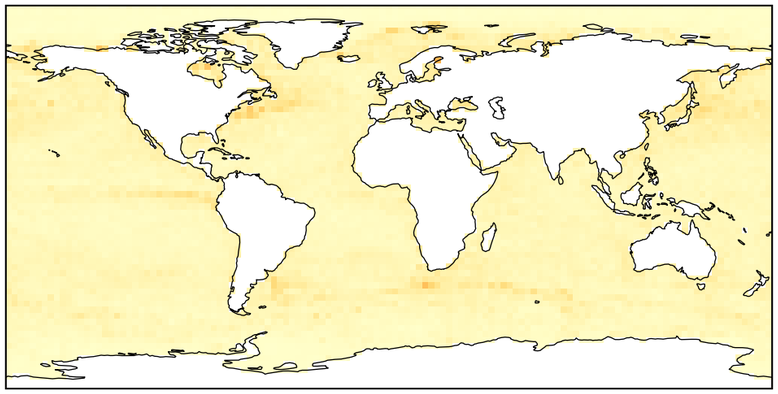


Spatial RMSE (ocean only) — Test

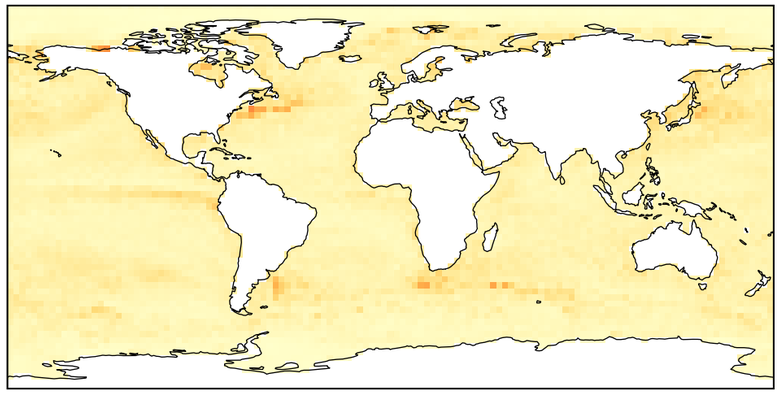
Step 1



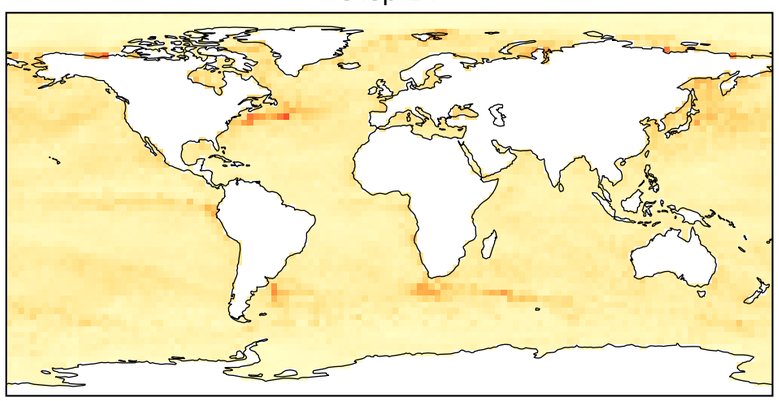
Step 3



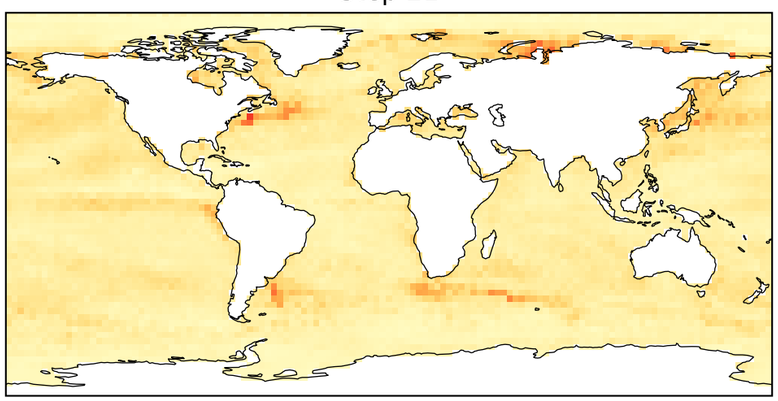
Step 7



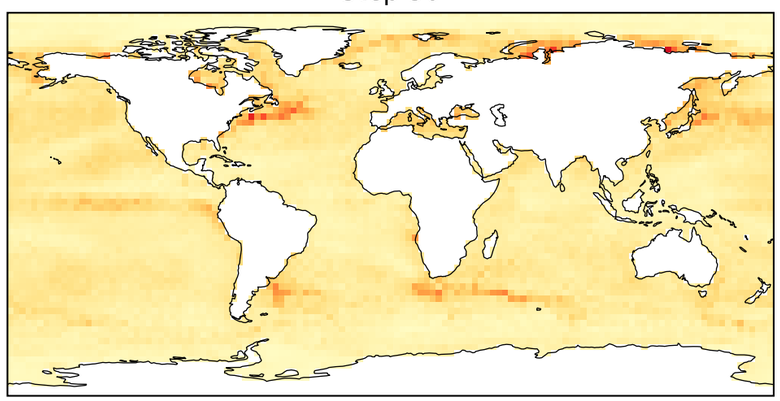
Step 14



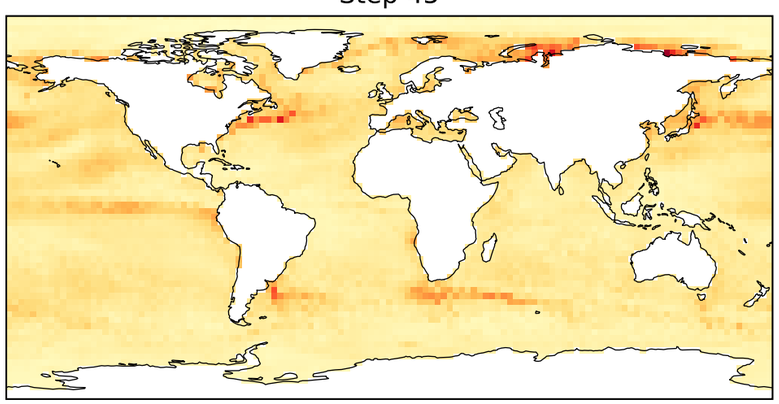
Step 21



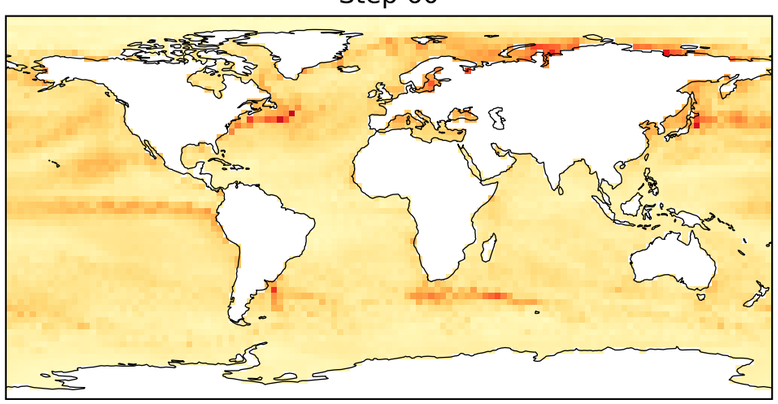
Step 30



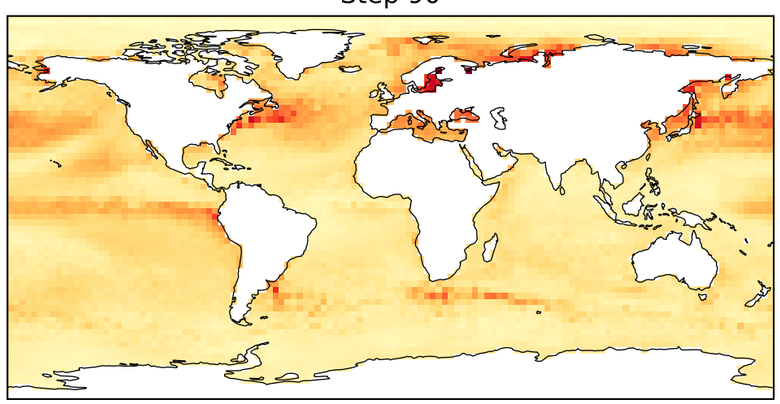
Step 45

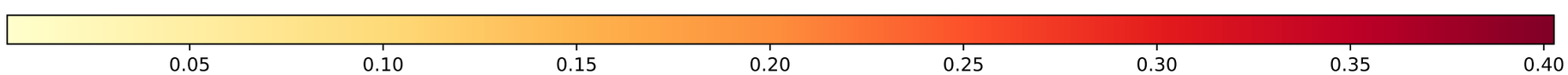


Step 60



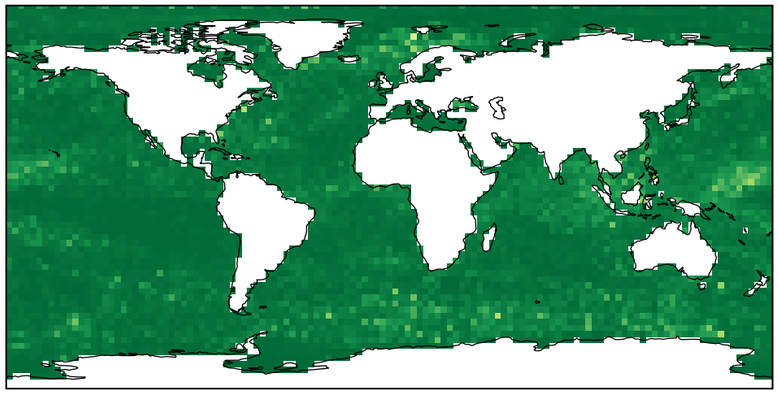
Step 90



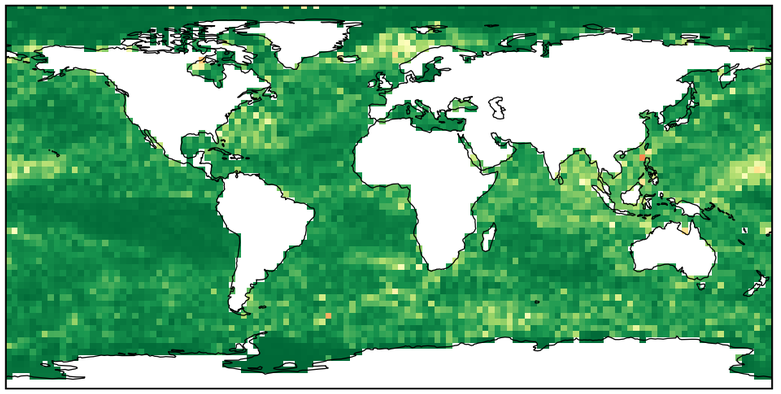


Spatial ACC (ocean only) — Test

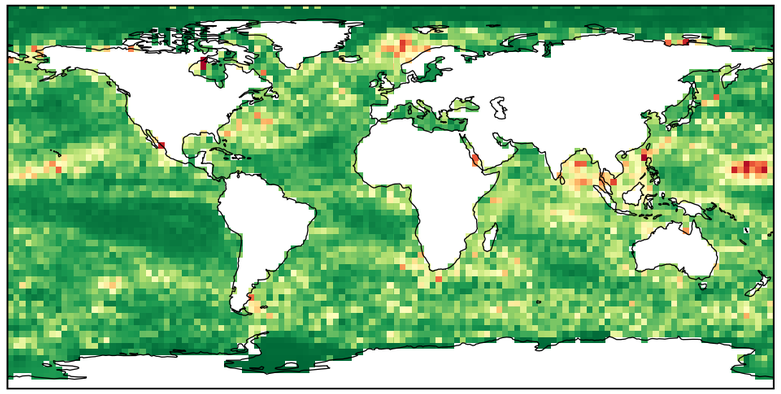
Step 1



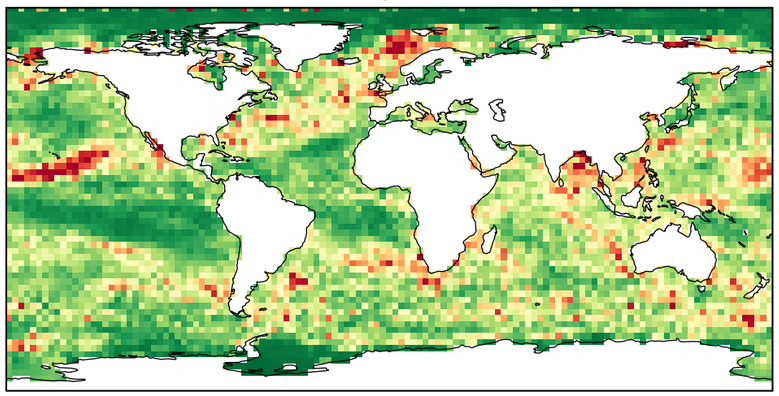
Step 3



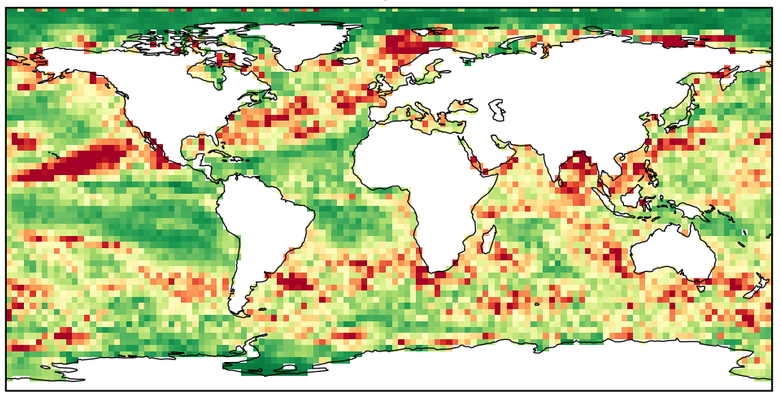
Step 7



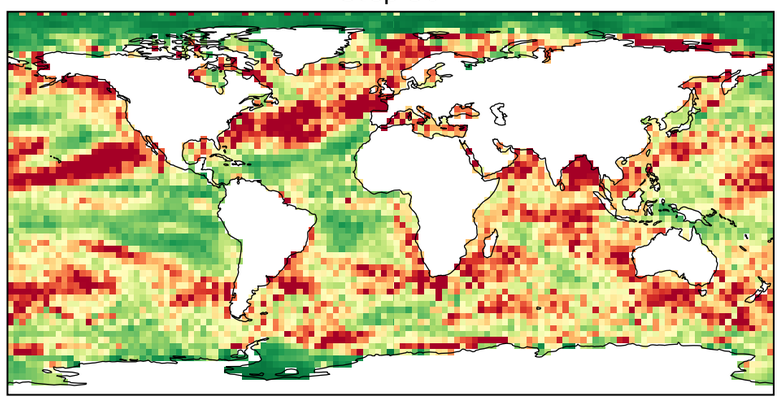
Step 14



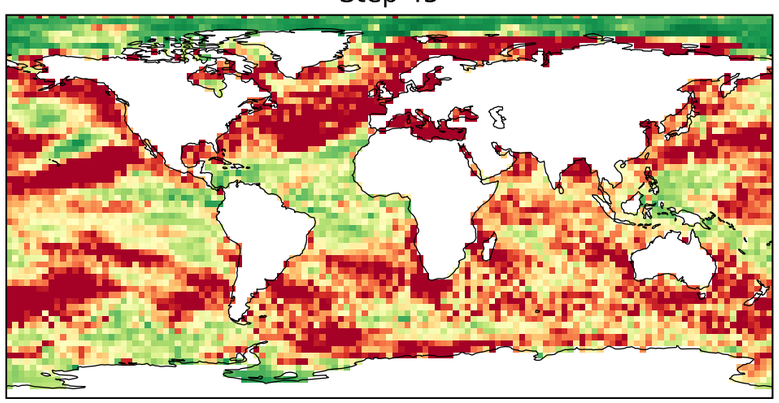
Step 21



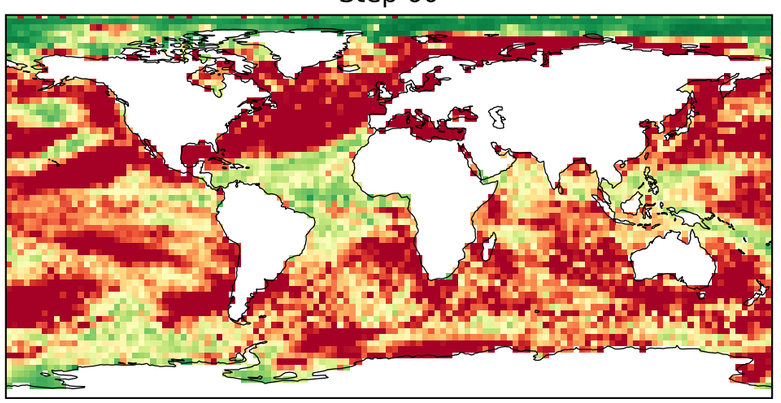
Step 30



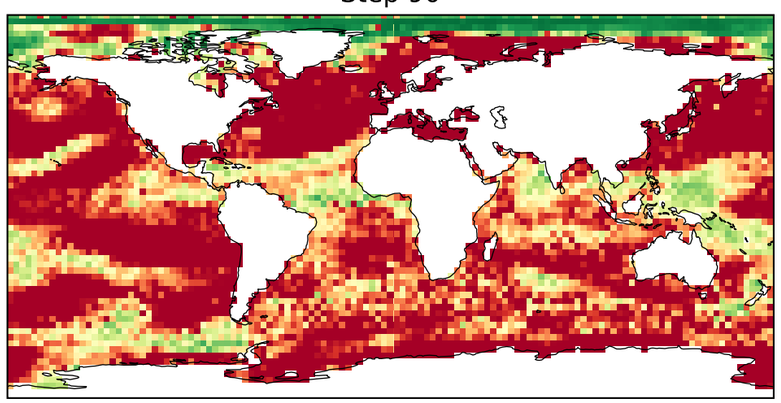
Step 45



Step 60

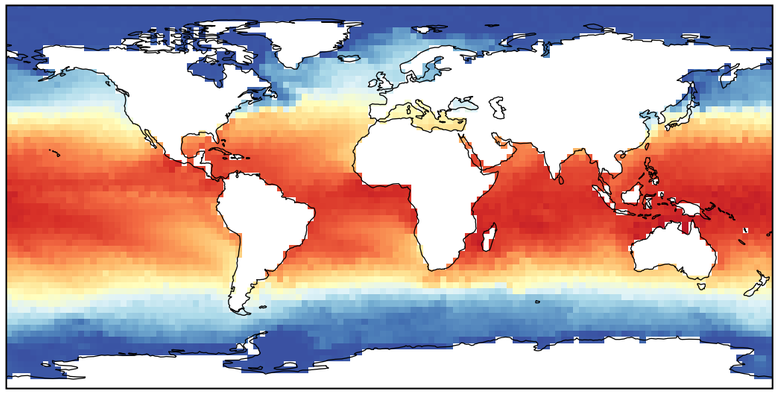


Step 90

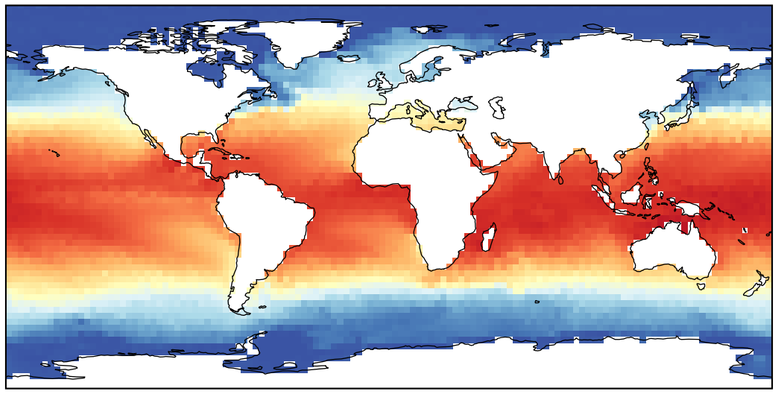


Chain init: 2020-01-11 — Test (°C)

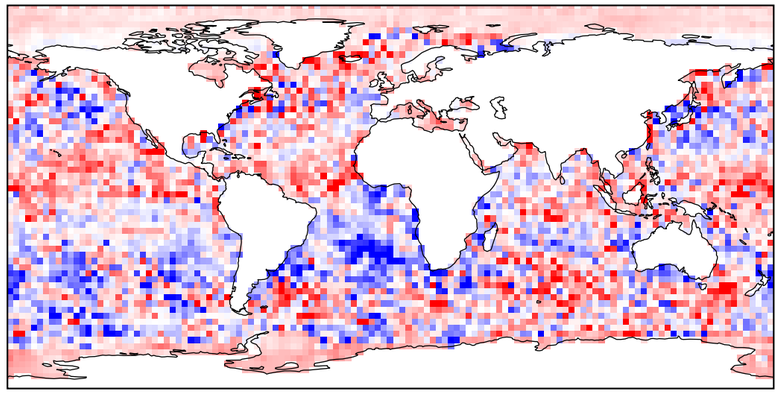
Actual — Step 1 (°C)



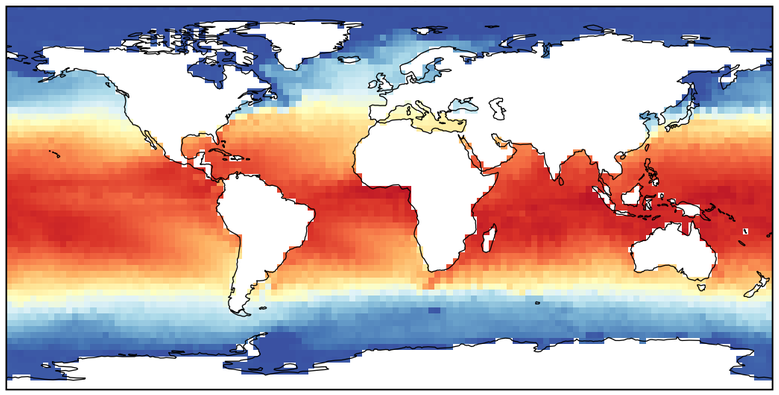
SFNO — Step 1 (°C)



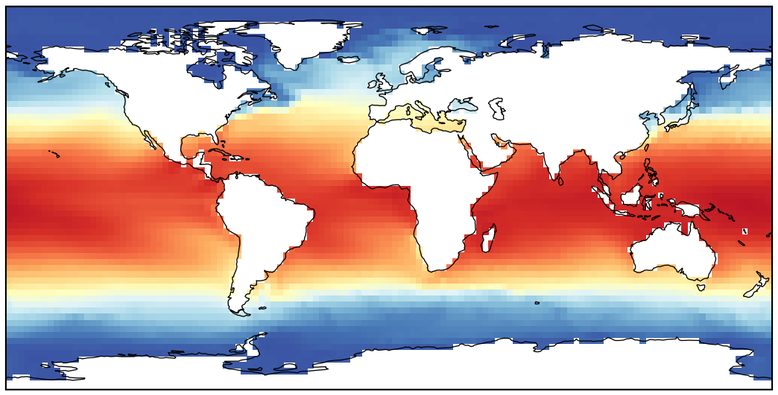
Error — Step 1 (°C)



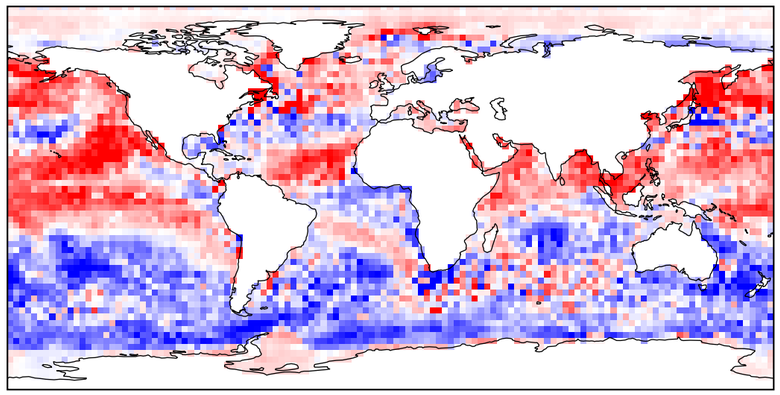
Actual — Step 46 (°C)



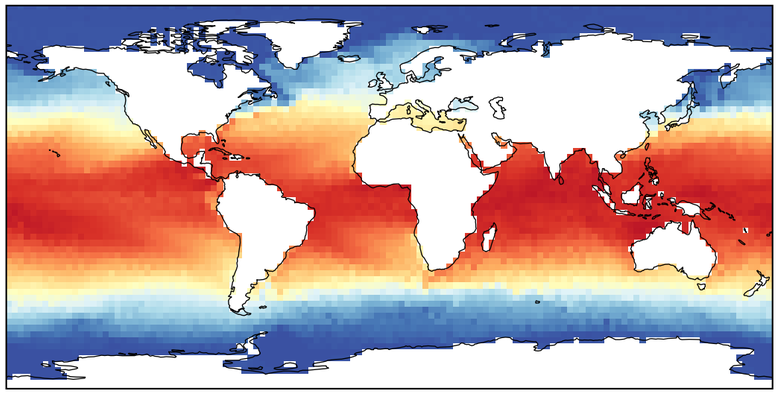
SFNO — Step 46 (°C)



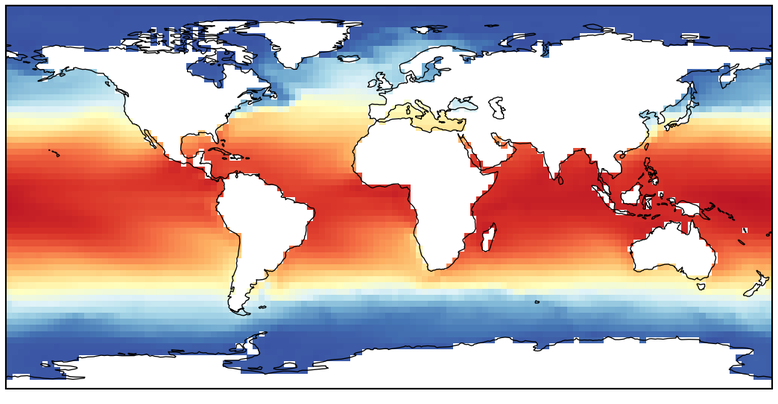
Error — Step 46 (°C)



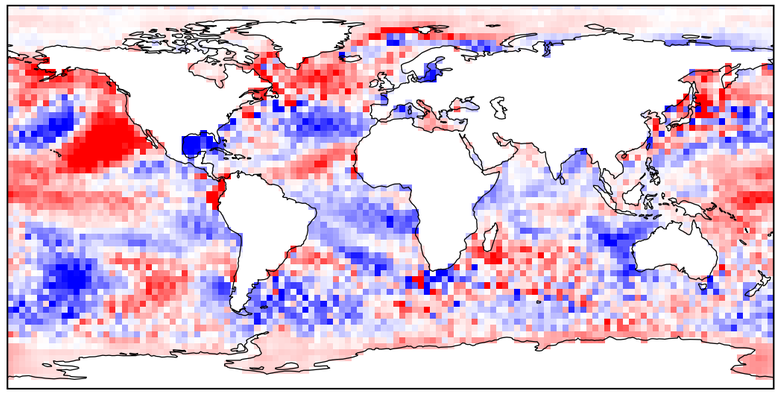
Actual — Step 90 (°C)

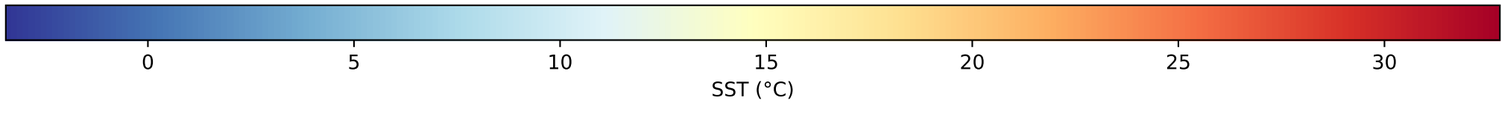


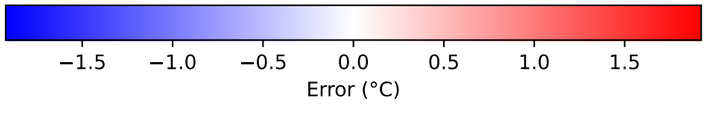
SFNO — Step 90 (°C)



Error — Step 90 (°C)

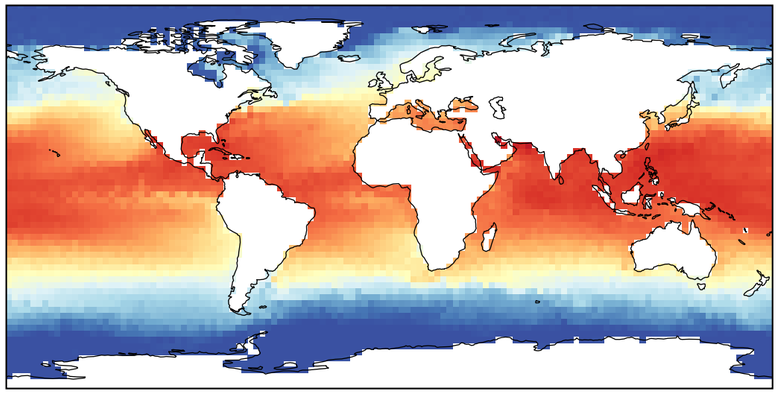




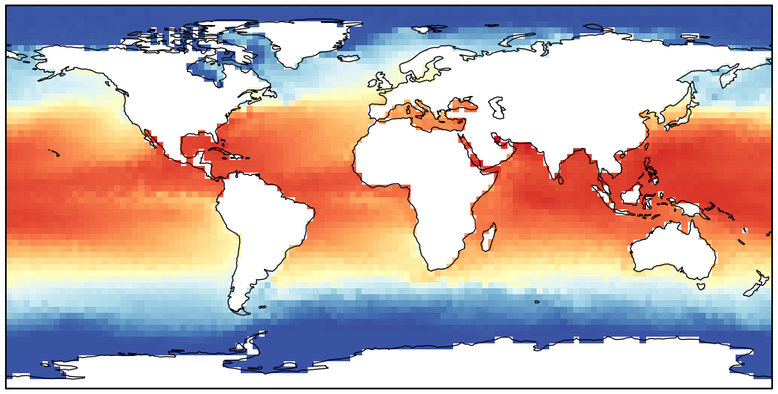


Chain init: 2020-07-09 — Test (°C)

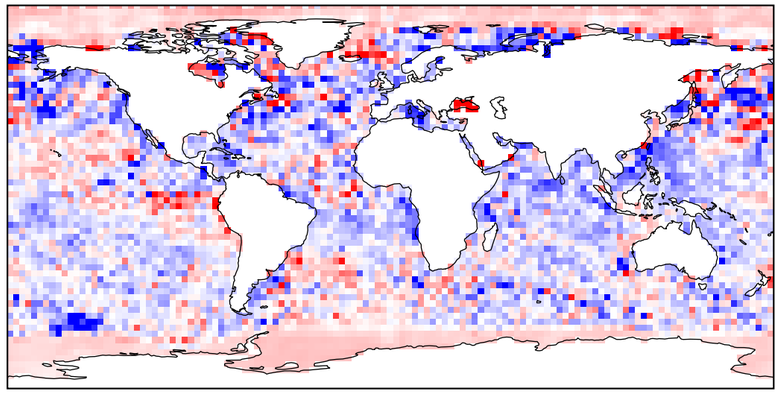
Actual — Step 1 (°C)



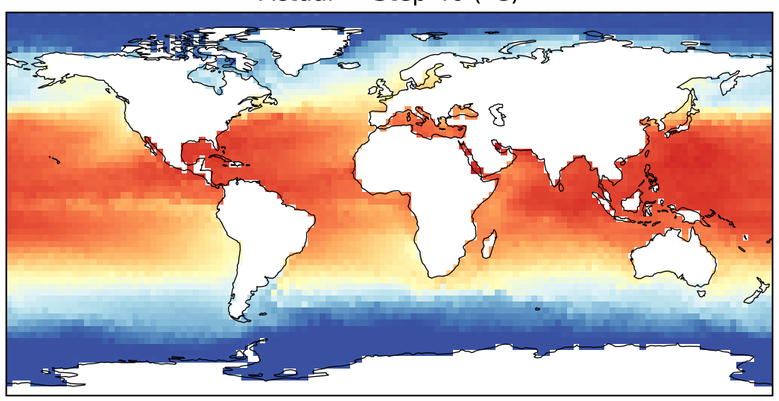
SFNO — Step 1 (°C)



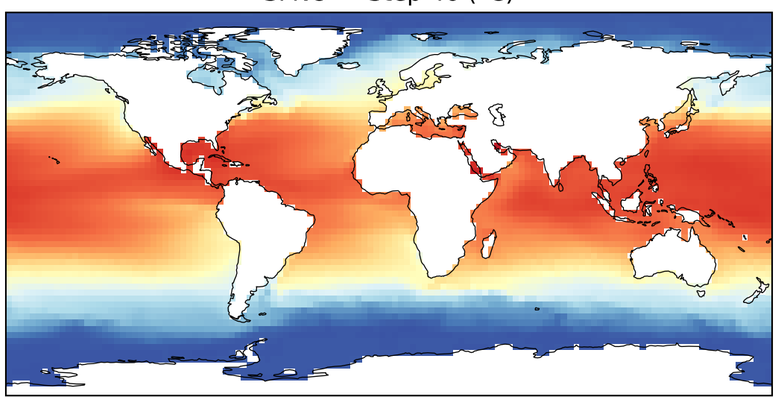
Error — Step 1 (°C)



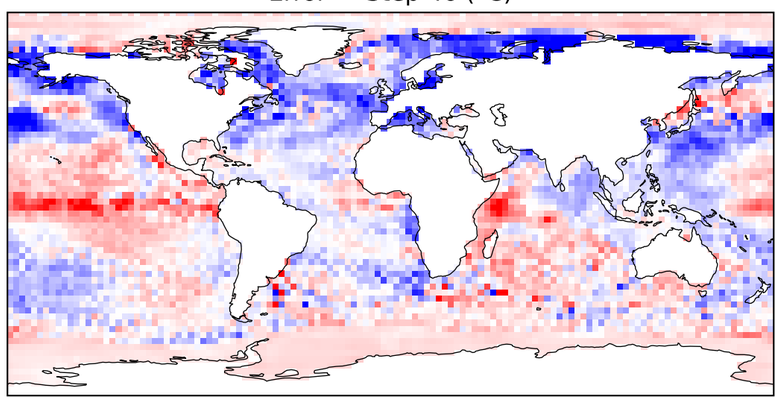
Actual — Step 46 (°C)



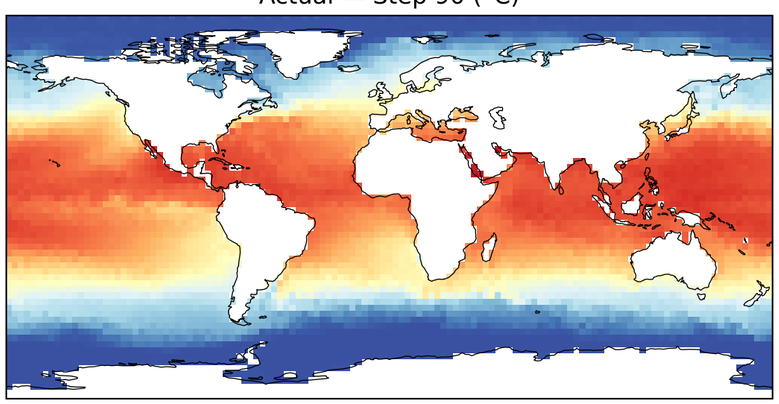
SFNO — Step 46 (°C)



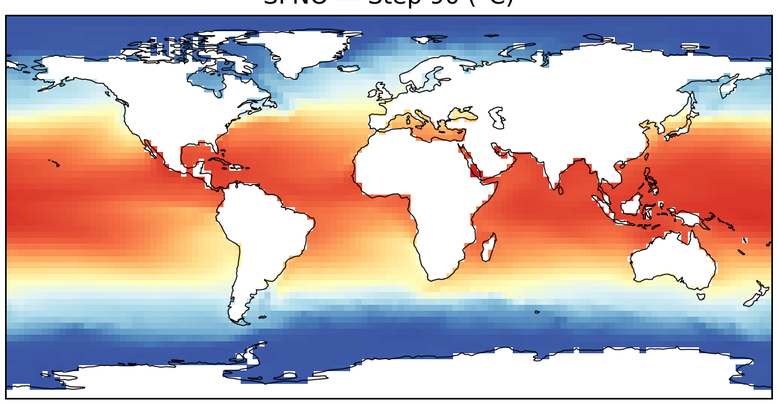
Error — Step 46 (°C)



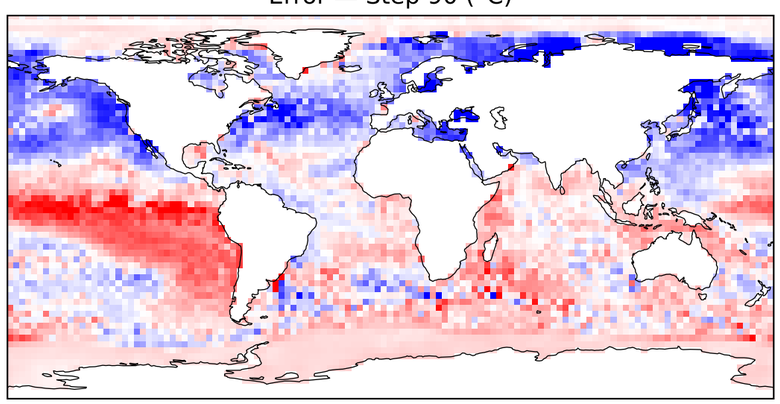
Actual — Step 90 (°C)

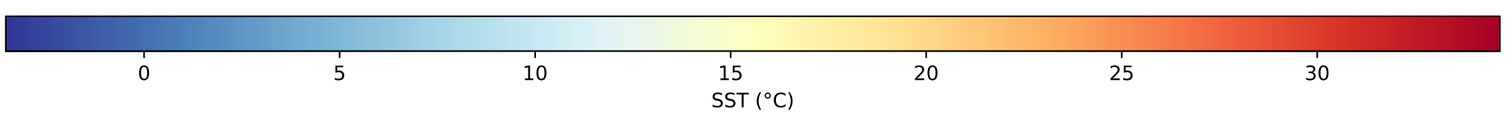


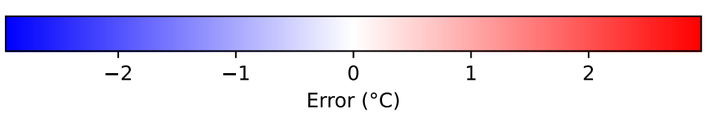
SFNO — Step 90 (°C)



Error — Step 90 (°C)

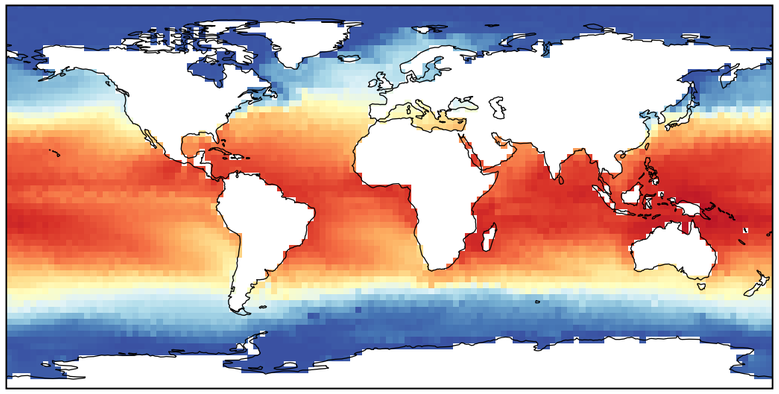




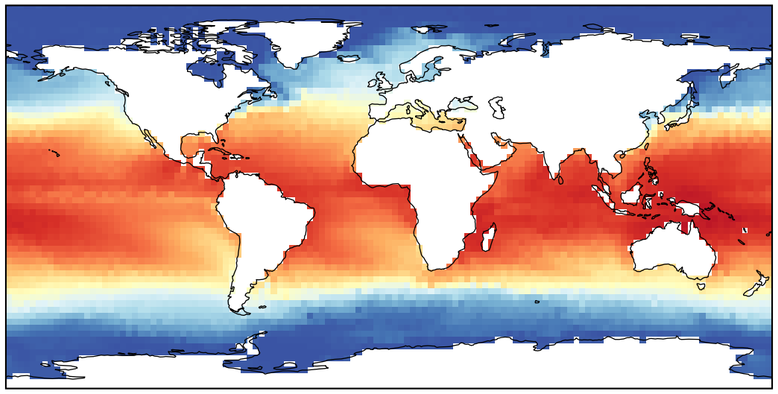


Chain init: 2021-01-06 — Test (°C)

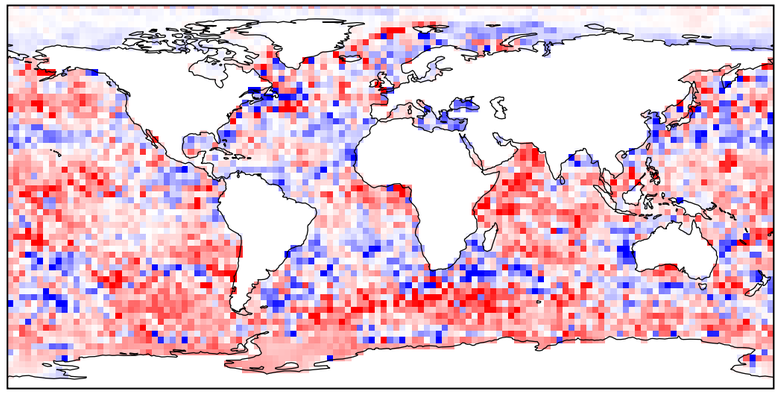
Actual — Step 1 (°C)



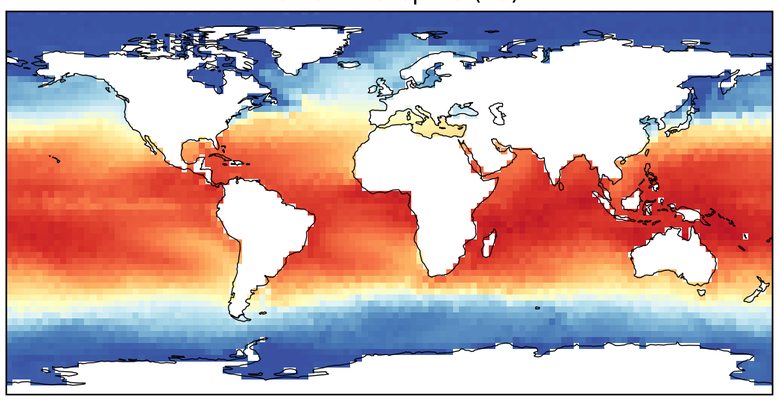
SFNO — Step 1 (°C)



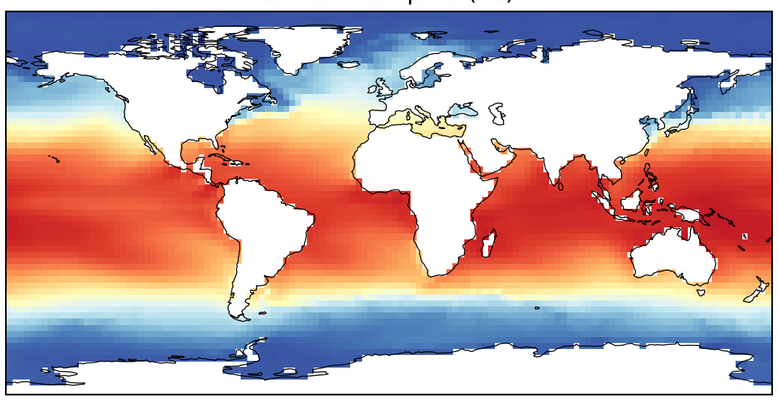
Error — Step 1 (°C)



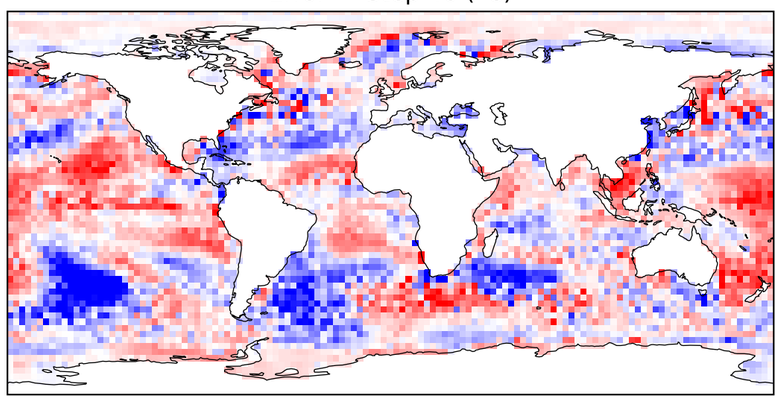
Actual — Step 46 (°C)



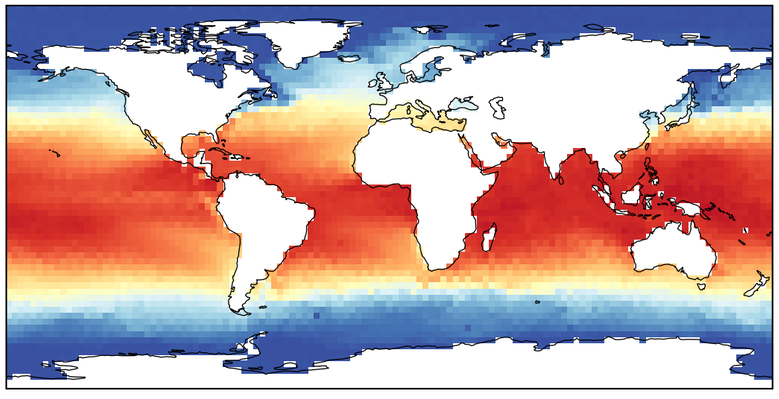
SFNO — Step 46 (°C)



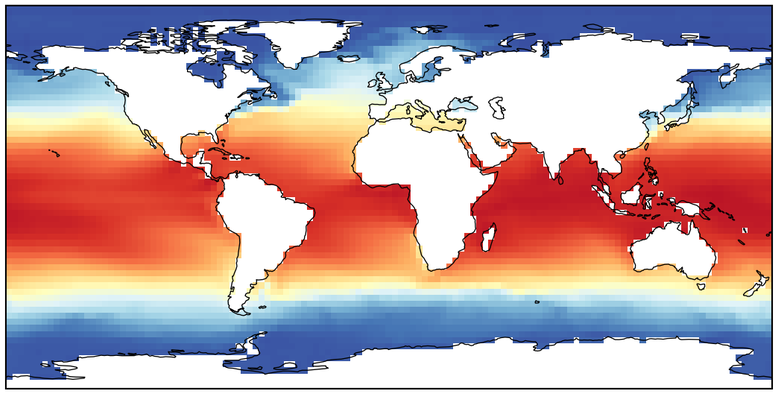
Error — Step 46 (°C)



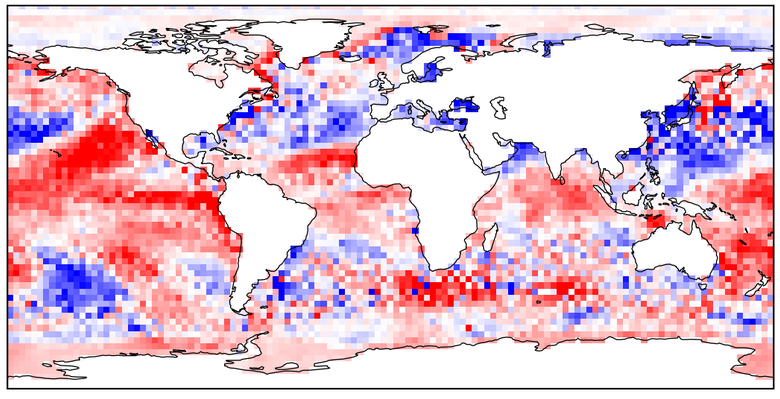
Actual — Step 90 (°C)

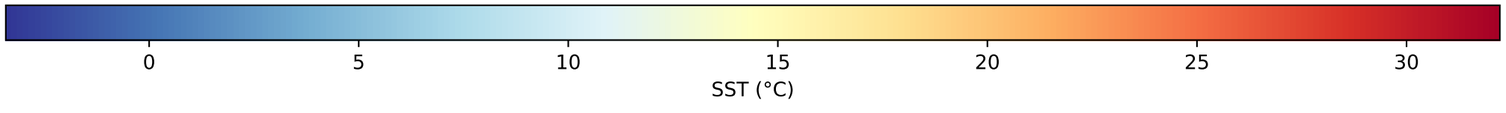


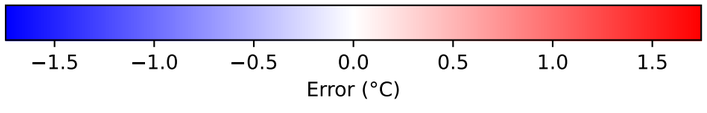
SFNO — Step 90 (°C)



Error — Step 90 (°C)

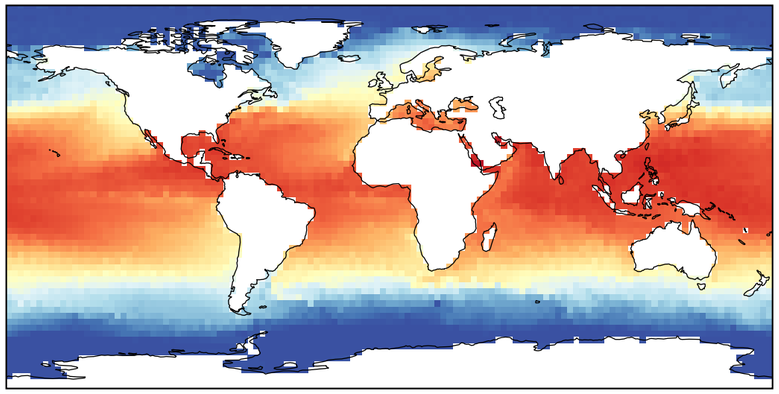




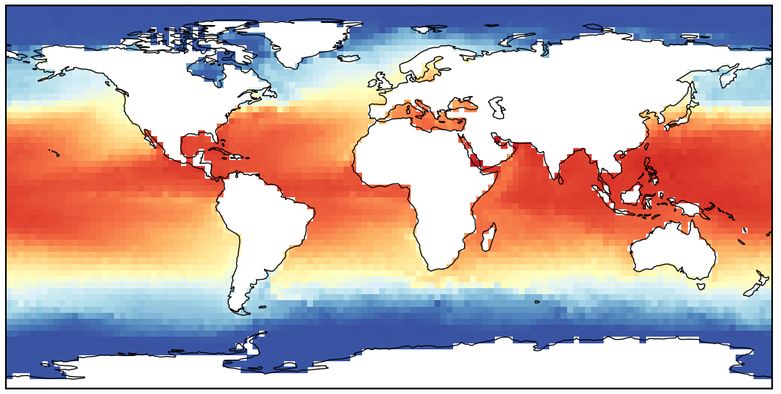


Chain init: 2021-07-05 — Test (°C)

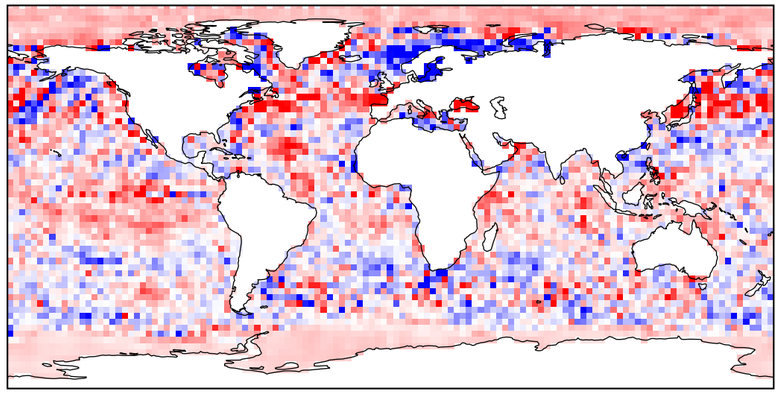
Actual — Step 1 (°C)



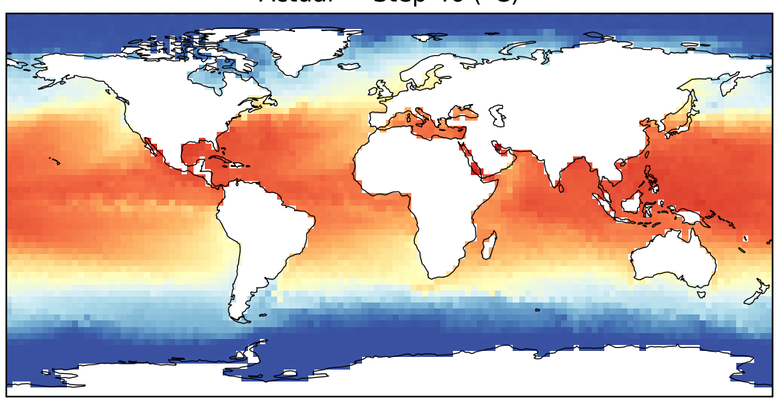
SFNO — Step 1 (°C)



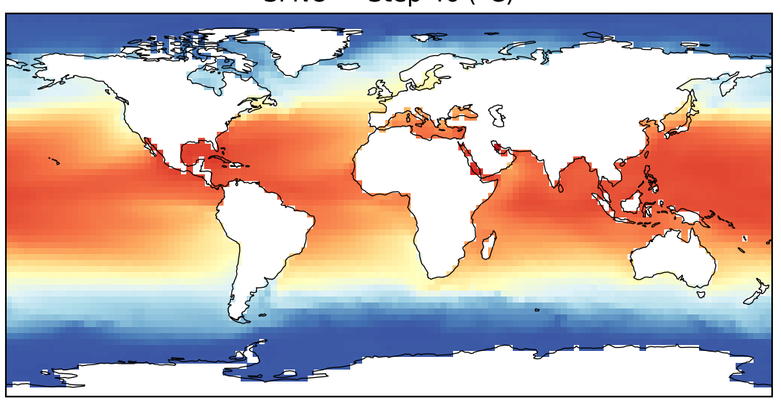
Error — Step 1 (°C)



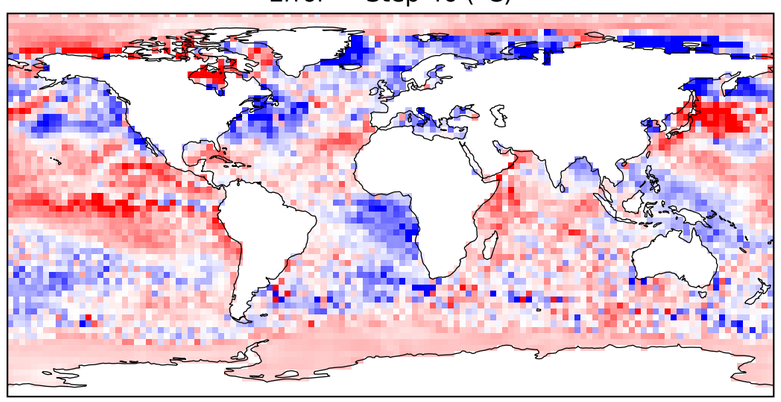
Actual — Step 46 (°C)



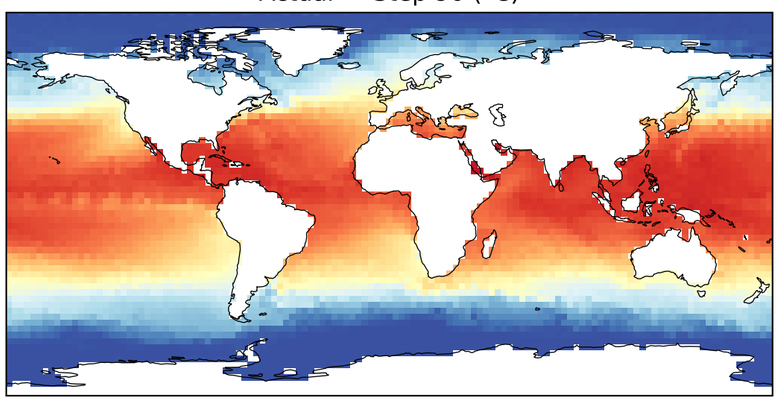
SFNO — Step 46 (°C)



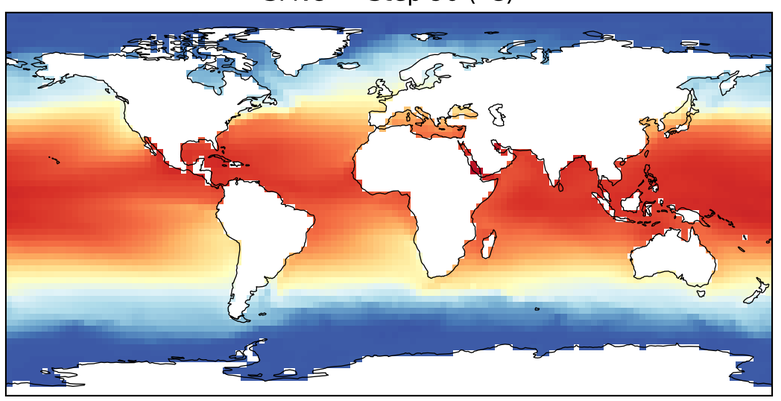
Error — Step 46 (°C)



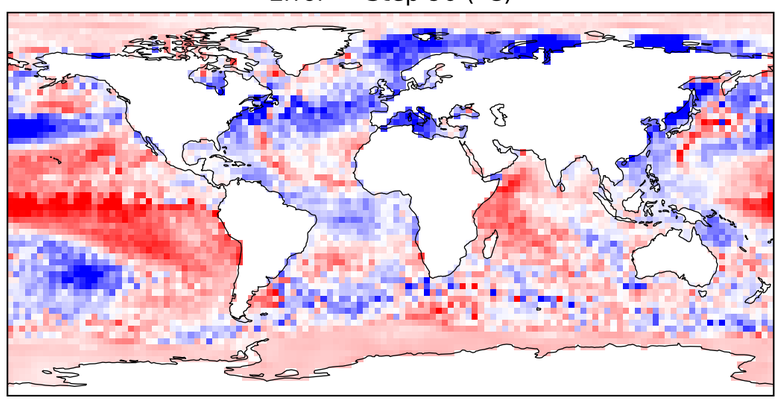
Actual — Step 90 (°C)

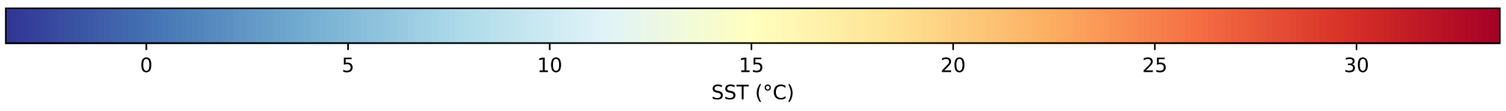


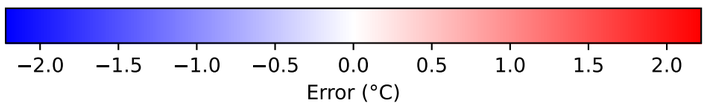
SFNO — Step 90 (°C)



Error — Step 90 (°C)

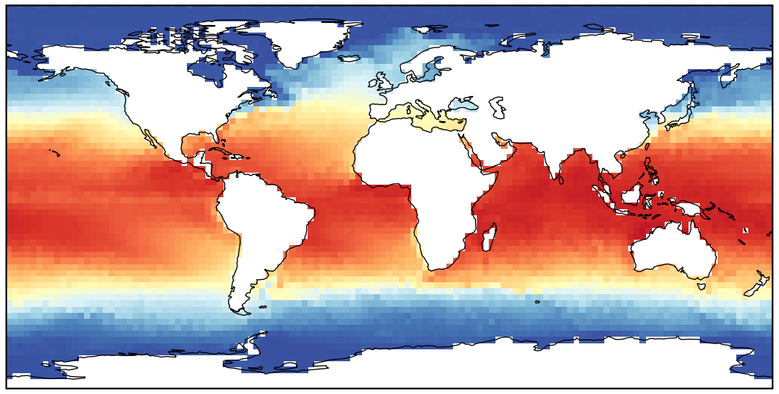




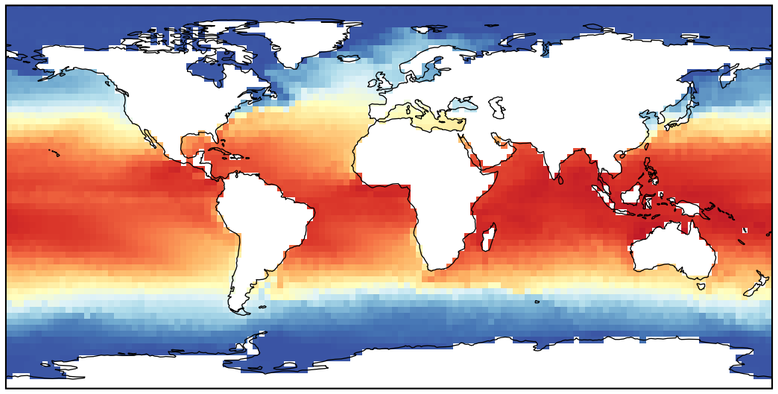


Chain init: 2022-04-01 — Test (°C)

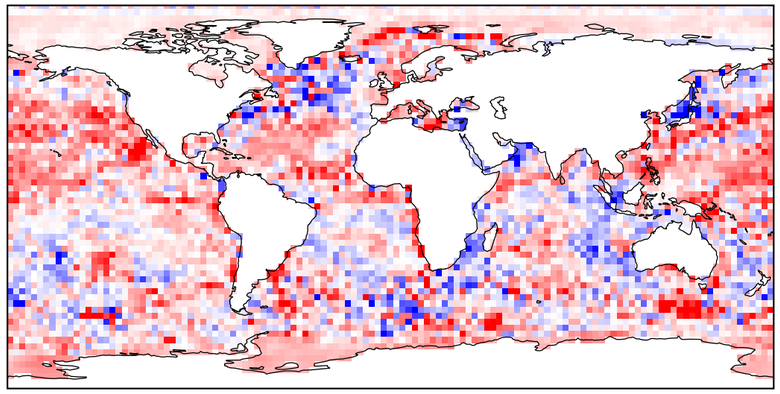
Actual — Step 1 (°C)



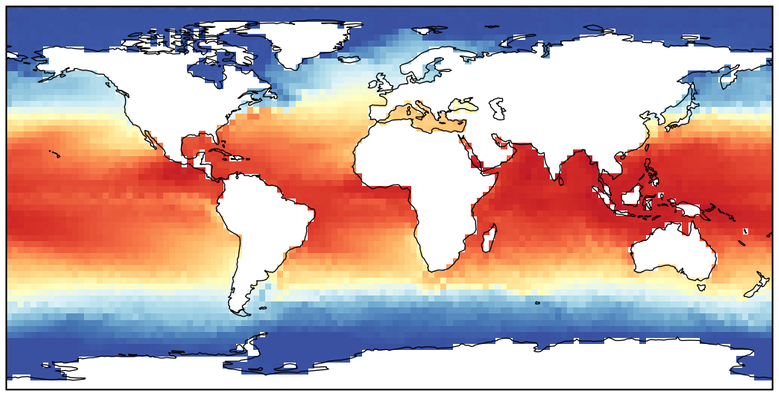
SFNO — Step 1 (°C)



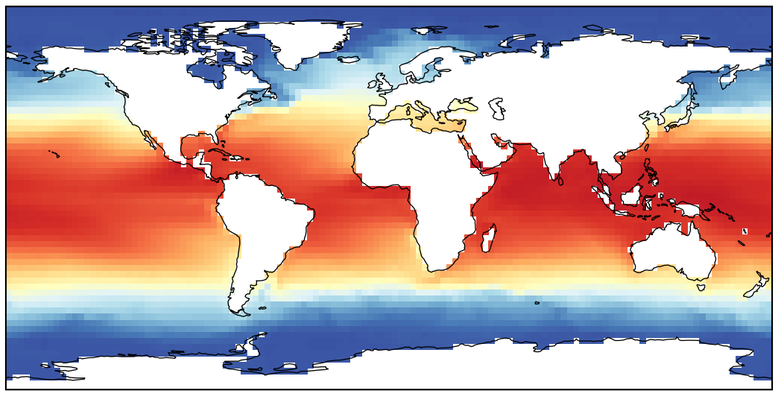
Error — Step 1 (°C)



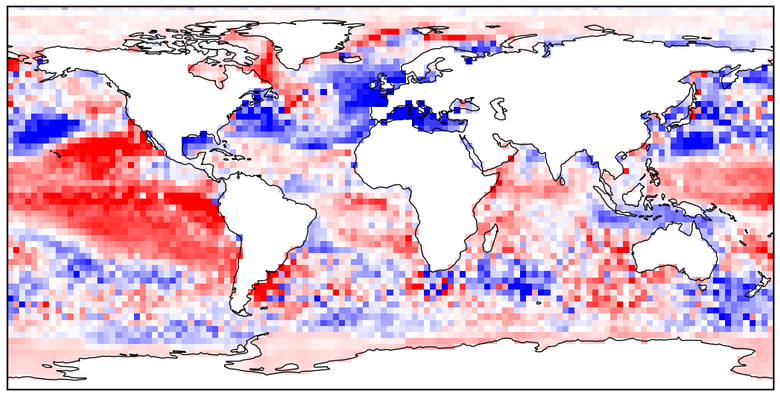
Actual — Step 46 (°C)



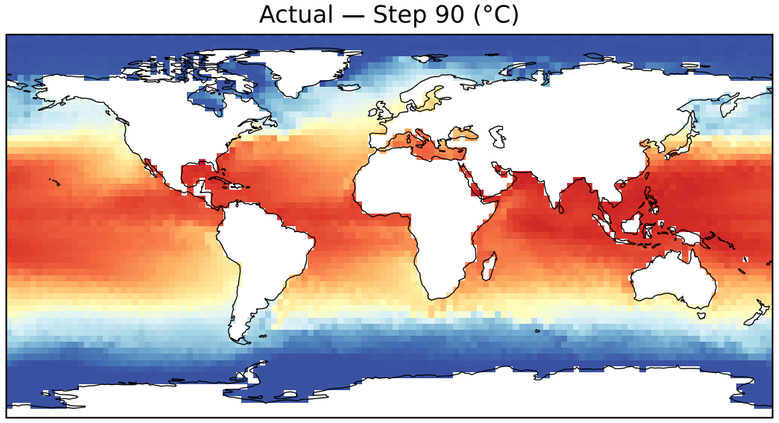
SFNO — Step 46 (°C)



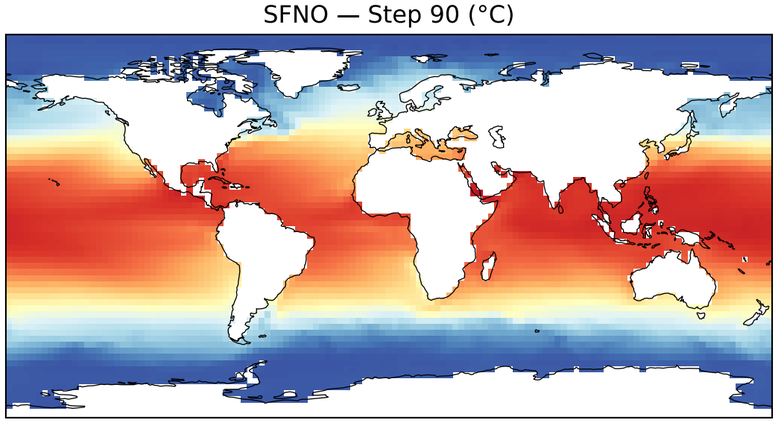
Error — Step 46 (°C)



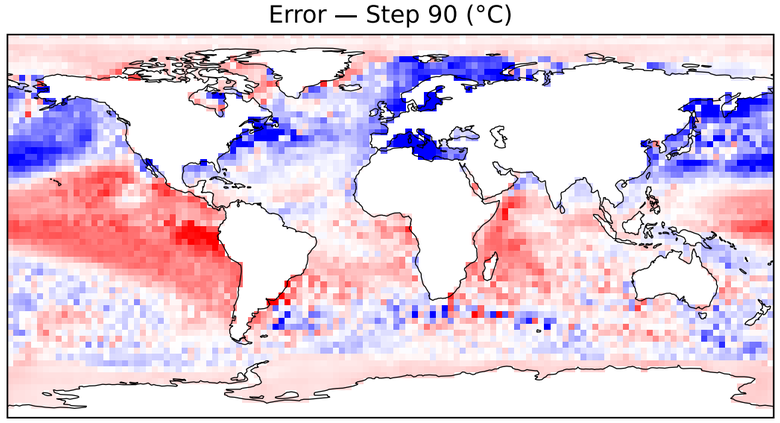
Actual — Step 90 (°C)

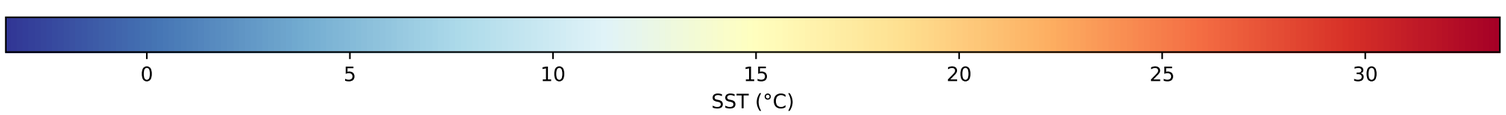


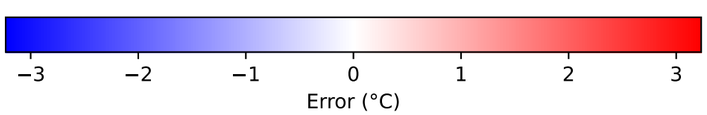
SFNO — Step 90 (°C)



Error — Step 90 (°C)

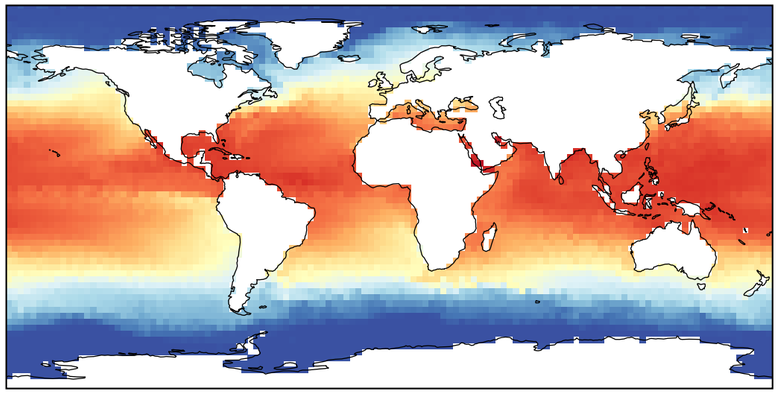




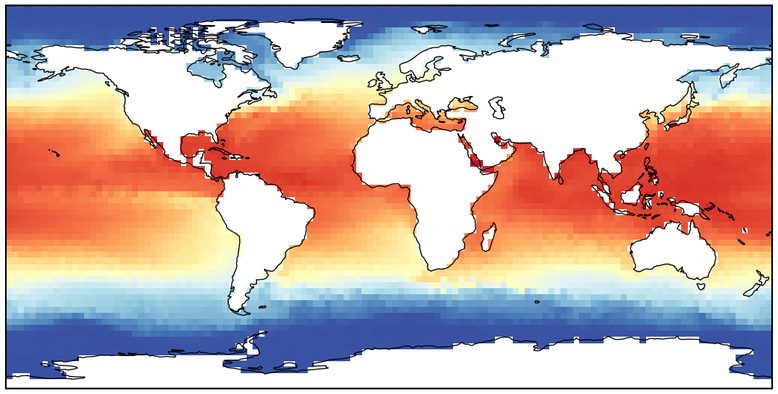


Chain init: 2022-09-28 — Test (°C)

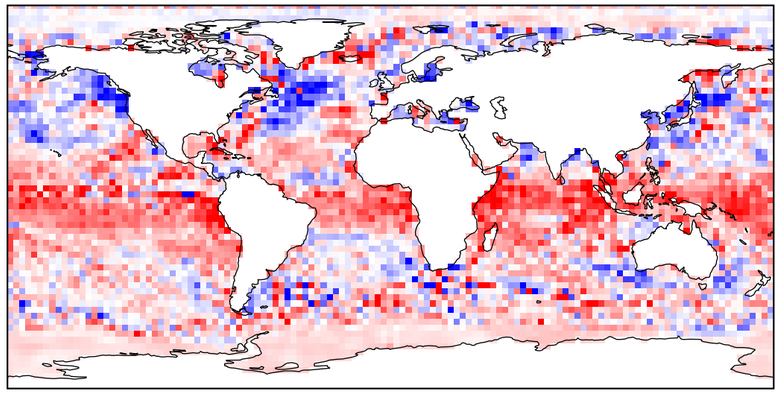
Actual — Step 1 (°C)



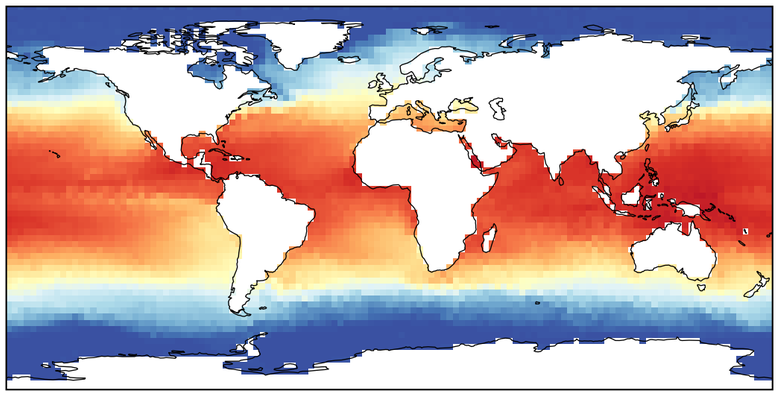
SFNO — Step 1 (°C)



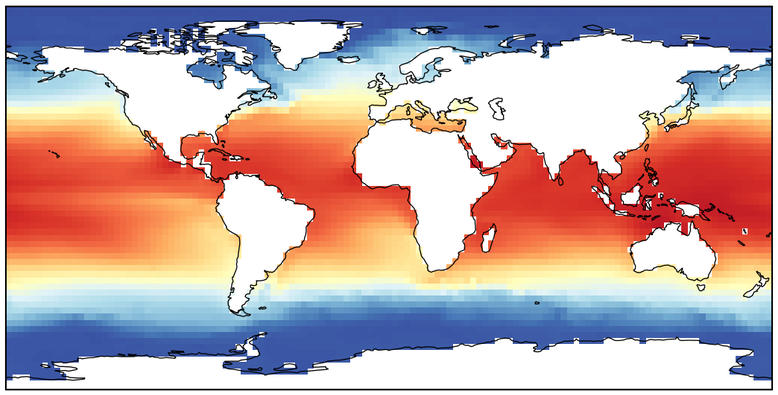
Error — Step 1 (°C)



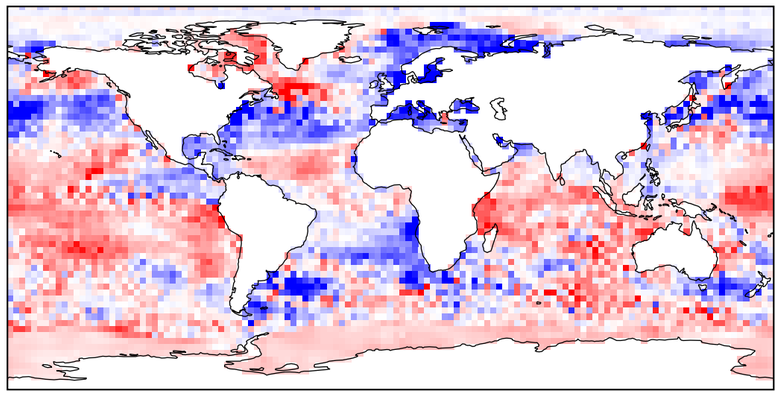
Actual — Step 46 (°C)



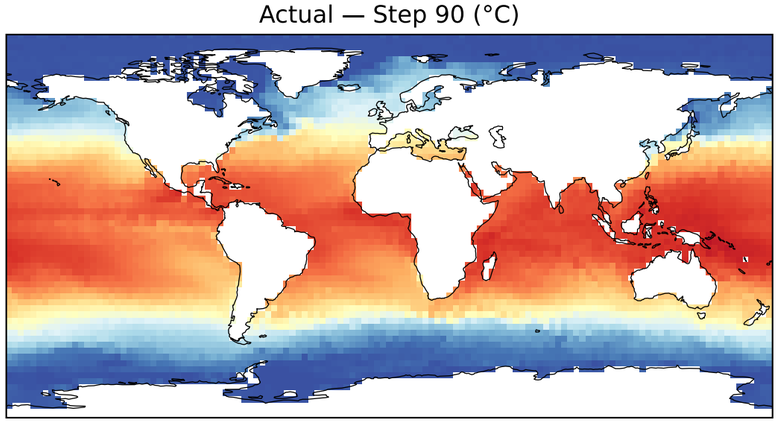
SFNO — Step 46 (°C)



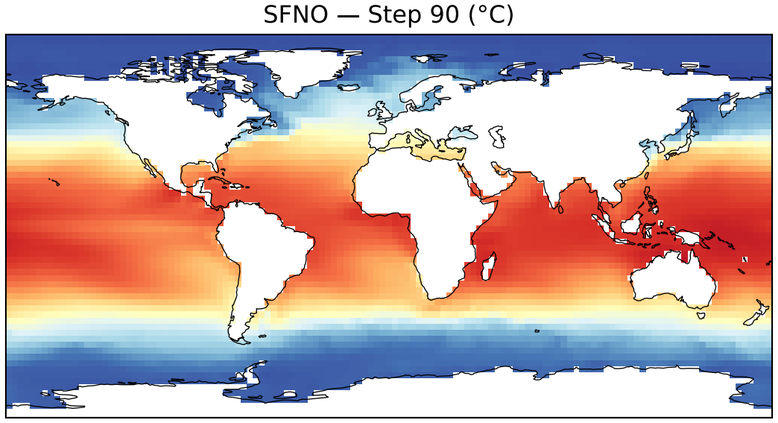
Error — Step 46 (°C)



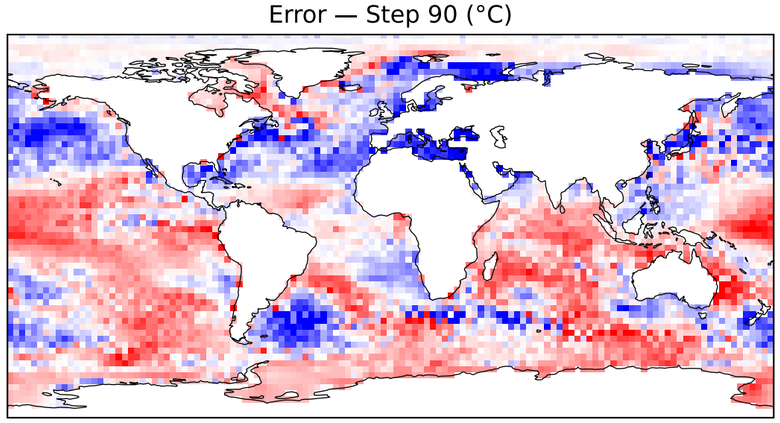
Actual — Step 90 (°C)

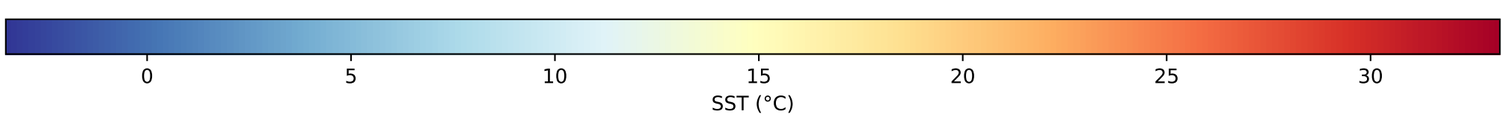


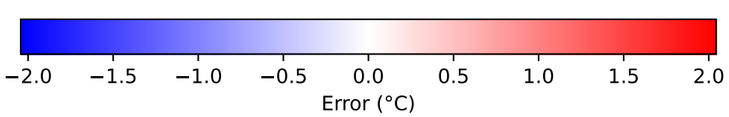
SFNO — Step 90 (°C)



Error — Step 90 (°C)

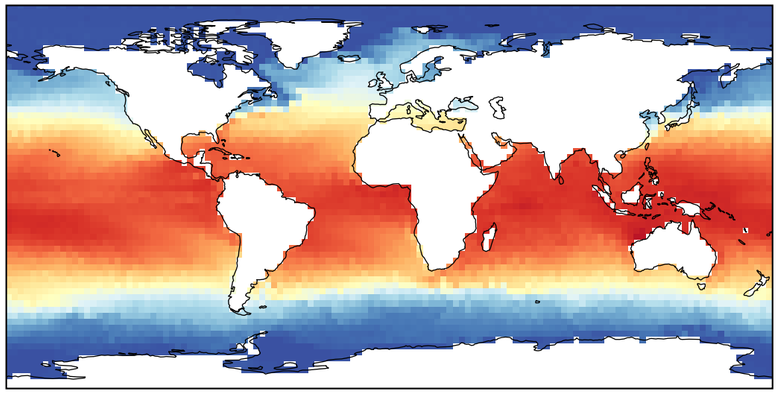




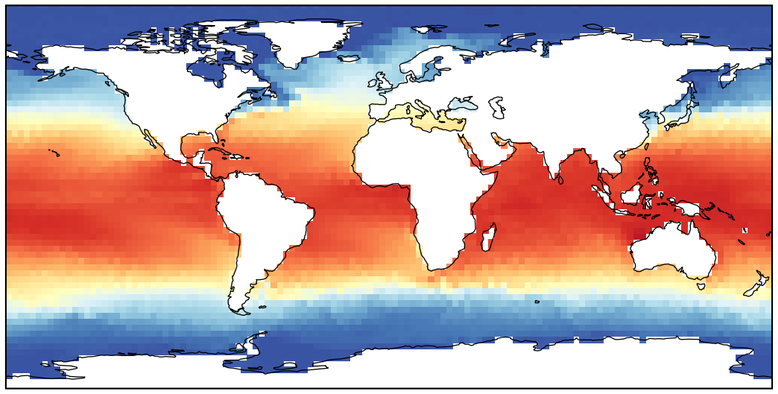


Chain init: 2023-03-27 — Test (°C)

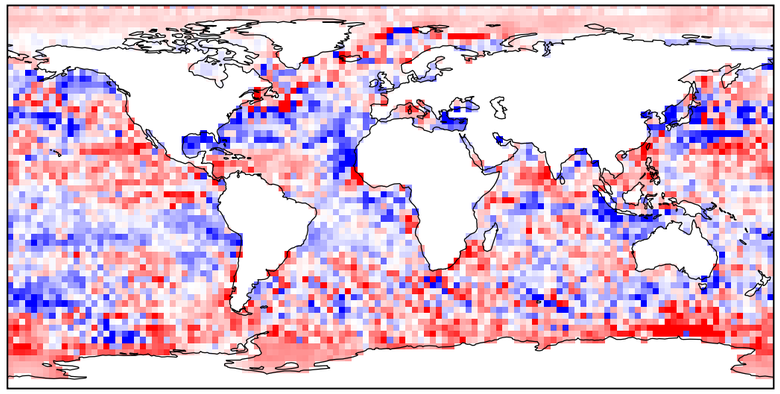
Actual — Step 1 (°C)



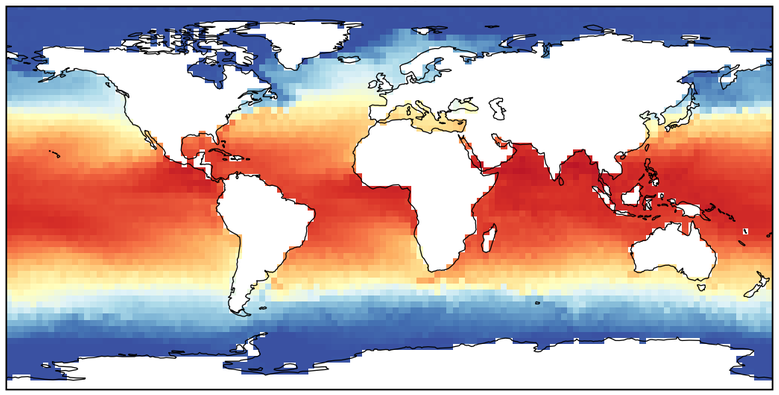
SFNO — Step 1 (°C)



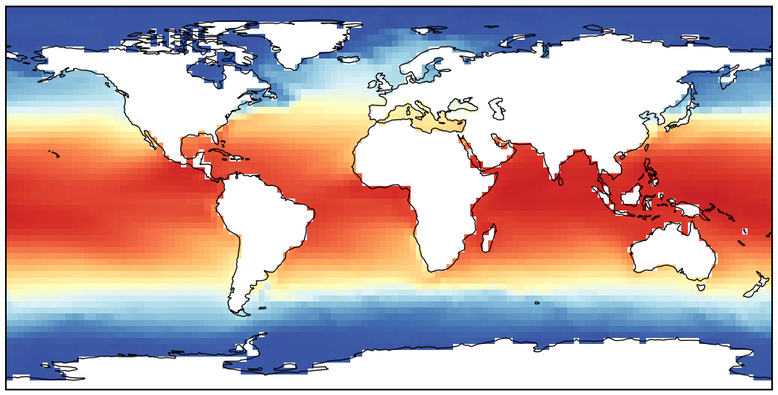
Error — Step 1 (°C)



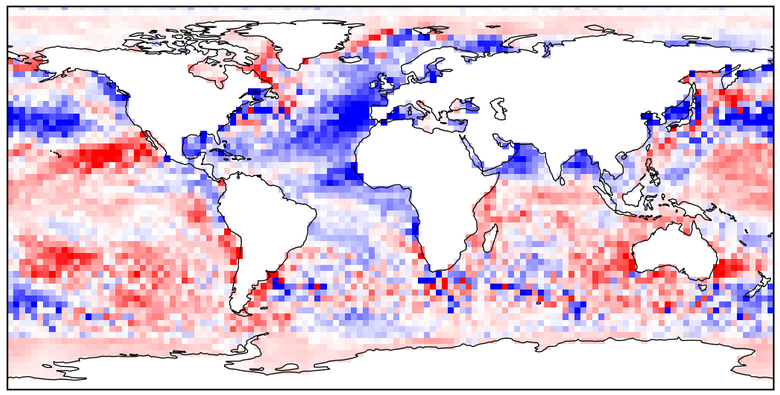
Actual — Step 46 (°C)



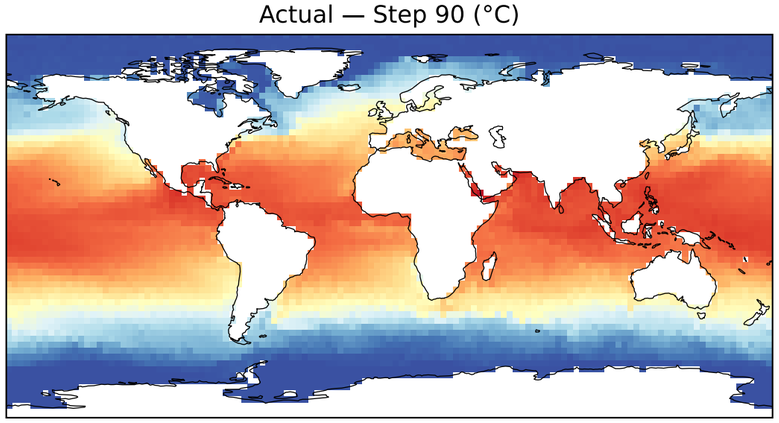
SFNO — Step 46 (°C)



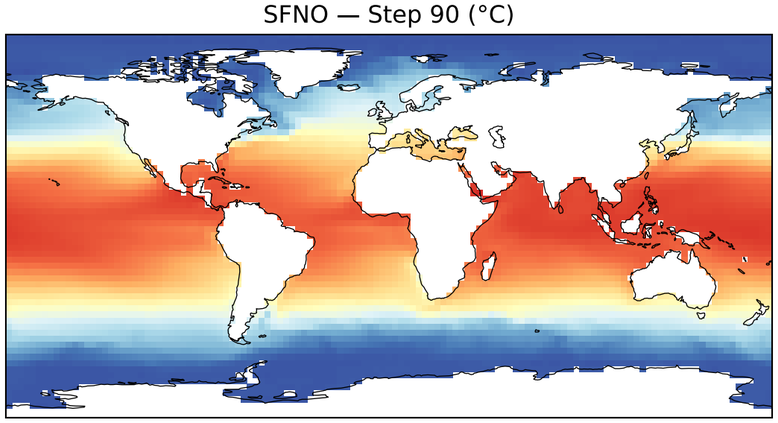
Error — Step 46 (°C)



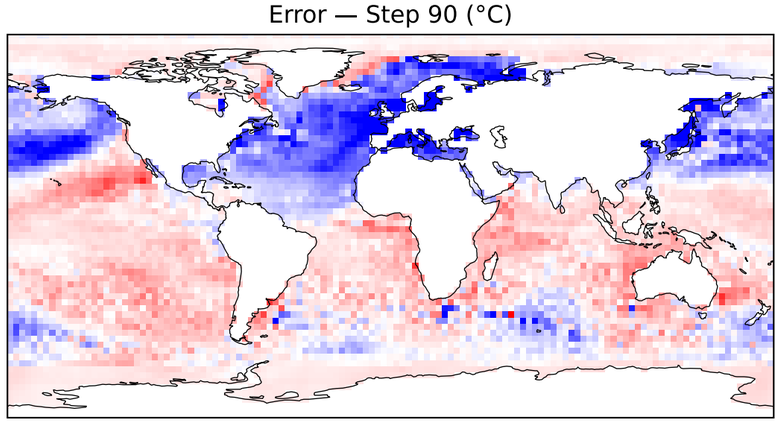
Actual — Step 90 (°C)

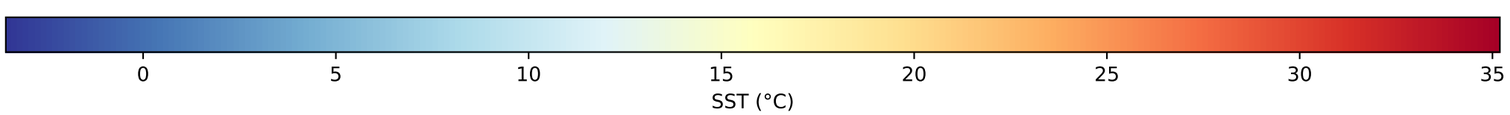


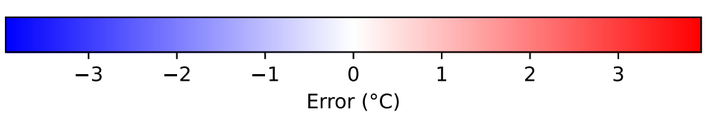
SFNO — Step 90 (°C)



Error — Step 90 (°C)

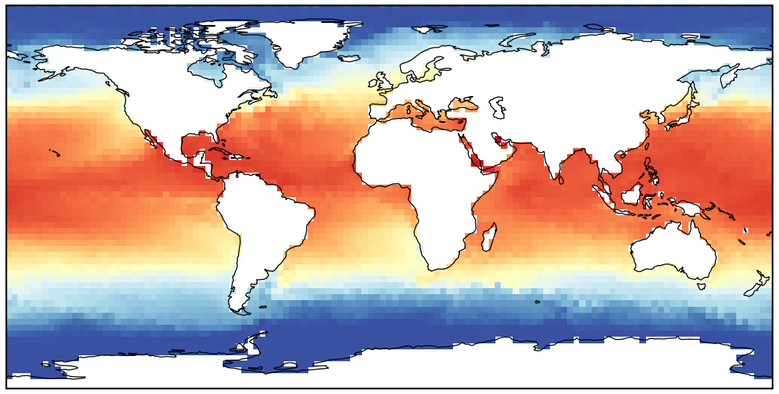




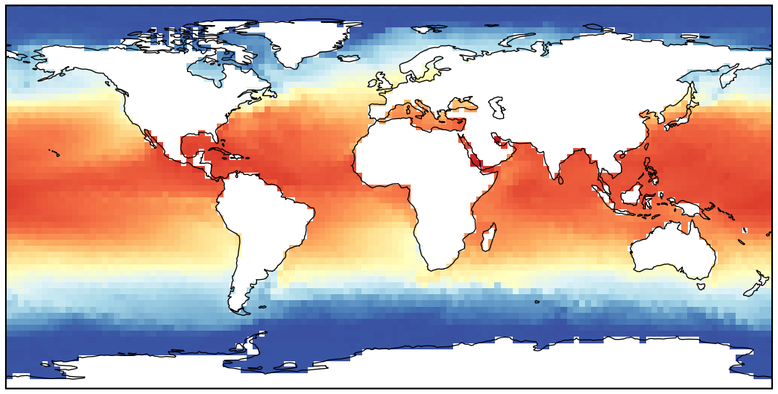


Chain init: 2023-09-23 — Test (°C)

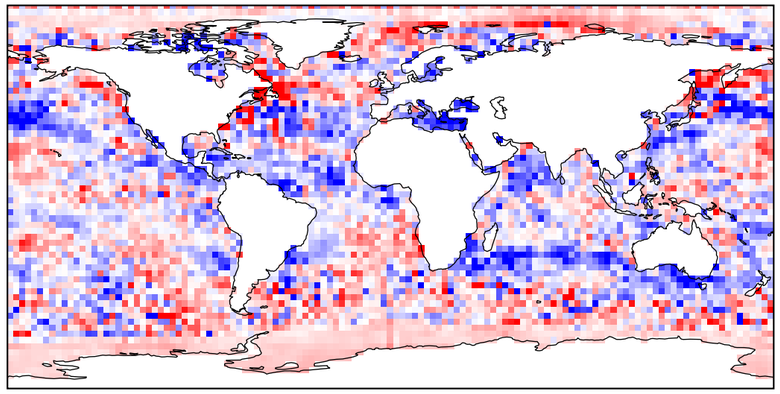
Actual — Step 1 (°C)



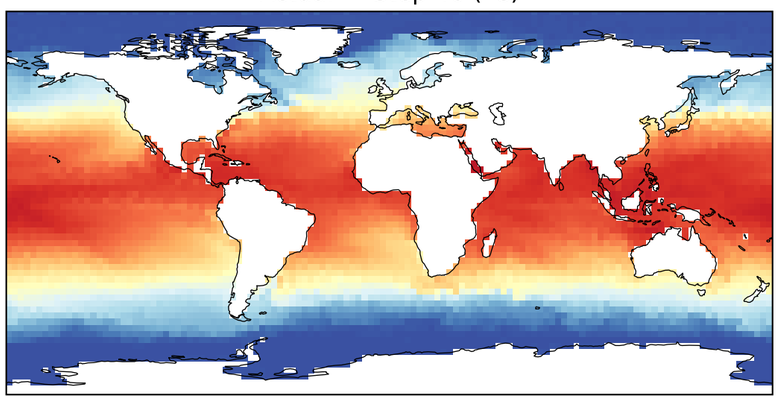
SFNO — Step 1 (°C)



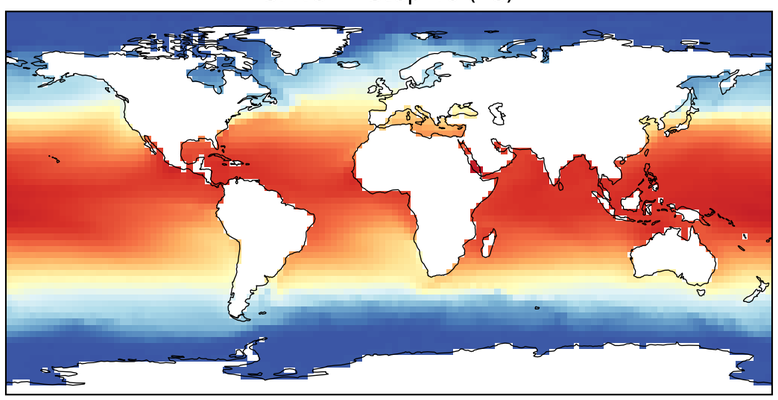
Error — Step 1 (°C)



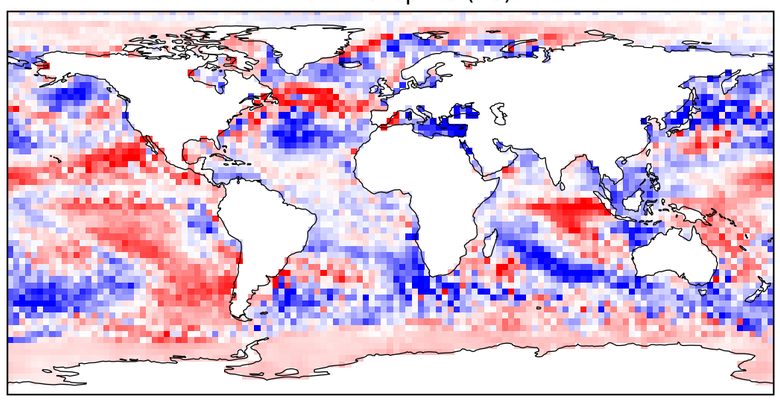
Actual — Step 46 (°C)



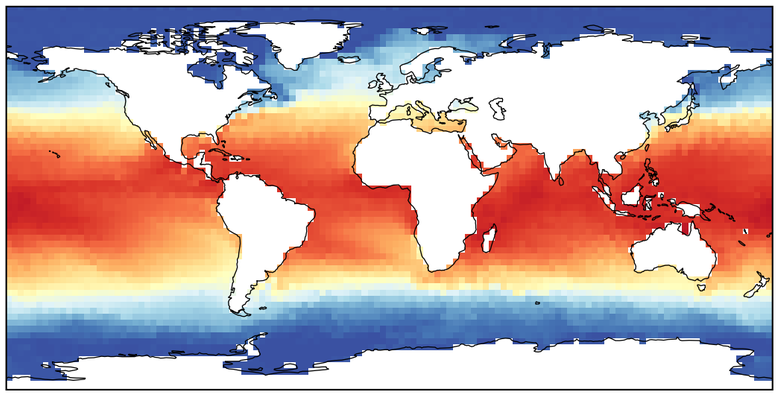
SFNO — Step 46 (°C)



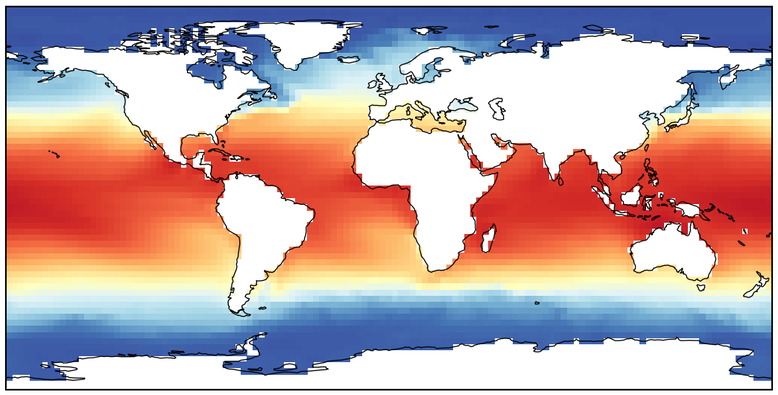
Error — Step 46 (°C)



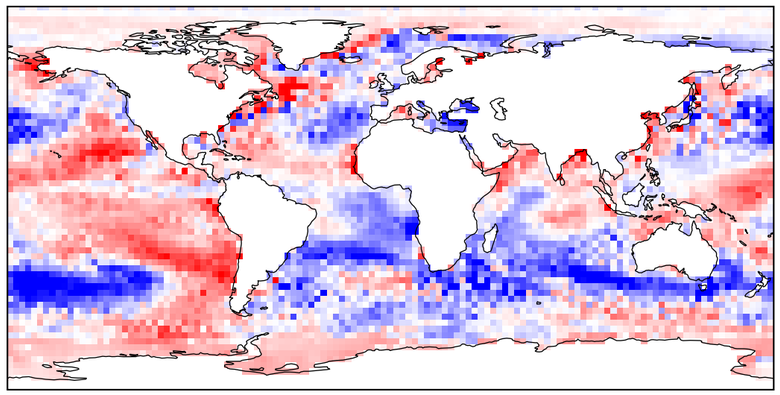
Actual — Step 90 (°C)

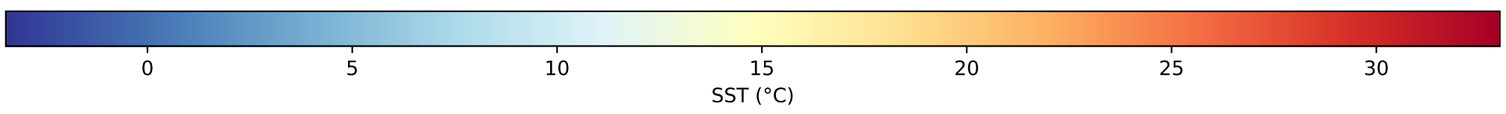


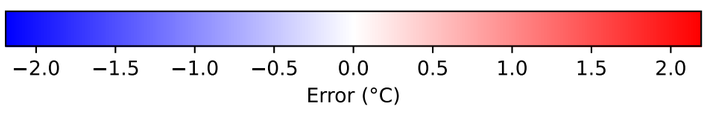
SFNO — Step 90 (°C)



Error — Step 90 (°C)

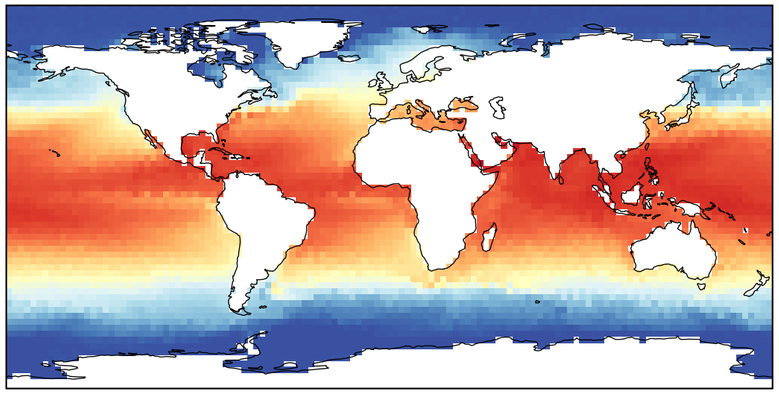




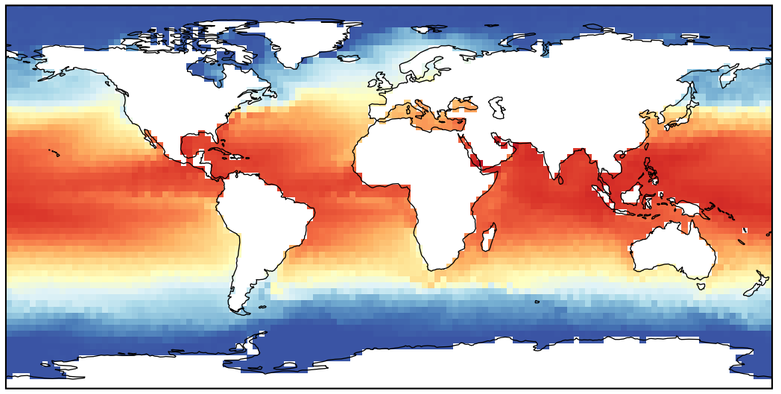


Chain init: 2024-06-19 — Test (°C)

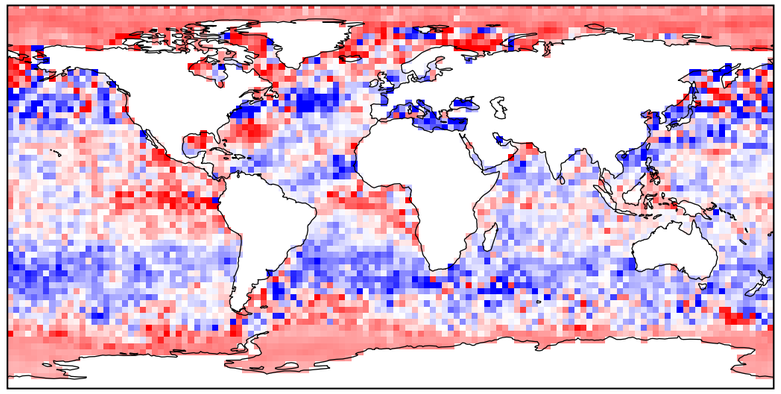
Actual — Step 1 (°C)



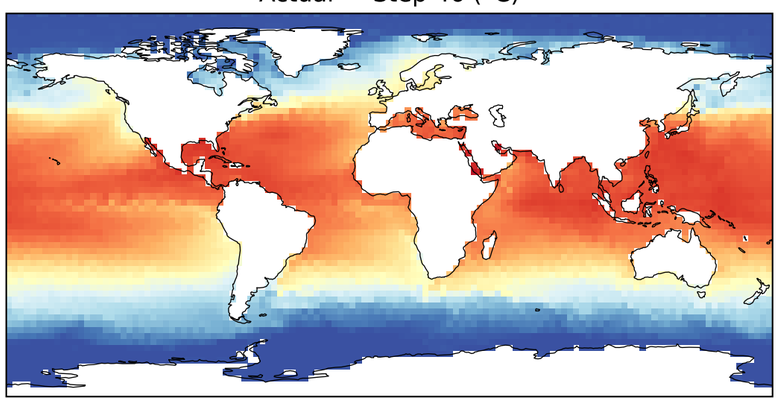
SFNO — Step 1 (°C)



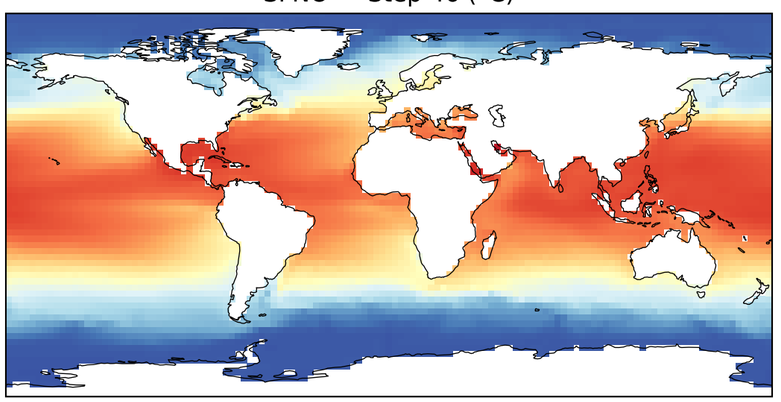
Error — Step 1 (°C)



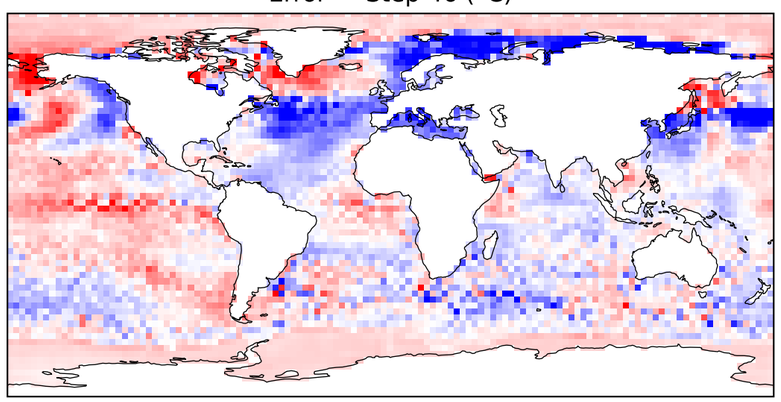
Actual — Step 46 (°C)



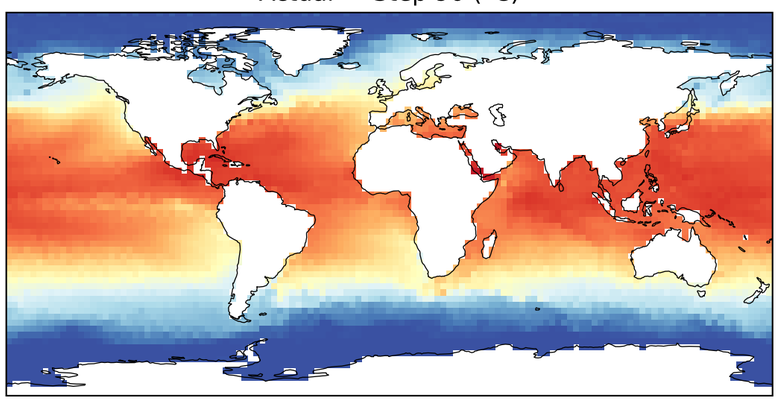
SFNO — Step 46 (°C)



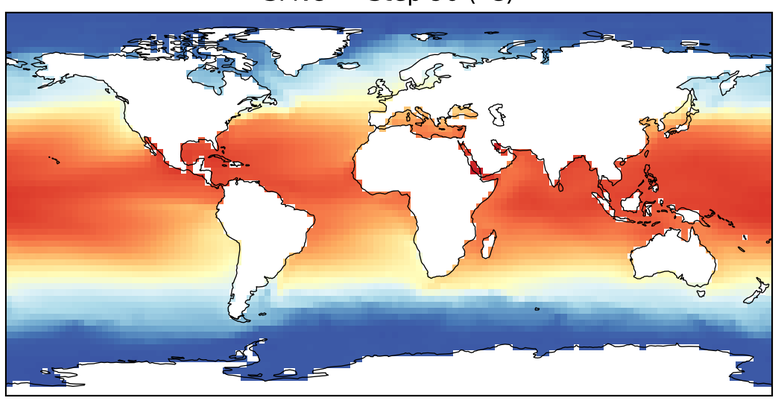
Error — Step 46 (°C)



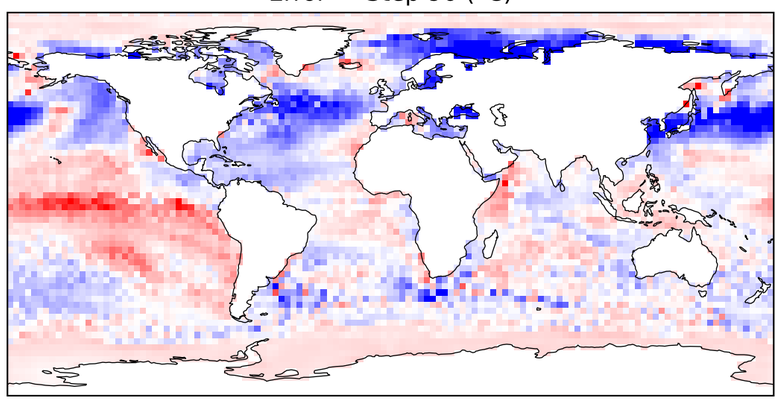
Actual — Step 90 (°C)

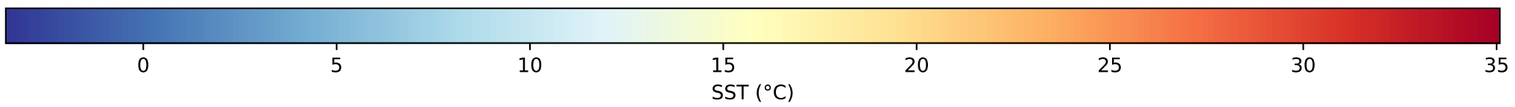


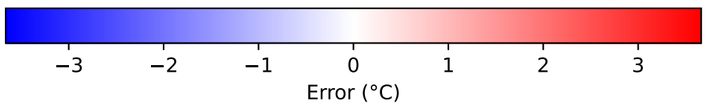
SFNO — Step 90 (°C)



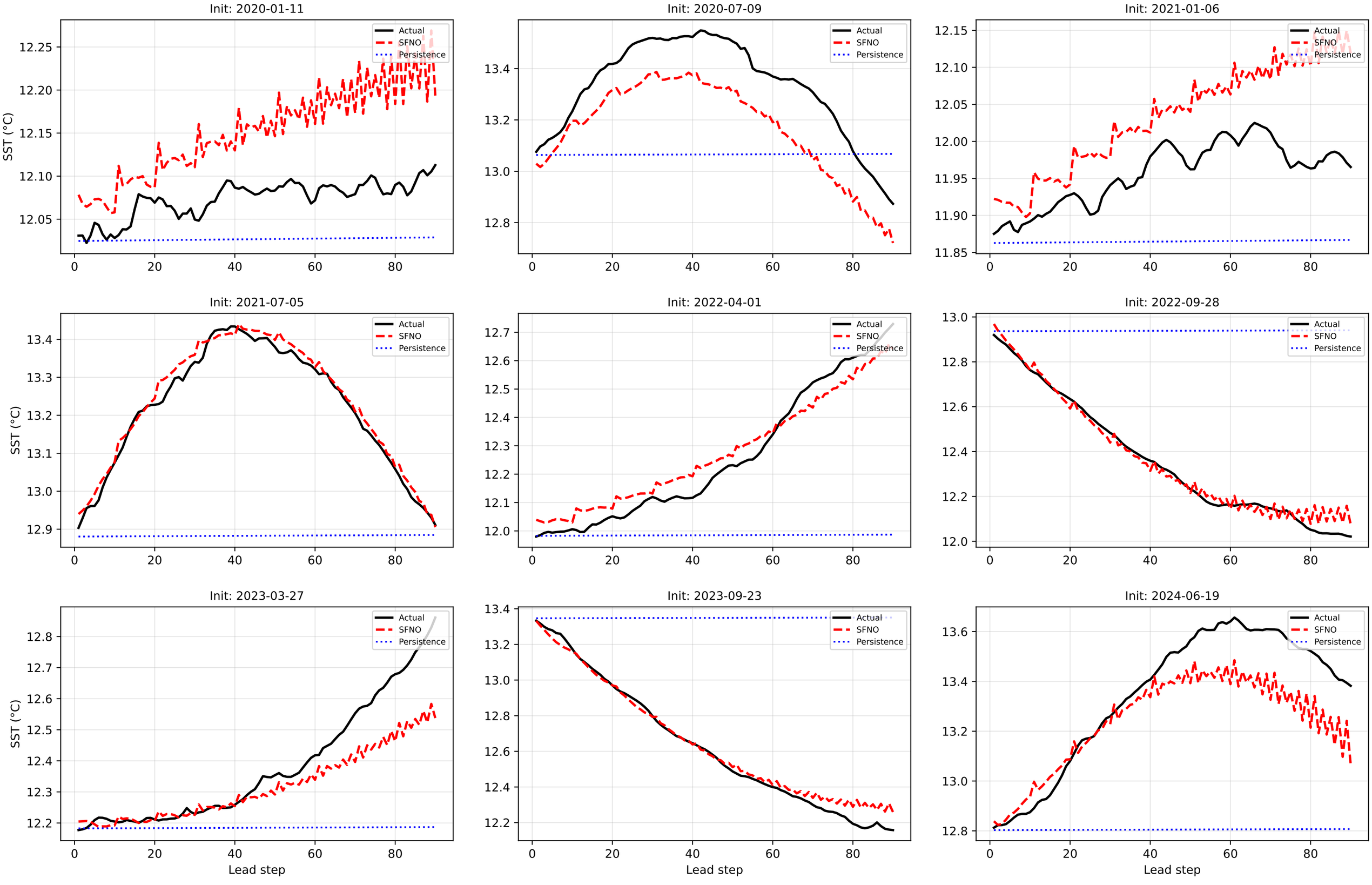
Error — Step 90 (°C)







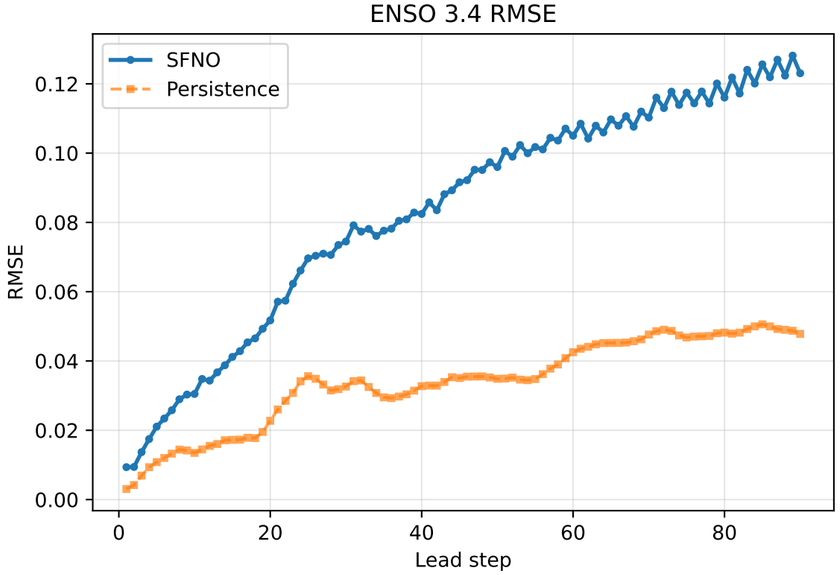
Global-Mean SST — Test (°C)

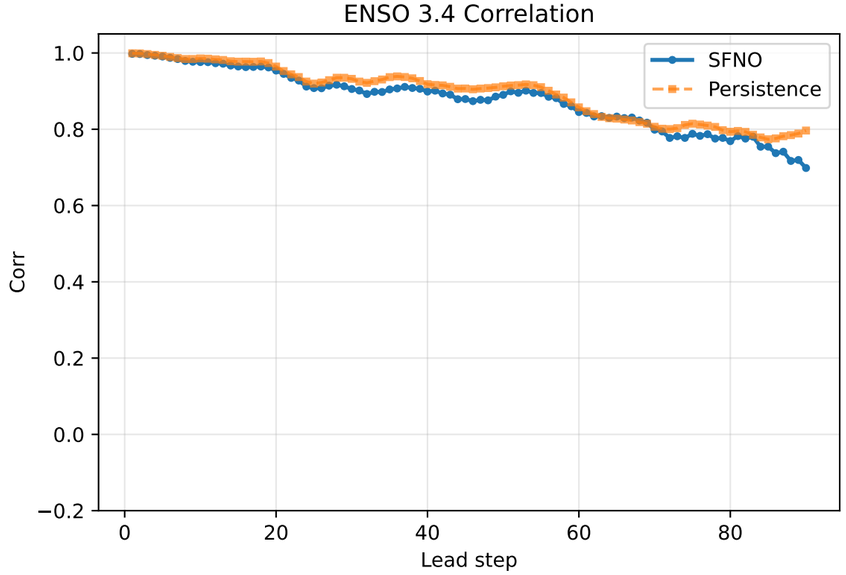


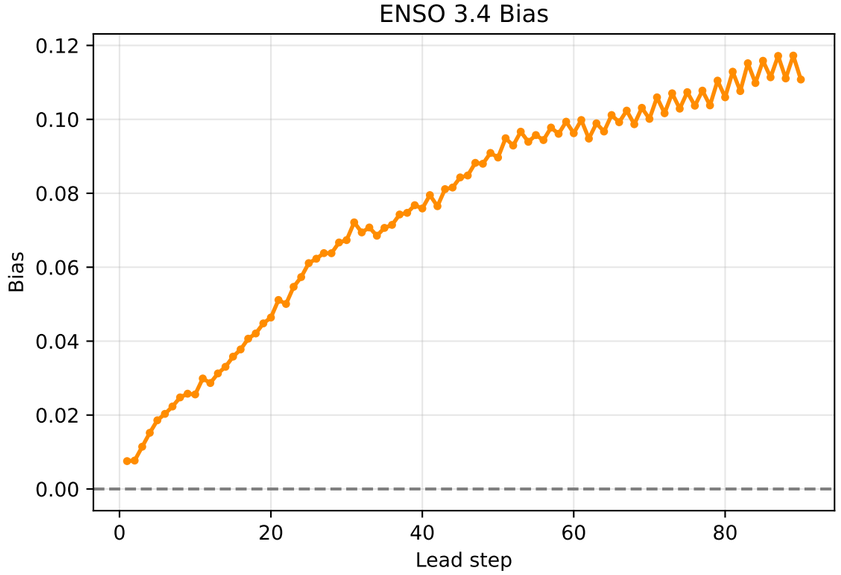
ENSO 3.4 Metrics — Test

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0094	0.0031	0.9977	0.9992	0.0075
2	0.0094	0.0042	0.9973	0.9989	0.0077
3	0.0137	0.0069	0.9946	0.9972	0.0114
4	0.0175	0.0093	0.9929	0.9950	0.0152
5	0.0210	0.0108	0.9907	0.9927	0.0186
6	0.0234	0.0120	0.9873	0.9896	0.0203
7	0.0258	0.0133	0.9841	0.9867	0.0223
8	0.0289	0.0144	0.9791	0.9842	0.0248
9	0.0303	0.0142	0.9770	0.9843	0.0258
10	0.0305	0.0135	0.9762	0.9857	0.0256
11	0.0348	0.0145	0.9759	0.9850	0.0299
12	0.0343	0.0155	0.9732	0.9833	0.0287
13	0.0367	0.0160	0.9719	0.9817	0.0313
14	0.0388	0.0171	0.9671	0.9784	0.0331
15	0.0412	0.0172	0.9646	0.9772	0.0358
16	0.0428	0.0173	0.9630	0.9774	0.0378
17	0.0454	0.0178	0.9634	0.9770	0.0407
18	0.0465	0.0177	0.9645	0.9777	0.0421
19	0.0493	0.0195	0.9621	0.9736	0.0448
20	0.0517	0.0227	0.9540	0.9644	0.0464
21	0.0571	0.0260	0.9452	0.9526	0.0511
22	0.0574	0.0285	0.9347	0.9440	0.0501
23	0.0623	0.0308	0.9277	0.9369	0.0547
24	0.0661	0.0341	0.9120	0.9251	0.0573
25	0.0696	0.0356	0.9080	0.9197	0.0611
26	0.0704	0.0349	0.9075	0.9226	0.0623
27	0.0710	0.0332	0.9143	0.9287	0.0638
28	0.0706	0.0315	0.9168	0.9351	0.0638
29	0.0735	0.0319	0.9127	0.9350	0.0667
30	0.0745	0.0326	0.9056	0.9322	0.0673
31	0.0792	0.0342	0.9015	0.9251	0.0721
32	0.0773	0.0344	0.8924	0.9217	0.0694
33	0.0781	0.0325	0.8982	0.9261	0.0707
34	0.0761	0.0308	0.8977	0.9305	0.0685
35	0.0776	0.0295	0.9044	0.9367	0.0706
36	0.0782	0.0293	0.9070	0.9386	0.0714
37	0.0805	0.0297	0.9111	0.9370	0.0743
38	0.0809	0.0304	0.9082	0.9336	0.0747
39	0.0829	0.0314	0.9061	0.9265	0.0768
40	0.0824	0.0327	0.8988	0.9182	0.0759
41	0.0858	0.0329	0.9012	0.9161	0.0795
42	0.0835	0.0329	0.8939	0.9151	0.0765
43	0.0881	0.0339	0.8908	0.9113	0.0811
44	0.0893	0.0353	0.8788	0.9066	0.0816
45	0.0916	0.0351	0.8793	0.9069	0.0843
46	0.0922	0.0355	0.8737	0.9049	0.0848
47	0.0952	0.0355	0.8772	0.9067	0.0882
48	0.0951	0.0355	0.8758	0.9080	0.0880
49	0.0974	0.0352	0.8856	0.9099	0.0909
50	0.0960	0.0349	0.8903	0.9127	0.0897
51	0.1006	0.0349	0.8995	0.9148	0.0949
52	0.0990	0.0352	0.8956	0.9157	0.0929
53	0.1023	0.0346	0.9012	0.9176	0.0967
54	0.1000	0.0345	0.8953	0.9155	0.0939
55	0.1018	0.0348	0.8950	0.9104	0.0957
56	0.1011	0.0362	0.8852	0.9010	0.0944
57	0.1044	0.0378	0.8817	0.8912	0.0978
58	0.1036	0.0390	0.8665	0.8831	0.0961
59	0.1071	0.0408	0.8603	0.8706	0.0994
60	0.1050	0.0425	0.8452	0.8575	0.0963
61	0.1085	0.0435	0.8429	0.8473	0.0998
62	0.1042	0.0441	0.8336	0.8398	0.0948
63	0.1079	0.0448	0.8343	0.8324	0.0989
64	0.1060	0.0452	0.8298	0.8290	0.0967
65	0.1097	0.0452	0.8332	0.8279	0.1012
66	0.1079	0.0452	0.8293	0.8256	0.0992
67	0.1107	0.0453	0.8308	0.8230	0.1024
68	0.1076	0.0457	0.8230	0.8187	0.0987
69	0.1120	0.0462	0.8173	0.8146	0.1031
70	0.1103	0.0476	0.7989	0.8065	0.1001
71	0.1160	0.0486	0.7943	0.8014	0.1060
72	0.1130	0.0490	0.7773	0.8005	0.1017
73	0.1177	0.0487	0.7817	0.8030	0.1071
74	0.1139	0.0474	0.7773	0.8113	0.1029
75	0.1175	0.0468	0.7881	0.8146	0.1074
76	0.1144	0.0470	0.7827	0.8124	0.1037
77	0.1178	0.0471	0.7871	0.8097	0.1078
78	0.1143	0.0472	0.7756	0.8062	0.1038
79	0.1201	0.0480	0.7773	0.7976	0.1105
80	0.1160	0.0482	0.7689	0.7935	0.1060
81	0.1218	0.0479	0.7821	0.7956	0.1129
82	0.1172	0.0482	0.7756	0.7931	0.1077
83	0.1241	0.0492	0.7805	0.7855	0.1152
84	0.1201	0.0500	0.7541	0.7786	0.1098
85	0.1256	0.0506	0.7544	0.7738	0.1158
86	0.1219	0.0500	0.7374	0.7761	0.1114
87	0.1270	0.0492	0.7411	0.7818	0.1172
88	0.1224	0.0490	0.7169	0.7847	0.1111
89	0.1282	0.0487	0.7195	0.7888	0.1173
90	0.1231	0.0478	0.6984	0.7967	0.1108

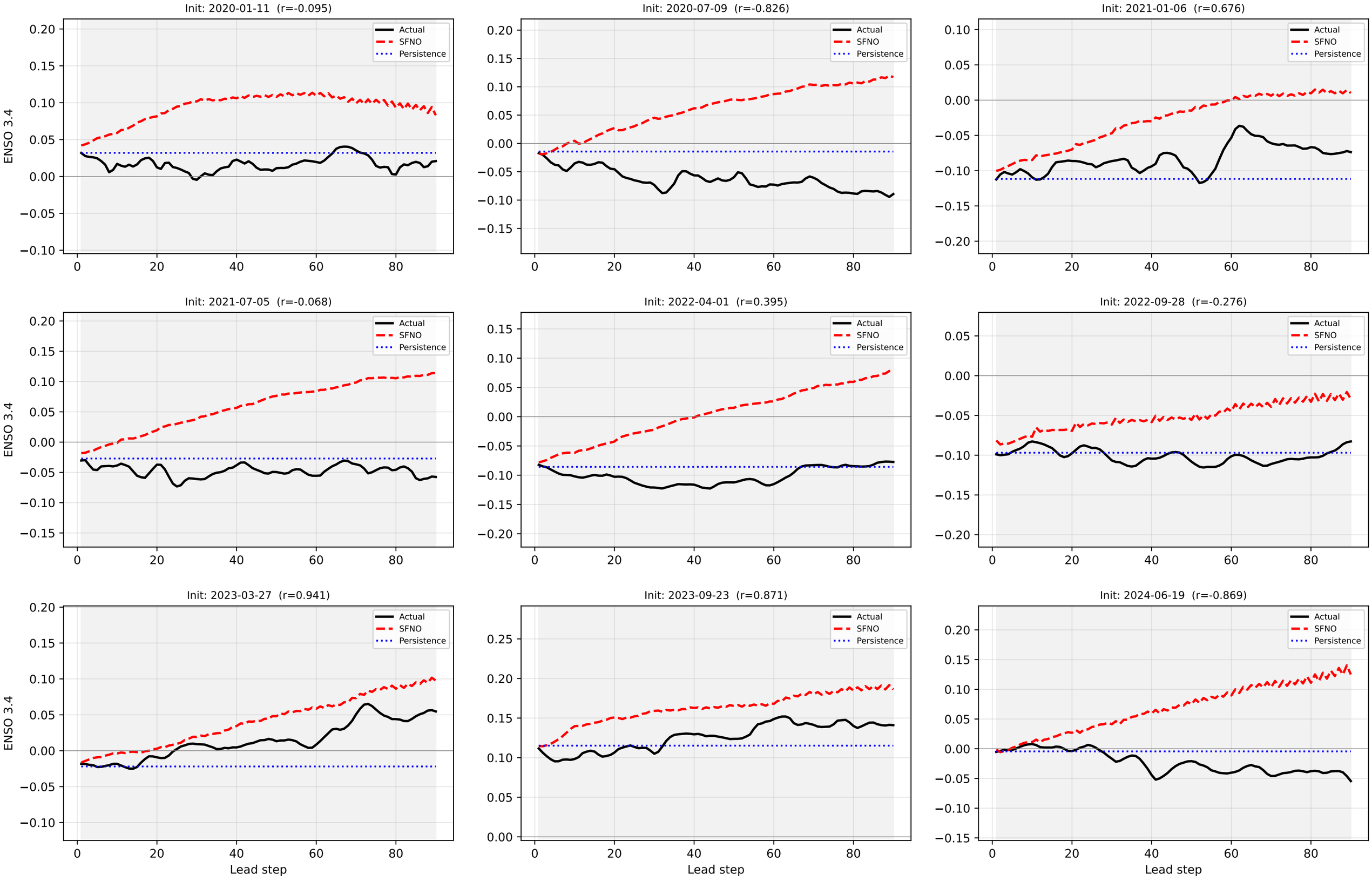
ENSO 3.4 Forecast Skill — Test







ENSO 3.4 Index — Test



ENSO 3.4 Scatter — Test